IPO Spillovers: Evidence from Private-Firm Networks*

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Abstract

I study how Initial Public Offering (IPO) decisions propagate through firm networks formed by common investors. Using a dynamic network constructed from Crunchbase data for 2000-2023, I show that firms are more likely to go public after directly connected peers complete IPOs, with a 10-percentage-point increase in peer IPO exposure associated with an approximately 65% increase relative of the average annual IPO probability. These spillovers are stronger when firms share a lead investor, but weaker when their peers have already been exposed to IPO activity through more distant network connections. Firms with prior peer IPO exposure also tend to go public at higher valuations and share prices, subsequently attract new investors and receive more private funding. To rationalize the findings, I develop a dynamic learning model in which peer IPOs provide connected private firms with information about how the public market value their common investors. This learning raises the price that firms expect to obtain from going public and can account for the main empirical patterns, including stronger spillovers through lead investors and weaker effects when similar information has already arrived through the network.

Keywords: IPOs, Corporate Finance, Financial Networks, Private Markets, Information Spillovers.

JEL Codes: G14, G24, G32, D22, D85.

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1 Introduction

Over the past few decades, firms have gained access to a wider range of private financing options. The expansion of venture capital and private equity has allowed high-growth firms to raise substantial amounts of capital while remaining private. At the same time, Initial Public Offering (IPO) activity has declined, with fewer firms making the transition from private to public markets (Ewens and Farre-Mensa, 2020; Kwon et al., 2020). Despite the increasing supply of private financing, going public remains an important decision for firms. An IPO allows firms to raise capital from public investors, but the decision also depends on the valuation they expect to obtain in the public market. Before going public, firms may be uncertain about this valuation, and this can influence whether and when they decide to undertake an IPO.

Prior to reaching this stage, firms often spend several years raising capital from private investors. During this period, they establish relationships with investors who provide not only financing but also advice and expertise. These relationships are relevant for a firm's development, particularly when it grows and consider whether to enter public markets.

An important feature of this environment is that private investors rarely finance a single company. Venture capitalists and other investors hold stakes in several firms at the same time, supporting them through different stages of their development. This creates links between different firms through these common investors, often long before any of them go public. Therefore, when a portfolio company decides to go public, its decision may affect other firms backed by the same investors. However, whether these links across firms generate spillovers in IPO decisions is not obvious given that firms differ in their financing needs, growth opportunities, ownership structures, and private information, while companies that share investors may also face similar market conditions. This raises the central question of the paper: does the IPO of one firm affect the subsequent IPO decisions of other firms connected to it through common investors?

I study this question using a large-scale dynamic network of firms constructed from Crunchbase financing data covering 2000–2023. Two firms are connected whenever they share at least one investor, with these connections evolving as firms raise more capital and form new investor relationships. Combining this network with information on IPO events, I measure each private firm's exposure to previous IPOs among its peers in each year. I then examine whether firms become more likely to go public following IPOs by their peers. The empirical analysis exploits within-firm variation in peer IPO exposure over time, controlling for time-invariant firm characteristics, common shocks at the year-by-industry-by-country level, and time-varying characteristics of both focal firms and their peers.

I find that firms become significantly more likely to go public following IPO activity

among their directly connected peers. In the baseline specification, a 10-percentage-point increase in the share of peers going public in the previous year is linked to a 0.17-percentage-point increase in the focal firm's IPO probability, compared with an unconditional annual IPO probability of about 0.26 percent. This represents an increase of about 65% of the average annual probability, representing a economically meaningful effect given the rarity of IPO events. The result is robust across alternative specifications and measures of network exposure.

I also show that the strength of these spillovers depends on the nature of the connections across firms. Specifically, a 10-percentage-point increase in peer IPO exposure is associated with a 0.60-percentage-point increase in the focal firm's IPO probability when at least one IPO peer shares a lead investor with the focal firm, compared with a 0.13-percentage-point increase when no IPO peer shares a lead investor. The difference is statistically significant suggesting that the role of common investors matters for how IPO activity spreads through the network. To examine the role of product-market competition, I distinguish between IPOs by peers operating in the same industry and those operating in different industries. Although the estimated spillovers differ, the difference is not statistically significant, providing no clear evidence that IPO spillovers are stronger among firms operating in the same industry.

The structure of the surrounding network also matters, with the relationship between a directly connected peer's IPO and the focal firm's IPO probability being weaker when that peer has previously been exposed to IPO activity through more distant network connections. Beyond IPO decisions, firms with prior peer IPO exposure also tend to go public at higher valuations and share prices, attract new investors, and raise more private funding. In particular, I find that prior peer IPO exposure is linked to approximately 72% higher IPO valuations, conditional on firm characteristics and year, industry, and country fixed effects.

However, it is difficult to determine the reasons for these IPO spillovers since they may be related to similarities in their IPO timing. Operating in the same industry, facing similar financing conditions, having access to similar investment opportunities and being subject to common changes in market conditions might explain these patterns. To address these concerns, the baseline specifications include firm fixed effects and year-by-industry-by-country fixed effects. I further examine the results using alternative fixed effects based on firms' investor and lead-investor relationships, as well as placebo networks that randomly rewire connections between firms while preserving their number of eligible peers. I also control for investor characteristics and exploit policy-induced variation in peer IPO exposure using an instrumental-variable strategy.

The question then is what mechanisms might explain the relationship between IPO decisions across firms that share investors. Investors holding stakes in multiple firms may

coordinate the timing of their portfolio companies' IPOs, encouraging some to go public earlier while delaying others depending on financing needs and market conditions. Beyond coordination, product-market competition may also influence IPO timing, as going public provides firms with resources for investment and acquisitions, potentially encouraging their competitors to reconsider when to go public. Alternatively, firms connected through common investors may learn from a peer's IPO decision and update their expectations about the valuation they could obtain if they went public themselves.

While the empirical evidence does not fully rule out investor coordination or product-market competition, some of the findings are particularly consistent with an informational explanation. The absence of a statistically significant difference between same-industry and different-industry peer IPO spillovers provides no clear evidence that product-market competition is the primary mechanism. Moreover, stronger spillovers through common lead investors and weaker effects when IPO peers have previously been exposed to more distant IPO activity are consistent with information transmission through firm networks.

To rationalize my findings, I develop a dynamic learning framework in which private firms are uncertain about how the public market values the investors backing them. The public market knows these valuations, while private firms only observe noisy signals. When a connected portfolio company chooses whether to go public, its decision partially reveals information about the value the public market assigns to its investors. Then, peer firms observe this decision and update their beliefs about how the public market would value that investor in their own case. These updates affects their expected valuations and, consequently, their incentives to go public. The model illustrates how an IPO decision can convey information beyond the firm going public, where connected firms use what they observe to reassess their own prospects in public markets. The information that is transmitted also depends on which investors firms have in common and on what has already been learned from earlier IPO decisions.

Finally, I extend the theoretical analysis to explore the implications of learning for IPO underpricing and consider how information transmission operates in more general network structures.

Related Literature

Prior work emphasizes firm life-cycle considerations, ownership and control, access to capital, and the trade-off between public and private financing (Zingales, 1995; Pagano et al., 1998; Thomas J Chemmanur and Fulghieri, 1999; Ritter and Welch, 2002; Brau and Fawcett, 2006). More recent work examines how developments in private capital markets have changed firms' financing opportunities and the decision to remain private or access public markets (Ewens and Farre-Mensa, 2020; Stulz, 2020; Bernstein, 2022). In

particular, Ewens and Farre-Mensa (2020) show that increased access to private capital changed the going public decision versus staying private trade-off for late-stage startups, allowing them to remain private for longer, and associated with founders having more control to decide on exit decisions.

A closely related literature studies clustering and fluctuations in IPO activity. IPO waves may reflect changes in market conditions, uncertainty, investor sentiment, or the arrival of firms with similar incentives to go public (Lowry, 2003; Pástor and Veronesi, 2005; He, 2007; Yung et al., 2008). Particularly relevant to my paper are models in which earlier IPOs generate information that affects subsequent IPO decisions. Hoffmann-Burchardi (2001) studies sequential information revelation in IPO markets, while Lowry and Schwert (2002) and Altı (2005) examine learning and market timing during IPO cycles. Aghamolla and Guttman (2021) develop a dynamic model in which firms strategically time their IPOs in the presence of information externalities and may delay going public to observe the market reception of earlier offerings.

My paper contributes to this literature by studying the network dimension of IPO spillovers and showing that IPO activity among firms connected through common private investors predicts subsequent IPO decisions. The theoretical framework provides an interpretation of these findings based on sequential learning. Rather than learning only from aggregate IPO market conditions or the general market reception of previous offerings, firms learn from IPO decisions occurring within their private-financing network about how the public market values investors that are common with other firms.

The paper also contributes to the literature on corporate peer effects and the diffusion of firm decisions through networks. Peer effects have been documented in corporate investment, and financing, and other firm decisions in peer effects model (Leary and Roberts, 2014; Chen and Ma, 2017). Chen and Ma (2017) sheds light on how firms are affected by the corporate policies (e.g. changes in leverage) taken by peers. Firms may learn from the market valuations of their peers and increase investment while remaining private, as in Foucault and Fresard (2014). In another work by Helmers et al. (2017) connections through shared board directors can facilitate information transmission across firms, influencing their investment and innovation decisions. The literature on peer effects also emphasizes the difficulty of distinguishing endogenous peer effects from common shocks, correlated characteristics, and endogenous network formation (Manski, 1993; Bramoullé et al., 2009; De Giorgi et al., 2010; Bramoullé et al., 2020; Bailey et al., 2022; Foroughi et al., 2022).

Within IPO markets, Hsu et al. (2010) study the competitive effects of IPOs on industry rivals and the impact on firm's performance, while Aghamolla and Thakor (2022) show that private firms' IPO decisions respond positively to recent IPOs by direct product-market competitors in the pharmaceutical industry. My paper instead defines peer re-

lationships through common private investors rather than industries. These connections arise from firms' financing histories rather than solely from product-market similarity and characterize a network through which IPO events can be related across portfolio firms. This also allows me to go beyond documenting a direct peer effect and examine how the strength of IPO spillovers varies with the economic importance of the shared investor and with prior IPO activity in the surrounding network.

My paper also relates to the literature on common ownership, although it differs in the underlying mechanism. For instance, Li et al. (2023) examine how common venture-capital ownership influences innovation decisions across portfolio firms, while López and Vives (2019) study how overlapping ownership affects firms' strategic incentives through the internalization of rivals' profits. Unlike these studies, my paper does not focus on the effects of common ownership on firms' objectives or competitive behavior. In my paper, common investors create a network through which IPO decisions convey information about public-market valuations, influencing connected firms' decisions to go public.

The paper is also linked to the literature on venture capital and the role of investors in firms' development. Venture capitalists provide not only financing, but also monitoring, governance, expertise, and access to professional networks (Hellmann and Puri, 2002; Thomas J Chemmanur, K. Krishnan, et al., 2011; Bernstein et al., 2016; Bottazzi et al., 2008), while connections among venture capitalists affect deal flow and investment performance (Lerner, 1994; Hochberg et al., 2007).

A closely related work, Eldar and Grennan (2024) show that common venture capital investors affect startup growth and outcomes. They find that same-industry startups sharing VC investors raise more capital, fail less often, and experience more successful exits, with common VC directors playing an important role, consistent with information exchange within portfolios. My paper examines a broader set of private firms and investors, including connections across industries and countries, and focuses on whether IPO activity propagates through a dynamic common-investor network. I also examine how these spillovers depend on the structure of investor connections, including common lead investors and prior IPO exposure among second-order peers. Unlike information exchange through common board representation, my theoretical framework shows how publicly observable IPO decisions can transmit information across connected firms. In my empirical part, the spillovers remains positive after controlling for investors' historical IPO experience and is stronger, and not limited to, connections through common lead investors.

The paper also relates to work studying how information generated in public markets affects decisions in private capital markets. P. Gompers et al. (2008) show that venture-capital investment responds to signals from public equity markets, while Liu and Tian (2022) provide evidence that venture capitalists learn information contained in public

stock prices when making financing decisions. Closely related, Yan (2024) shows that private firms learn from the stock-market valuations of public firms in the same industry and adjust their investment decisions accordingly. I complement this literature by studying a different source of public-to-private-market spillovers. Rather than focusing on information contained in the stock prices of public firms, I study whether the IPO decisions of firms connected through common investors transmit information to other firms about how the market values investors and influences subsequent IPOs.

Finally, the paper is related to the literature on the certification and reputational role of financial intermediaries and investors in IPOs. Reputable venture capitalists and other intermediaries can reduce informational frictions and are linked to IPO outcomes (Megginson and Weiss, 1991; Thomas J. Chemmanur and Fulghieri, 1994; P. A. Gompers, 1996; Nahata, 2008; Bajo et al., 2016; Nanda et al., 2020). In particular, Nahata (2008) link VC reputation based on prior IPO success to subsequent portfolio-company outcomes, Nanda et al. (2020) examine the persistence of VC performance following early success.

This literature helps explain why public markets may assign different values to the investors backing a firm, particularly through their reputation and certification role. My paper builds on this idea by examining how private firms learn about the market's valuation of their investors through IPO decisions within a common-investor network.

Road Map The rest of the paper is organized as follows. Section 2 describes the data and the sample construction. Section 3 shows the empirical framework and identification strategy. Section 4 reports the main empirical results. Section 5 performs robustness tests and identification challenges. Section 6 introduces the theoretical model and derives its main results. Section 7 comments on different policy implications. Section 8 concludes.

2 Data and Network Construction

2.1 Data sources

The data comes from Crunchbase.com's academic API. Crunchbase combines records on organizations, funding rounds, investments, investment partners, investors, and IPOs. This database is widely used in research on firms and private capital markets and provides unusually broad coverage of financing relationships, investor identities, firm characteristics, and exit events Moon and Suh (2021), Abakah et al. (2024), Zava (2024), and Bonelli (2026).

Although Crunchbase provides extensive information on startups, typically characterized as young, high-growth firms often backed by external investors, the database also includes established companies and other organizations. Throughout the paper, however,

I use the more general term *firm*, since the empirical sample is not restricted to companies satisfying a particular firm definition. The sample includes all firms reported in Crunchbase that received external financing. The coverage becomes materially more complete after 2000, and my empirical analysis covers the period 2000–2023.

Crunchbase also identifies the investors financing these firms. The investors database contains information on investor characteristics, including investor type, location, and investment activity. Investors include individuals and organizations, ranging from angel investors and venture capital firms to corporate investors and private equity firms. I use these investor–firm relationships to construct the network analyzed in the paper.

The funding rounds database provides information on financing events, including the type of funding round, amount raised, and lead investors. These rounds range from early-stage financing, such as angel and seed investments, to venture capital and later-stage private equity financing. I combine these records with the investments and investment partners databases, which identify the investors and investment partners participating in each financing round.

I merge investor-firm relationships with financing information and exclude post-IPO funding rounds to focus on investment relationships established while firms are private. The resulting dataset contains 208,918 total number of firms and 155,524 investors. IPO information is matched to the corresponding firms, and firms with no recorded IPO are treated as remaining private in the data. The subsequent analysis focuses on firms connected to at least one other firm through a common investor.

Crunchbase does not report every private transaction and often lacks financing amounts for smaller or less visible rounds. Missing amounts are therefore retained as missing rather than coded as zero. Appendix A.1 provides further details on the underlying databases, missingness, investor types, financing-round classifications, and additional summary statistics.

2.2 Dynamic Link Formation

The empirical environment is represented by a dynamic network in which links between firms are formed through common investors. Investors participate in funding rounds over time, and the network is observed annually over the period 2000–2023.

Let $\mathcal{K}_{it}$ denote the set of investors that have financed firm i by year t. Two firms i and j are connected in year t if at least one investor has financed both firms by that date. The corresponding adjacency element is

$$g_{ij,t} = \mathbf{1}\left\{\mathcal{K}_{it} \cap \mathcal{K}_{jt} \neq \emptyset\right\}.$$

Equivalently, $g_{ij,t} = 1$ if there exists an investor k that has invested in both firms i and j by year t, and $g_{ij,t} = 0$ otherwise. Thus, if investor k first finances firm i in year t and subsequently finances firm j in year $t + T$, the link between i and j becomes active in year $t + T$.

Because the data do not report when investor–firm relationships are dissolved, links are treated as persistent: once two firms become connected through a common investor, the corresponding edge remains active in all subsequent periods. This construction captures the idea that common investors can generate persistent informational, reputational, and governance relationships across their portfolio firms, even after their direct involvement has ended. Given that firm's previous investors are publicly observable, these connections are informative even when firms no longer share that investor. Therefore, the resulting adjacency matrix, $G_t = (g_{ij,t})$ is dynamic and cumulative. As new funding rounds take place, additional common-investor relationships create new edges. The set of first-order (or direct) peers of firm i in year t is

$$N_{i,t} = \{j \neq i : g_{ij,t} = 1\}, \qquad d_{i,t} = |N_{i,t}|,$$

where $d_{i,t}$ denotes the time-varying degree of firm i at period t equivalent to $|N_{i,t}|$ that shows the cardinality (size) of the neighborhood i at period t. $N_{i,t}$ denotes the set of firms connected to firm i through common investors in period t.

Second-order neighborhoods evolve accordingly. In each year t, the second-order peers of firm i consist of the first-order peers of firms in $N_{i,t}$, excluding firm i itself and its direct peers. Then, if a firm j is a peer of i and $k \in N_{jt}$, but $k \notin N_{it}$, k is a second-order (indirect) peer of i because it does not belong to the first-order (direct) peers of i, but it belongs to the direct peers of j.

The dynamic structure of the network is constructed using the date information available in the funding rounds database through two variables: the announced_on variable, and in the investments and investment partners databases through created_at variable. Investments that occurred before 2000 are treated as pre-existing links and are therefore included in the initial network structure at the beginning of the sample period. Incorporating these pre-2000 connections is important because some investor–firm relationships were already established before the start of the analysis window, and excluding them would artificially understate the degree and connectivity of firms in the early years of the sample.

The investment creation date is used as a proxy when the exact date of the funding round is unavailable. As a result, some links that were built close to year 2000 may be recorded with delays if the corresponding information was added to the database several some years later. While this may introduce some measurement error in the timing of a

subset of links, the overall network structure and the persistence of investor relationships remain largely unaffected.

Once a firm goes public, it exits the risk set and is excluded from the set of eligible peers in subsequent years. The network contains 198,421 firms connected to at least one other firm through a common investor, of which 198,243 remain in the sample after excluding firms that had previously completed an IPO. Hence, some firms that are in the dataset have no peers, and thus, do not enter in the analysis. Table 15 in Appendix A.2 reports the annual distribution of connected firms. In my dataset, Level is the firm with the highest number of peers in year 2023. However, up to when Uber went public, it was Uber the most-connected firm. The total number of connected firms before excluding IPO firms was 198421 firms.

The estimation sample is restricted to firm-years with at least one active first-order peer. Section 3 describes this restriction when defining peer exposure. Regarding second-order connections, a total of 193,937 firms are connected to at least one second-order neighbor at some point during the sample period. This network structure generates partially overlapping peer groups across firms, creating variation in peer exposure through both direct and indirect investor connections.

3 Empirical Framework

3.1 Empirical Framework and Identification

This section studies whether IPO activity among connected firms affects a firm 's probability of going public. The empirical analysis focuses on firms connected through common investors within the dynamic firms network described in the previous section. The objective is to examine whether exposure to IPO activity among connected firms is linked to a higher likelihood that a focal firm undertakes an IPO.

The baseline specifications include firm fixed effects and year-by-industry-by-country fixed effects. Firm fixed effects absorb time-invariant firm characteristics, including the firm's industry, employee category, total number of funding rounds, and total amount raised over the full observation period. These variables are therefore not entered separately in the regressions. The year-by-industry-by-country fixed effects control for aggregate IPO conditions, including the hot markets phenomena, while allowing common shocks to vary flexibly across industry-country cells in each year.

In addition, I construct a set of time-varying controls capturing firm age, investor reputation, financing history, investor industry specialization, and the participation of lead investors. Throughout the empirical analysis, I refer to firm i as the *focal firm*, that is, the firm whose IPO outcome is being explained. Its connected firms are referred to

as its peers and are used to construct the peer-IPO exposure measures and peer-level characteristics. Unless otherwise indicated, these variables enter the regressions lagged by one year.

For each investor j and year t, I define the investor score as the number of portfolio firms that went public before year t, divided by the cumulative number of distinct firms financed by that investor up to year t. At the firm-year level, I average this score across the investors financing the focal firm.

I distinguish between two measures of cumulative financing activity. The cumulative number of unique investors counts each investor only once for a given firm, even when the same investor participates in several funding rounds or invests repeatedly in different years. Investor relationships established before 2000 are included in the initial stock of investors observed at the beginning of the estimation period. By contrast, the cumulative number of investments counts every observed investment event separately, including repeated investments by the same investor within or across years. This measure uses investment records from 1901 onward and therefore captures the cumulative intensity of financing activity received by the firm.

I also construct two controls describing the composition of the firm's investor portfolio. First, the leave-one-out investor-industry specialization measure captures the extent to which the firm's investors finance other firms in the same industry. For every investor connected to focal firm i, I calculate the cumulative share of the investor's other portfolio firms that belong to firm i's industry, excluding firm i itself, and then average this share across the focal firm's investors. Excluding the focal firm prevents its own industry classification from mechanically increasing the measured degree of investor specialization. Second, the cumulative lead-investor share is the proportion of the firm's cumulative investment observations involving investors classified as lead investors.

Firm age is measured in each year as the number of years elapsed since the firm's founding year. Thus, for a firm that completes an IPO, its final observed age in the private-firm panel is its age at IPO; for a firm that remains private, it is observed through 2023. For the focal firm, age enters through the log-transformed squared-age measure used in the empirical specifications. This transformation allows IPO propensity to vary nonlinearly with the firm's stage of development and reduces the influence of the highly skewed age distribution.

The resulting vector of focal-firm controls is then the transformed firm age, cumulative number of investments, cumulative number of unique investors, average investor reputation score, cumulative lead-investor share, and leave-one-out investor-industry specialization measure. I construct analogous controls for the firms' network peers. For every focal firm and year, I average the corresponding characteristics across its active first-order peers: age, investor score, cumulative unique investors, cumulative investments, investor-

industry specialization, and lead-investor share. Unlike the focal-firm control, peer age enters as the average level of age across the relevant peers rather than through the focal firm's age transformation. This procedure constructs these peer averages using characteristics measured in calendar year t, and I subsequently lag the completed averages when estimating the regressions.

Finally, I construct the main variable of interest. This one is the endogenous peer-exposure variable based on the average number of connected firms that experienced an IPO in the previous year. To account for differences in network size across firms and over time, these measures are normalized by the firm's degree in the corresponding year, that is, by the total number of network connections the firm has in that period.

The empirical analysis uses an annual firm-level panel, where each observation corresponds to a firm-year. Firm i's first-order peers are the firms connected to i through at least one common investor in that year. The empirical analysis uses firm-year observations for which the focal firm has at least one active first-order peer. When the focal firm has degree zero, normalized peer-exposure measures are undefined because their denominator is the size of the focal firm 's neighborhood. These firm-year observations are therefore coded as missing for peer variables and are excluded from specifications requiring peer exposure.

The baseline regression, presented in Section 4, relates a firm's IPO decision to lagged IPO activity among its first-order peers, controlling for time-varying focal-firm and peer characteristics, firm fixed effects (FEs), and year-by-industry-by-country fixed effects (FEs). Standard errors are clustered at the firm level.

Assumption 1 (IPO as an absorbing event)**.** *Each firm can undertake an IPO at most once. Let T_i^{IPO} denote firm i's IPO year. The IPO indicator is defined as*

$$IPO_{it} = 1\{t = T_i^{IPO}\},$$

so that

$$IPO_{it} = 0 \qquad \text{for all } t < T_i^{IPO},$$

and $IPO_{iT_i^{IPO}} = 1$.

Assumption 1 means that after the IPO, firm i exits the risk set and all observations with $t > T_i^{IPO}$ are discarded. Thus, going public is an absorbing event: once a firm completes an IPO, it cannot return to the private-firm sample.

The objective is to study how recent peer IPO activity is related with the decision of still-private firms to go public. By excluding post-IPO observations, the estimation sample contains only firms that remain at risk of going public in a given period. The empirical specification therefore follows the spirit of a discrete-time hazard framework

estimated using pooled OLS with within-firm variation. The dependent variable captures the probability that a firm undertakes an IPO in a given year. The model is estimated by OLS with firm fixed effects (within estimator) and additional fixed effects, i.e., as a linear probability model (LPM).[1], allowing the coefficients to be interpreted as changes in the probability of going public. Firms exit the sample after going public, so the specification captures the probability of an IPO conditional on not having previously gone public.

One potential concern is the endogenous formation of the network, since firms and investors may select each other based on characteristics that also influence IPO decisions. Another concern is the reflection problem identified by Manski (1993), which arises in peer-effects models when firms' decisions and those of their peers are determined simultaneously, making it difficult to identify peer effects separately from other influences. However, in my model, the firm i's IPO decision in year t is related to IPO activity among its peers in year $t-1$. This temporal ordering avoids the classic simultaneous reflection problem in which a firm's outcome and its peers' outcomes are jointly determined in the same period. The partially overlapping structure[2] of the network additionally provides variation in peer exposure across firms that mitigates reflection Bramoullé et al. (2009) and De Giorgi et al. (2010). However, this does not eliminate correlated effects or endogenous network formation, such as connected firms facing common shocks or may have selected into similar investor relationships. The fixed effects, controls, robustness exercises, and instrumental-variable strategy described below are designed to address these separate identification concerns.

4 Results

4.1 Baseline peer IPO spillovers

I begin by examining whether IPO activity among directly connected peers is associated with a firm's likelihood of going public. Specifically, I estimate whether IPOs completed by first-order peers in the previous year predict the firm's probability of going public in the current year.

All specifications include firm fixed effects, which absorb time-invariant characteristics of firms that may affect their likelihood of going public, such as persistent differences in business models, founder characteristics, or initial investor composition. Identification

[1]The firm fixed effects OLS model is my main specification rather than a non-linear model such as logit or probit. OLS allows the inclusion of fixed effects and control for time-invariant firm heterogeneity as well as common shocks. Previous literature, such as Greene (2004), has discussed potential biases in non-linear fixed-effects models. I cluster standard errors at the firm level. In the robustness analysis, I estimate non-linear models to compare with the main specification.

[2]As an example, two firms i and j whose peers of i is j and k, and peer of j is i and l. Therefore, they both have themselves as peers, but they also have other peers that are not peers of the other.

therefore comes from within-firm changes in peer IPO exposure over time. In addition, all specifications include year-industry-country fixed effects, which absorb common shocks affecting firms in the same industry and country in a given year.

The baseline specification follows a linear-in-means framework that relates a firm's IPO outcome to the average IPO activity of firms in its network neighborhood.

$$IPO_{i,t} = \alpha + \delta \sum_{j \in N_{i,t-1}} \tilde{g}_{ij,t-1}\, IPO_{j,t-1} + \gamma X_{i,t-1} + \beta \sum_{j \in N_{i,t-1}} \tilde{g}_{ij,t-1} X_{j,t-1} + \rho_i + \mu_{t \times r \times s} + \varepsilon_{i,t}, \tag{1}$$

where $g_{ij,t} \in \{0, 1\}$ indicates whether firms i and j are connected at time t through at least one common investor; $\tilde{g}_{ij,t} = \frac{g_{ij,t}}{d_{i,t}}$ is the row-normalized weight, such that $\sum_j \tilde{g}_{ij,t} = 1$ for each firm i with positive degree in year t; $N_{i,t}$ is the set of first-order neighbors of firm i at time t; $X_{i,t-1}$ represents time-varying firm characteristics and dynamic controls; $X_{j,t-1}$ represents time-varying i's peers characteristics and dynamic controls; ρ_i denotes firm fixed effects; and $\mu_{t \times r \times s}$ denotes the interaction fixed effects by year, region r, and sector s. To estimate this Linear Probability Model, I initially assume a sequential exogeneity condition of $\mathbb{E}[\varepsilon_{i,t} \mid G_{t-1}, X_{i,t-1}, X_{j,t-1}, Y_{j,t-1}] = 0$. This baseline assumption implies that, conditional on firm fixed effects, there are no unobserved, time-varying correlated effects or endogenous self-selection (sorting) into peer groups that anticipate future IPO shocks. Then, lagging the peer variables to $t-1$ ensures they are predetermined, bypassing the simultaneous reflection problem. I explicitly relax these assumptions and address remaining identification concerns in the Appendix.

Remark. Region r is characterized by the geographic location of the firm , and sector s by its sector (industry group). The fixed effects $\mu_{t \times r \times s}$ flexibly absorb shocks that vary simultaneously across time, region, and sector.

The first empirical exercise examines whether recent IPO activity among connected peers predicts the focal firm's subsequent probability of going public. I therefore test whether the coefficient on lagged peer IPO exposure, δ, differs from zero.

Table 1 reports the baseline estimates. Across specifications, the coefficient on the lagged first-order peer-IPO share is positive and statistically significant. In the preferred specification, the coefficient δ is approximately 0.017. Thus, a ten-percentage-point increase in the share of peers completing an IPO rises in about 0.17 percentage points the focal firm's annual IPO probability.[3] The three columns coefficients vary in their fourth decimal, and then are almost identical.

Apart from this, I also examine whether effect of lagged peer IPO exposure varies with the intensity of that exposure by showing non-linearity. In a specification including both

[3]Relative to the average annual IPO probability of approximately 0.26% among connected firm-year observations, this represents an increase of about 65%.

the lagged first-order peer IPO share and its square, the coefficient on the linear term is positive while the coefficient on the squared term is negative. This pattern indicates a concave relationship: greater peer IPO exposure is is linked to a higher probability of going public, but the marginal association becomes smaller as the share of peers completing an IPO increases. The full specification is reported in Appendix B.

Table 1: First-Order Peer IPO Exposure and IPO Likelihood

	IPO No controls	IPO Focal controls	IPO All controls
Lagged peer IPO ratio	0.017***	0.017***	0.017***
	(0.006)	(0.006)	(0.006)
Focal-firm controls	No	Yes	Yes
First-order peer controls	No	No	Yes
Firm fixed effects	Yes	Yes	Yes
Industry-Country-Year FEs	Yes	Yes	Yes
Observations	1,059,860	991,388	989,915
Clusters (firm)	163,379	154,617	154,484
R-squared	0.254	0.265	0.265
Adjusted R-squared	0.105	0.116	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's eligible first-order peers that complete an IPO in year $t-1$. Column (1) excludes additional time-varying controls. Column (2) adds the focal-firm controls, comprising transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (3) additionally includes first-order peer averages of age, cumulative investments, cumulative unique investors, investor score, lead-investor share, and investor-industry specialization. All time-varying controls are measured at $t-1$. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

As a common-sample check, Appendix Table 16 re-estimates the specification without the additional time-varying controls using the same estimation sample as Column (3). The estimated coefficient on the lagged peer IPO ratio is very similar to that reported in Column (3), indicating that the result is not driven by either the inclusion of these controls or the associated change in sample composition.

The effect is concentrated at short horizons. Peer IPO activity in the two preceding years has the strongest association with the focal firm's decision. Then, estimates at longer lags are smaller and eventually statistically indistinguishable from zero. This timing is consistent with recent IPO outcomes providing information relevant for a current market window. It is also consistent with temporary coordination or investor shocks, so the lag

pattern alone does not identify the mechanism. We can check Appendix B that shows this empirical estimation and its results.

4.2 Which connections transmit the effect?

The baseline results establish that IPO activity among connected firms predicts the focal firm's IPO decision. I next examine which types of connections are most important for transmitting this effect. In particular, I study whether the estimated spillover varies with the role of the common investor and with the fact of peers belonging to the same industry. These dimensions are informative about the channels underlying the baseline relationship. Stronger effects through more involved investors would be consistent with information being transmitted through investor connections, as one of the mechanisms. However, concentration of the effect among same-industry peers would be more consistent with product-market competition or common industry-level shocks.

I first examine whether the effect between peer IPO exposure and the focal firm's IPO probability differs depending on whether the exposure includes a common-lead IPO connection. Lead investors are the ones who invest the highest amount of money in a funding round. Given this, they play a more active role in financing rounds, including financing terms, governance, and preparation for subsequent financing or exit decisions. Thus, they may be relevant for the transmission of information, although their greater involvement could also facilitate coordination.

Prior research documents that investors differ in their involvement and influence on firm outcomes. Venture capitalists can add value to portfolio companies through monitoring, strategic advice, and governance, while investor experience and reputation are also related to firms' financing and performance outcomes (P. Gompers et al., 2008; Thomas J Chemmanur, K. Krishnan, et al., 2011; Bernstein, 2022). Related work emphasizes the role of venture capital networks and shows that connections to better-connected investors are aligned to the access to resources and improved portfolio-company outcomes (Hochberg et al., 2007). These findings motivate examining whether connections operating through lead investors differ from those operating through other investors.

Importantly, investor–firm relationships and investor leadership roles are treated differently over time. Investor-firm relationships are persistent: once an investor is observed financing a firm, the investor remains connected to that firm in all subsequent years. Leadership status, however, is defined separately in each year. An investor is classified as a leader for a firm in year t only if the investor is categorized as a leader for that firm in that year. If the investor is not classified as leader, then it remains connected to the firm but is not classified as a leader in that year. Consequently, common-lead connections are reconstructed annually and can change even when the common-investor relationships

remain unchanged. This is justified by the idea that being a leader in one year, does not mean that you exhibits the same role next year, since it might have appeared another investor whose equity share is high enough that now is more involved in the firm than the previous leader.

I then construct a binary indicator equal to one for focal firm i in year t if there exists at least one first-order peer that completes an IPO in year t and shares with firm i an investor classified as a leader for both firms in that year. The indicator equals zero when no such IPO peer exists. This includes both firm-years in which none of the focal firm's peers completes an IPO and firm-years in which peer IPOs occur but none of the IPO peers shares a common lead investor with the focal firm. Therefore, the indicator is not cumulative and can change from one year to the next.

This indicator is used to decompose the focal firm's total first-order peer IPO exposure according to whether the firm-year contains a common-lead IPO connection. I define $L_{i,t}$ as the indicator defined above and let $PeerIPO_{i,t}$ denote the overall first-order peer IPO ratio. I construct $LeadIPO_{i,t} = PeerIPO_{i,t} \times L_{i,t}$, and $NonLeadIPO_{i,t} = PeerIPO_{i,t} \times (1 - L_{i,t})$.[4] Thus, when at least one IPO peer shares a common lead investor with the focal firm, the total peer IPO exposure is assigned to $LeadIPO_{i,t}$. When no IPO peer shares a common lead investor with the focal firm, the total exposure is assigned to $NonLeadIPO_{i,t}$. By construction, $LeadIPO_{i,t} + NonLeadIPO_{i,t} = PeerIPO_{i,t}$.

This decomposition is defined at the focal-firm and year level. In particular, a firm-year classified in the common-lead category may contain both IPO peers that share a common lead investor with the focal firm and IPO peers that do not. To examine this heterogeneity, I estimate the following specification:

$$\begin{aligned} IPO_{i,t} = \alpha + \delta_1 LeadIPO_{i,t-1} + \delta_2 NonLeadIPO_{i,t-1} + \\ +\gamma X_{i,t-1} + \beta \sum_{j \in N_{i,t-1}} \tilde{g}_{ij,t-1} X_{j,t-1} + \rho_i + \mu_{t \times r \times s} + \varepsilon_{i,t}. \end{aligned} \tag{2}$$

The coefficient δ_1 captures the effect of lagged peer IPO exposure on the focal firm's IPO probability in firm-years in which at least one IPO peer shares a common lead investor with the focal firm. The coefficient δ_2 captures the corresponding effect of no IPO peer shares a common lead investor on the focal firm. The comparison of interest is therefore whether peer IPO exposure is stronger on the focal firm's IPO decision when the exposure includes at least one IPO peer connected through a common lead investor.

The time-varying controls include firm age, the cumulative number of investments received, the cumulative number of unique investors, and the investor score. For a more complete analysis, I also include the corresponding averages among first-order peers to

[4]where $LeadIPO_{i,t} \equiv \tilde{g}_{ij,t} IPO_{j,t} \times L_{i,t}$ and $NonLeadIPO_{i,t} \equiv \tilde{g}_{ij,t} IPO_{j,t} \times (1 - L_{i,t})$.

account for observable differences in the characteristics and financing histories of firms in the focal firm's network. I therefore examine whether peer IPO exposure is linked to the focal firm's IPO probability in each type of firm-year and whether the two estimated coefficients differ from one another.

Table 2: Peer IPO Spillovers Through Lead and Non-Lead Investors

	IPO No peer controls	IPO Peer controls
Lead-investor peer IPO ratio	0.060***	0.060***
	(0.022)	(0.022)
Non-lead-investor peer IPO ratio	0.013**	0.013**
	(0.006)	(0.006)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Lead − non-lead exposure	0.047	0.047
p-value of equality test	0.044	0.044
Observations	989,915	989,915
Clusters (firm)	154,484	154,484
Adjusted R-squared	0.115	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. The Lead-investor peer IPO ratio variable equals the overall lagged first-order peer IPO ratio when at least one IPO peer shares a common lead investor with the focal firm, and zero otherwise. The Non-lead-investor peer IPO ratio variable equals the overall lagged first-order peer IPO ratio when no IPO peer shares a common lead investor with the focal firm, and zero otherwise. The reported coefficient difference is the common-lead exposure coefficient minus the no-common-lead exposure coefficient. Both columns include lagged focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (2) includes lagged first-order peer averages of age, cumulative investments, cumulative unique investors, investor score, lead-investor share, and investor-industry specialization. All specifications use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 2 shows that the estimated coefficient is larger for peer connections operating through common lead investors. The estimates are almost identical, they only vary in their fourth decimal. The estimated difference between the two coefficients is approximately 0.047, with a p-value of 0.044. Thus, a ten-percentage-point increase in the share of peers

completing an IPO is related to an increase of about 0.60 percentage points in the focal firm's annual IPO probability when common lead investors (δ_1) while 0.13 in the presence of non-common lead investors (δ_2). The evidence is therefore suggestive of a stronger spillover through common lead investors. Given the greater involvement of lead investors in portfolio firms, this pattern is consistent with investor involvement playing a role in the transmission of IPO-related information, but it does not by itself distinguish information transmission from other mechanisms related to lead-investor involvement.

As a validation exercise, I examine whether the probability of being classified as a lead investor is related to investor experience. Specifically, I test whether the leader indicator is positively affected by the lagged investor score, measured using the investor's prior IPO exits relative to its previous firm investments. The results show that investors with higher lagged investor scores are more likely to be classified as lead investors. This relationship provides some validation that the classification identifies investors with greater prior exit experience. However, this should not be interpreted as establishing that the leader indicator itself measures investor quality.

I next examine whether the baseline relationship differs between same-industry and cross-industry peers. Among firms connected through shared investors, I separately construct peer IPO exposure for firms in the focal firm's primary industry and for firms in other industries. This distinction is informative because same-industry firms may face more similar market conditions and stronger product-market interactions. As in the baseline network construction, peer firms are identified through shared investor connections. The industry classification is based on the primary industry category assigned to each firm. This approach differs from simply controlling for industry-specific shocks. The year-by-industry-by-country fixed effects absorb common shocks affecting firms in the same industry, country, and year, but they do not distinguish whether the IPO activity observed through the network comes from peers operating in the same industry as the focal firm or from peers in different industries. Separating peer IPO exposure into same-industry and different-industry peers therefore allows the relationship between peer IPO activity and the focal firm's IPO decision to vary with industry.

I define the normalized first-order network weights separately for same-industry and different-industry peers as

$$\widetilde{g}^{\text{same}}_{ij,t-1} = \frac{g_{ij,t-1}1\{s_i = s_j\}}{\sum_{\ell \neq i} g_{i\ell,t-1}1\{s_i = s_\ell\}}, \qquad \widetilde{g}^{\text{diff}}_{ij,t-1} = \frac{g_{ij,t-1}1\{s_i \neq s_j\}}{\sum_{\ell \neq i} g_{i\ell,t-1}1\{s_i \neq s_\ell\}},$$

where $g_{ij,t-1} = 1$ if firms i and j are first-order peers at $t-1$, and s_i denotes firm i's industry. In the spirit of Patacchini et al. (2017) for a different analysis, ratios are

constructed over the total number of peers within each group.

$$IPO_{i,t} = \alpha + \delta_1 \sum_{j \neq i} \widetilde{g}^{\text{same}}_{ij,t-1} IPO_{j,t-1} + \delta_2 \sum_{j \neq i} \widetilde{g}^{\text{diff}}_{ij,t-1} IPO_{j,t-1} + \boldsymbol{\gamma}' X_{i,t-1} + \beta_1 \sum_{j \neq i} \widetilde{g}^{\text{same}}_{ij,t-1} X_{j,t-1} + \beta_2 \sum_{j \neq i} \widetilde{g}^{\text{diff}}_{ij,t-1} X_{j,t-1} + \rho_i + \mu_{t \times r \times s} + \varepsilon_{i,t}. \quad (3)$$

The parameters of interest are δ_1 and δ_2, which capture the effect of the lagged share of peer IPOs in the same and different industries, respectively, on firm i's probability of going public. The vectors β_1 and β_2 contain the coefficients on the average lagged characteristics of firm i's same-industry and different-industry peers, respectively.

The comparison of interest is whether the effect of recent peer IPO activity and the focal firm's IPO probability differs depending on whether the connected peers operate in the same industry or in a different industry. The results are shown in Table 3 below.

Table 3: Influence of Lagged First-Order IPOs on IPO Likelihood by Industry

	IPO No peer controls	IPO Peer controls
Same-industry peer IPO ratio	0.019***	0.018***
	(0.004)	(0.004)
Other-industry peer IPO ratio	0.013**	0.011**
	(0.006)	(0.006)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Same-industry − other-industry	0.006	0.007
p-value of equality test	0.349	0.304
Observations	841,956	841,956
Clusters (firm)	135,693	135,693
Adjusted R-squared	0.114	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. The peer IPO ratio variables measure the lagged share of first-order peers completing an IPO, separately for same- and other-industry peers. The reported difference is the same-industry coefficient minus the other-industry coefficient. Focal-firm controls include lagged age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of lead investors, and same-industry investor specialization. Column (2) additionally includes lagged first-order peer controls. Both columns use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Thus, a ten-percentage-point increase in the share of peers completing an IPO is linked to an increase of about 0.19 (δ_1) percentage points in the focal firm's annual IPO probability when belonging to the same industry versus 0.13 (δ_2) when being from a different industry. If the baseline relationship were driven mainly by product-market competition or unobserved industry shocks, same-industry peer IPOs should have a stronger effect. Table 3 does not show a statistically significant difference between the two groups. This result does not rule out product-market channels, but it indicates that the relationship is not confined to firms operating in the same industry.

Finally, I examine connection intensity in Appendix B.4. Weighted-peer specifications assign more importance to peers sharing a larger fraction of the focal firm's investors. These estimates remain positive. The comparison between peers connected by exactly one investor and peers connected by multiple investors is stronger for the former but is not statistically different in every specification. The full tables are reported in Appendix B.4.

4.3 Heterogeneity by Second-Order IPO Exposure

The next analysis examines whether the relationship between a first-order peer IPO and the focal firm's IPO decision depends on the peer's prior exposure to IPO activity elsewhere in the network. I divide first-order peers in $t-1$ into those whose own neighborhoods contained at least one second-order IPO in period $t-2$ and those with no such exposure. Then for firm i the neighbors of the peers that did IPO are indirect connections or second-order peers, and will not be connected to i at $t-2$.

For each group, I construct row-normalized first-order network weights according to whether peer j had previously been exposed to an IPO in its own neighborhood. To examine this, I define

$$\widetilde{g}^{S=1}_{ij,t-1} = \frac{g_{ij,t-1}1\{S_{j,t-1}=1\}}{\sum_{\ell\neq i\in N_{i,t-1}} g_{i\ell,t-1}1\{S_{\ell,t-1}=1\}}, \qquad \widetilde{g}^{S=0}_{ij,t-1} = \frac{g_{ij,t-1}1\{S_{j,t-1}=0\}}{\sum_{\ell\neq i\in N_{i,t-1}} g_{i\ell,t-1}1\{S_{\ell,t-1}=0\}},$$

where $S_{j,t-1}=1$ if peer j's own first-order neighborhood contained at least one IPO in period $t-2$, and $S_{j,t-1}=0$ otherwise.

$$\begin{aligned} IPO_{i,t} = \alpha &+ \delta_1 \sum_{j\in N_{i,t-1}} \widetilde{g}^{S=1}_{ij,t-1} IPO_{j,t-1} + \delta_2 \sum_{j\in N_{i,t-1}} \widetilde{g}^{S=0}_{ij,t-1} IPO_{j,t-1} + \boldsymbol{\gamma}' X_{i,t-1} \\ &+ \beta_1 \sum_{j\in N_{i,t-1}} \widetilde{g}^{S=1}_{ij,t-1} X_{j,t-1} + \beta_0 \sum_{j\in N_{i,t-1}} \widetilde{g}^{S=0}_{ij,t-1} X_{j,t-1} + \rho_i + \mu_{t\times r\times s} + \varepsilon_{i,t}. \end{aligned} \tag{4}$$

The weights $\widetilde{g}^{S=1}_{ij,t-1}$ and $\widetilde{g}^{S=0}_{ij,t-1}$ are row-normalized network weights constructed separately

for previously exposed and previously unexposed first-order peers. A peer is classified as previously exposed, $S_{j,t-1} = 1$, if its own first-order neighborhood contained at least one IPO in period $t-2$, and otherwise $S_{j,t-1} = 0$.

Hence, $\sum_{j \neq i} \widetilde{g}_{ij,t-1}^{S=1} IPO_{j,t-1}$ and is the lagged share of IPOs among previously exposed peers, while $\sum_{j \in N_{i,t-1}} \widetilde{g}_{ij,t-1}^{S=0} IPO_{j,t-1}$. is the corresponding share among previously unexposed peers. Peer characteristics are aggregated separately for the two groups using the same normalized network weights. The coefficients δ_1 and δ_2 therefore capture how the effect of a peer IPO on the focal firm's IPO probability differs according to whether the peer had already been exposed to prior IPO activity in its own neighborhood. The coefficient vectors β_1 and β_0 allow the characteristics of previously exposed and previously unexposed peers to enter the specification separately. Finally, ρ_i denotes firm fixed effects and $\mu_{t \times r \times s}$ denotes year–industry–country fixed effects.

I therefore study whether the impact of first-order peer IPO activity and the focal firm's IPO probability differs depending on prior indirect IPO exposure. By doing so, I compare δ_1 and δ_2. The results are shown in the Table below.

Table 4: First-Order Peer IPOs by Prior Second-Order IPO Exposure

	IPO No peer controls	IPO Peer controls
Peer IPO ratio: no prior second-order exposure	0.023***	0.023***
	(0.008)	(0.008)
Peer IPO ratio: prior second-order exposure	0.005***	0.004**
	(0.002)	(0.002)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
No prior exposure − prior exposure	0.018	0.019
p-value of equality test	0.023	0.021
Observations	818,536	818,536
Clusters (firm)	139,436	139,436
Adjusted R-squared	0.115	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. The first variable measures IPO exposure through first-order peers without prior second-order IPO exposure, while the second measures IPO exposure through first-order peers with such prior exposure. The reported coefficient difference is the coefficient for peers without prior second-order exposure minus the coefficient for peers with prior second-order exposure. Both columns include focal-firm lagged controls comprising transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (2) additionally includes the corresponding lagged first-order peer controls. All specifications use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4 shows that the coefficient is larger for IPOs by first-order peers with no prior second-order IPO exposure. A test of equality between the two coefficients indicates that the difference is statistically significant. The estimated difference is 0.019.

The no-exposure category should be interpreted carefully. It includes both first-order peers that have no additional peers other than the focal firm and first-order peers that are connected to other firms but did not experience an IPO among those connections in period $t-2$. In addition, firms that went public in earlier periods are no longer part of the at-risk sample. Thus, the distinction captures the presence or absence of recent second-order IPO exposure rather than the presence or absence of second-order connections at any point. More generally, the result indicates that the effect between a firm's IPO decision and IPO activity among its direct peers varies with the broader network environment.

This attenuation is also useful for distinguishing among possible mechanisms. Common market conditions alone, or generic pressure to accelerate portfolio exits, do not by themselves imply that the effect of a first-order peer IPO should become weaker simply because that peer was previously exposed to an IPO elsewhere in its network. Neither mechanism, without additional assumptions, generates a clear prediction of this attenuation pattern. Under these mechanisms, once a direct peer goes public, its IPO may generate a transmitted effect on firm i, but there is no clear reason why this effect should depend on whether that peer had previously been exposed to an IPO through an indirect connection. Thus, one would expect the effect of the direct peer IPO to remain positive and, in particular, not to differ systematically with prior second-order exposure. Despite IPOs being relatively rare events, the estimated coefficients exhibit a statistically significant difference across these two groups.

One possible interpretation of the pattern is informational redundancy. When a first-order peer has already been exposed to an IPO elsewhere in its neighborhood, part of the information is related to the subsequent IPO of that peer may already have been available through the network. Its own IPO may therefore provide less incremental information to the focal firm i. By contrast, an IPO by a previously unexposed first-order peer may constitute a relatively less redundant event. This interpretation is consistent with my finding. Then, I formalize this possibility in the learning framework developed in Section 6. However, at this stage, the evidence should be interpreted as suggestive, since other investor-level mechanisms could also generate attenuation if previous exits alter investors' incentives or constraints.

Additional exercises examine whether similar attenuation arises in network structures characterized by greater overlap among IPO peers. The estimated effect is more pronounced in less-overlapping structures, while structures in which the relevant connections operate through the same investors exhibit a smaller marginal association. Conditional measures of the strength of IPO peers' indirect connections yield similar patterns. Be-

cause these measures involve more complex network classifications and are conditional on realized IPO events, I report their construction and full results in Appendix D.

4.4 IPO pricing and financing outcomes

While the previous analysis focuses on firms' decisions to go public, I next examine whether prior IPO activity within the firm network also affects outcomes observed at the time of the focal firm's IPO and with its financing outcomes.

I begin by comparing the IPO share prices of firms with at least one prior first-order peer IPO to those of firms whose peers have not previously gone public. Given that the share price depends partly on the number of shares outstanding, I interpret this outcome as a pricing-related measure rather than a direct measure of firm valuation. To measure differences in firm valuation (and market capitalization) we need to include information on the total outstanding shares.

Among the 5,422 firms completing an IPO in the network sample, I estimate cross-sectional specifications for firms with available IPO outcome information. All regressions include year, industry, and country fixed effects to account for differences in IPO outcomes across time, industries, and locations. I start by estimating the following[5]

$$log(\text{IPOSharePrice}_i) = \alpha + \beta\, \text{PriorPeerIPO}_i + X_{it}'\boldsymbol{\gamma} + \lambda_{t(i)} + \mu_{s(i)} + \nu_{c(i)} + \varepsilon_i, \quad (5)$$

where the share price is for the stock at the time of IPO $\mathbf{X}_{it}$ is a vector of predetermined firm characteristics measured at the IPO, including firm age, cumulative number of investments, cumulative number of unique investors, and the average prior IPO performance of the firm's investors. The specification also includes year fixed effects $\lambda_{t(i)}$, industry fixed effects $\mu_{s(i)}$, and country fixed effects $\nu_{c(i)}$. Table 5 reports two specifications examining the relationship between prior first-order peer IPO exposure and the focal firm's IPO share price. Column (1) excludes additional firm-level covariates, whereas Column (2) includes the focal-firm controls. The estimated coefficient on prior first-order peer IPO exposure is positive in both specifications and remains positive after the inclusion of these controls.

We observe in Table 5 that firms with prior first-order peer IPO exposure raises approximately while 58% the IPO share price, calculated as $100[\exp(0.459) - 1]$ compared with firms without this exposure. This difference decreases to 28% after controlling for firm characteristics.

[5] As a robustness check, I winsorized at the 1st and 99th percentiles to reduce the influence of extreme observations and the results are consistent with the main specification.

Table 5: First-Order IPO Exposure and IPO Pricing

	$\log(\text{IPOSharePrice}_i)$ No firm controls	$\log(\text{IPOSharePrice}_i)$ Firm controls
First-order IPO exposure before focal-firm IPO	0.459***	0.247***
	(0.077)	(0.079)
Focal-firm controls	No	Yes
Industry Fixed Effects	Yes	Yes
Country Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	1,667	1,667
Adjusted R-squared	0.344	0.367

Notes: The dependent variable is the logarithm of the focal firm's IPO share price. Prior first-order IPO exposure is an indicator equal to one if at least one of the focal firm's first-order peers completed an IPO before the focal firm's listing. Column (1) excludes additional firm-level controls. Column (2) adds transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Both columns use the same estimation sample. Because the dependent variable is expressed in logarithms, the exact percentage difference is related to prior peer IPO exposure is calculated as $100[\exp(\widehat{\beta}) - 1]$. Heteroskedasticity-robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Now I will focus on examining the firm valuation at the time of the IPO. This outcome captures the value attributed to the firm at listing than the share-price measure considered above. For this purpose, I estimate the following specification:

$$log(\text{IPOValuation}_i) = \alpha + \beta\, \text{PriorPeerIPO}_i + X'_{it}\boldsymbol{\gamma} + \lambda_{t(i)} + \mu_{s(i)} + \nu_{c(i)} + \varepsilon_i, \qquad (6)$$

where $\mathbf{X}_{it}$ is a vector of predetermined firm characteristics measured before the IPO, including firm age, cumulative number of investments, cumulative number of unique investors, and the average prior IPO performance of the firm's investors. It also includes year fixed effects $\lambda_{t(i)}$, industry fixed effects $\mu_{s(i)}$, and country fixed effects $\nu_{c(i)}$.

Table 6 reports the relationship between prior first-order peer IPO exposure and firm valuation at the time of listing. Column (1) excludes additional firm-level covariates, whereas Column (2) includes the focal-firm controls. The estimated coefficient on prior first-order peer IPO exposure is positive in both specifications and remains positive after the inclusion of these controls. Thus, firms with prior first-order peer IPO exposure tend to have higher valuations when they subsequently list.

This finding is consistent with prior IPO activity within the network being related to the valuation environment faced by subsequently listed firms.

Table 6: First-Order IPO Exposure and Valuation

	$\log(\text{IPOValuation}_i)$ No firm controls	$\log(\text{IPOValuation}_i)$ Firm controls
First-order IPO exposure before focal-firm IPO	0.858^{***}	0.545^{***}
	(0.116)	(0.126)
Focal-firm controls	No	Yes
Industry Fixed Effects	Yes	Yes
Country Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	1,501	1,501
Adjusted R-squared	0.296	0.380

Notes: The dependent variable is the logarithm of the focal firm's valuation at the time of its IPO. Prior first-order IPO exposure is an indicator equal to one if at least one of the focal firm's first-order peers completed an IPO before the focal firm's listing. Column (1) excludes additional firm-level controls. Column (2) adds transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Both columns use the same estimation sample. Heteroskedasticity-robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Now I examine the effect of prior peer IPO exposure on the amount of capital raised by the focal firm at the time of its IPO. The dependent variable is the logarithm of IPO proceeds in USD. In the Crunchbase data, IPO proceeds correspond to the total amount raised from the IPO. In order to do so, I estimate the following regression

$$\log(\text{IPOProceeds}_i) = \alpha + \beta\,\text{PriorPeerIPO}_i + X_{it}'\boldsymbol{\gamma} + \lambda_{t(i)} + \mu_{s(i)} + \nu_{c(i)} + \varepsilon_i, \qquad (7)$$

where X_{it} is a vector of predetermined firm characteristics measured at the IPO, including firm age, cumulative number of investments, cumulative number of unique investors, and the average prior IPO performance of the firm's investors. This specification also includes year fixed effects $\lambda_{t(i)}$, industry fixed effects $\mu_{s(i)}$, and country fixed effects $\nu_{c(i)}$.

Table 7 shows a strong positive relationship between prior first-order peer IPO exposure and the proceeds raised by the focal firm at IPO. Column (1), which excludes additional firm-level controls, reports a coefficient of 0.723. This implies that firms with prior first-order peer IPO exposure raise approximately 106% more in IPO proceeds, calculated as $100[\exp(0.723) - 1]$. Column (2) adds the focal-firm controls. Although the estimated coefficient declines to 0.445, it remains statistically significant at the 1 percent level and economically substantial. The fully controlled estimate implies an approximately 56% difference in IPO proceeds between firms with and without prior first-order peer IPO exposure. Thus, prior IPO activity within the firm's network is associated not only with

its probability of going public but also with the amount of capital it raises upon listing.

Table 7: First-Order IPO Exposure and IPO Proceeds

	$\log(\text{IPOProceeds}_i)$ No firm controls	$\log(\text{IPOProceeds}_i)$ Firm controls
First-order IPO exposure before focal-firm IPO	0.723^{***}	0.445^{***}
	(0.086)	(0.093)
Focal-firm controls	No	Yes
Industry Fixed Effects	Yes	Yes
Country Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Observations	1,910	1,910
Adjusted R-squared	0.265	0.322

Notes: The dependent variable is the logarithm of the proceeds raised by the focal firm at IPO. Prior first-order IPO exposure is an indicator equal to one if at least one of the focal firm's first-order peers completed an IPO before the focal firm's listing. Column (1) excludes additional firm-level controls. Column (2) adds transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Both columns use the same estimation sample. Heteroskedasticity-robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

This analysis is motivated by Aghamolla and Thakor (2022), who identify the time between filing and IPO completion, IPO underpricing, and proceeds raised as offering outcomes that may reflect informational spillovers from peer IPOs. In their paper, they find no statistically significant effects on these outcomes, whereas I find that prior IPO exposure through the common-investor network is positively related to IPO proceeds. My data allows me to examine only proceeds and not the other two offering outcomes they proposed. The result is therefore consistent with an informational channel, but it cannot independently establish that mechanism or exclude alternative explanations.

Beyond IPO proceeds, IPO valuations, and IPO share price, private funding is an important source of financing before a firm execute an IPO. It can also affect a firm's incentives to remain private rather than go public. Hence, I examine whether prior peer IPO activity has an effect on private-market financing. The outcome is the cumulative funding amount reported by Crunchbase. This analysis asks whether firms exposed to prior peer IPOs also differ in the cumulative amount of financing they received before the IPO. The corresponding specification is

$$\log(\text{CumulativeFunding}_i) = \alpha + \beta\,\text{PriorPeerIPO}_i + \theta\log(\text{IPOValuation}_i) + X_{it}'\boldsymbol{\gamma} + \\ + \lambda_{t(i)} + \mu_{s(i)} + \nu_{c(i)} + \varepsilon_i. \tag{8}$$

Table 8 shows a strong positive relationship between prior first-order peer IPO exposure and the focal firm's cumulative funding. All three columns include year, industry, and country fixed effects and use the same sample of 1,136 IPO firms. Column (1) excludes additional covariates, Column (2) adds the focal-firm controls, and Column (3) additionally controls for the firm's valuation at IPO.

The coefficient declines from 1.389 in Column (1) to 0.597 after the focal-firm controls are included in Column (2), but remains statistically significant at the 1 percent level. The preferred estimate in Column (2) implies that firms with prior first-order peer IPO exposure have approximately 81% higher cumulative funding, calculated as $100[\exp(0.597)-1]$.

After additionally conditioning on IPO valuation in Column (3), the coefficient declines to 0.287 but remains statistically significant at the 1 percent level. This estimate corresponds to an approximately 33% difference in cumulative funding, calculated as $100[\exp(0.287)-1]$. Thus, observed differences in IPO valuation account for part, but not all, of the positive association between prior peer IPO exposure and cumulative funding. Because IPO valuation is contemporaneous and may itself respond to prior peer IPO exposure, Column (3) as an alternative robustness specification.

Table 8: First-Order IPO Exposure and Cumulative Funding

	$\log(\text{CumFunding}_i)$ No firm controls	$\log(\text{CumFunding}_i)$ Firm controls	$\log(\text{CumFunding}_i)$ Firm controls + valuation
First-order IPO exposure before focal-firm IPO	1.389***	0.597***	0.287**
	(0.149)	(0.153)	(0.124)
Focal-firm controls	No	Yes	Yes
IPO valuation control	No	No	Yes
Industry Fixed Effects	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	1,336	1,336	1,336
Adjusted R-squared	0.232	0.424	0.590

Notes: The dependent variable is the logarithm of the cumulative funding raised by the focal firm by the time of its IPO. Prior first-order IPO exposure is an indicator equal to one if at least one of the focal firm's first-order peers completed an IPO before the focal firm's listing. Column (1) excludes additional firm-level controls. Column (2) adds transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (3) controls for the logarithm of the focal firm's valuation at IPO. All columns use the same estimation sample. Heteroskedasticity-robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

This analysis aims to assess whether peer IPOs are associated with improved access to external financing, which could represent an additional channel through which IPO activity spreads across connected firms. However, this analysis does not distinguish between financing provided by new investors and additional investments by existing investor.

Thus, one possible interpretation of these results is that IPO events among connected firms provide information that is relevant beyond the decision of whether to go public. Observing a peer IPO may provide information public-market conditions or the valuation of firms connected through similar investors. The positive coefficients when estimating the impact of peer IPOs on valuation, IPO proceeds, and private funding are consistent with this interpretation.

Going beyond this, I distinguish financing provided by investors that are new to the focal firm from follow-on investment by existing investors. This distinction is useful for interpreting alternative mechanisms. When a firm is financed by new investors, the pattern is less directly consistent with a liquidity mechanism operating through the focal firm's existing investors, since these investors do not receive liquidity from the peer IPO. However, follow-on investment by existing investors is not that straightforward because it may reflect changes in expected firm value, coordination among existing investors, or liquidity released following an earlier portfolio company exit.

5 Identification and Robustness

The baseline specification addresses several identification concerns through firm fixed effects, lagged focal-firm and peer characteristics, and year-by-industry-by-country fixed effects. Nevertheless, identification remains challenging because the common-investor network is not randomly formed, but endogenous. Investor-firm relationships are created through bilateral matching or sorting since firms seek financing from particular investors, while investors select firms based on their characteristics and future prospects. As a result, firms connected through common investors may be similar in both observed and unobserved dimensions that are themselves related to IPO propensity. This type of assortative matching can generate correlated IPO timing. A related concern is the dynamic nature of the network is that firms establish new investor relationships over time, and these links may form in response to changing firm prospects or market conditions. If firms that are approaching an IPO become more likely to attract particular investors, or if investors expand their portfolios when market exit opportunities improve, changes in network structure and peer IPO exposure may themselves be endogenous to the same forces affecting the focal firm's IPO decision. For the latter concern, controlling for year-by-country-by-industry and firm fixed effects helps to control for common market shocks and unobserved quality in firms, respectively, but not for the changes in the firm's IPO prospects over time.

A further concern that arises is unobserved investor heterogeneity. We know that investors differ in reputation, experience, resources, sector specialization, portfolio strategy, and involvement in preparing firms for exit. Then, prominent or particularly successful

investors may select firms with strong IPO prospects and directly affect those firms' likelihood of going public. In addition, firms sharing investors may be exposed to common investor-level shocks, such as changes in fundraising conditions, portfolio liquidity, exit pressure, or investment strategy, that vary over time and are not fully absorbed by firm or year-by-industry-by-country fixed effects. Consequently, investors might want to coordinate the timing of the IPOs across their portfolio firms in response to liquidity needs or other strategic motives.

Therefore, this section presents a set of exercises designed to address these concerns. I use placebo networks to evaluate whether the estimated relationship is specific to the observed matching of firms through common investors, specifications that condition on the identity and composition of influential investors to absorb persistent investor-level heterogeneity, and an instrumental-variable design based on a network fixed prior to the policy change to limit concerns about endogenous post-policy network formation. These exercises handle different dimensions of the identification problem and should be interpreted as complementary analysis. The corresponding results are reported in Appendices B to E.

5.1 Placebo networks

I implement a conditional network-placebo exercise by directly rewiring the projected shared-investor peer network. For each of 100 draws, degree-preserving edge swaps randomly change peer identities while preserving each firm's exact number of eligible peers in every year. IPO outcomes are not randomized. The firm-year estimation sample, focal-firm controls, observed-network peer controls, and fixed effects are held constant across draws; only peer IPO exposure is recalculated using the randomized peer assignments [6]. Approximately 90.3% of focal-peer assignments change through the rewiring procedure.

The placebo coefficients are centered near zero, with a mean of 0.00036, and range from approximately -0.00634 to 0.00837. In contrast, the observed-network estimate is approximately 0.01682 and lies above the entire empirical placebo distribution. None of the 100 placebo estimates is as large in absolute value as the observed-network estimate. Appendix C describes the construction of the placebo networks and reports the placebo distribution.

[6] As an additional robustness check, I recompute the average peer controls using the randomized peer assignments in each draw. To ensure comparability, I restrict all regressions, including the actual-network regression, to firm-year observations with complete regressors in the actual network and all 100 placebo draws. This restriction reduces the estimation sample to 949,361 observations and yields an actual-network coefficient of 0.0279. The placebo coefficients remain centered close to zero, with a mean of 0.00051 and a standard deviation of 0.00368. None reaches the absolute magnitude of the actual-network coefficient estimated on this common sample, giving a two-sided empirical placebo tail probability of 0.0099.

5.2 VC and leader fixed effects

As an additional robustness exercise, I augment the baseline specification with fixed effects for the cumulative set of lead investors and for the cumulative set of investors that are simultaneously venture capitalists and lead investors. These specifications compare firms conditional on being backed by those set of investors and then, absorb persistent differences linked to the identity and composition of their lead (lead and VC) investors.

The coefficient on lagged first-order peer IPO exposure remains positive and is quantitatively similar to the baseline estimate under both specifications. This indicates that the baseline relationship is not driven by persistent differences across firms driven by the fact of being backed by different sets of lead investors or VC-lead investors. Consequently, this alleviates the unobserved investor heterogeneity concern.

These specifications are particularly demanding because the investor relationships that are used to construct the fixed effects also contribute to defining the network through which peer IPO exposure is measured. At the same time, they should be interpreted primarily as controls for persistent investor heterogeneity. The construction and complete results are reported in Appendix D.

5.3 Instrumental Variable: Policy-based instrument

A common approach in the network peer-effects literature is to use characteristics of peers-of-peers as instruments for the behavior of direct peers (Bramoullé et al., 2009). I do not follow this approach here because the exclusion restriction is particularly difficult to justify in my setting. One of my main results of this paper is precisely that prior IPO activity among second-order peers changes the effect of a subsequent first-order peer IPO. The theoretical mechanism that I will show next provides the same interpretation: prior second-order IPO exposure changes the information contained in the direct peer's IPO decision. Then, we know that second-order IPO activity affects the focal firm's beliefs and IPO incentives beyond its effect on the direct peer's decision, making it inappropriate as an excluded instrument in this setting. Also, I should not use IPOs from indirect peers as Aghamolla and Thakor (2022) because this is likely to violate the exclusion restriction in the presence of correlated shocks between the focal firm and the direct peers.

Instead, I exploit the 2012 JOBS Act as a source of variation in peer IPO activity.[7] The reform reduced disclosure and compliance requirements for firms qualifying as emerging growth companies. Because historical firm-level revenues are not observed in Crunchbase, I cannot identify the exact eligibility. Therefore, I use the number of U.S.-domiciled firms in firm i's first-order network as of 2011, interacted with the post-2012 period, as a proxy

[7]Policy-induced variation has also been used as an instrumental-variable strategy in studies of peer effects and common investors, see Foroughi et al. (2022) and Eldar and Grennan (2024)

for predetermined exposure to the reform. Fixing the network before the policy change prevents post-policy link formation from affecting the construction of the instrument. The specification also controls for the focal firm's own U.S. post-policy exposure, which allows U.S. focal firms to experience a different average change in IPO probability after the reform. Appendix E describes the construction, identifying assumptions, and robustness exercises in detail.

The first stage is strong, with a Kleibergen–Paap statistic of 115 in the preferred specification. The second-stage coefficient on lagged peer IPO exposure is positive and equal to 1.38 (SE = 0.45). Taking into account that the IV estimate is much larger than the OLS, I interpret the IV as alternative evidence that the positive relationship between peer IPO exposure and the focal firm's IPO decision is not driven solely by endogenous changes in the network. The design still relies on the exclusion restriction that, conditional on the included controls and fixed effects, own policy effect, predetermined exposure to U.S. peers affects the focal firm's IPO probability through peer IPO activity rather than through other post-2012 changes related to U.S.-oriented networks. Appendix E reports the result for this.

The IV design requires that predetermined US peer composition affects the focal firm's post-2012 IPO probability only through peer IPO activity after conditioning on own policy exposure and the fixed effects. This assumption could fail if US peer exposure proxies for changes in access to underwriters, capital flows, or investor strategy that directly affect the focal firm. Related to this, this concern will be present when the firm is not from the US, given that spillovers in IPOs exist.[8] Given this, I use the IV as complementary evidence rather than as the sole basis for the causal claim. Explanation and alternative fixed-effect structures are reported in Appendix E.

5.4 Additional robustness

In appendix B reports additional robustness exercises. The baseline relationship is larger among younger firms, remains positive after controlling for detailed network-type fixed effects, and is robust to weighted network exposure. Additionally, logit and probit estimates preserve the qualitative result, although rare events, as IPO choices, limit the feasibility of nonlinear specifications with the entire set of fixed effects. Placebo and other exercises to study attenuation in the spillovers confirm that the effect is tied to the observed network and is weaker when IPO signals are highly overlapping.

Together, these exercises strengthen the evidence of positive IPO spillovers through

[8]Restricting the sample to non-U.S. focal firms may mitigate concerns about direct policy effects but does not eliminate potential spillovers through common investors. The JOBS Act can influence their financing and exit strategies across U.S. and non-U.S. portfolio firms, affecting focal firms' IPO decisions independently of their peers' IPO activity.

common-investor networks. They help to address concerns about aggregate shocks, persistent firm and investor heterogeneity, and mechanical features of the network structure. However, they do not fully rule out time-varying mechanisms within investor portfolios, such as coordination, liquidity conditions, or changes in financing and exit strategies. The results therefore provide complementary evidence of network spillovers without uniquely identifying their causal mechanism.

Overall, the empirical analysis reveals three main patterns: firms are more likely to go public following an IPO by a directly connected peer; spillovers are stronger through common lead investors; it is weaker when the IPO peer has previously been exposed to IPO activity through more distant network connections. These findings suggest that IPO spillovers depend not only on network connections but also on their economic importance and prior history of those connections.

These patterns are consistent with a learning mechanism in which firms update their expectations about their potential IPO valuations by observing IPO decisions among connected firms.

6 A Model of Firm Learning

In this section, I develop a simple sequential learning model to rationalize the empirical results. Here, the key informational friction is that private firms do not perfectly observe how the public market values the investors backing them. Prior to its decision period, each firm receives noisy private signals about the relevant public-market valuation components. Firms subsequently act sequentially. Thus, when an earlier connected firm chooses whether to go public, its decision reflects its private information about the public-market valuation of the common investor. A later firm does not observe the earlier firm's signal, but it does observe the IPO decision and can use it to update its own posterior.

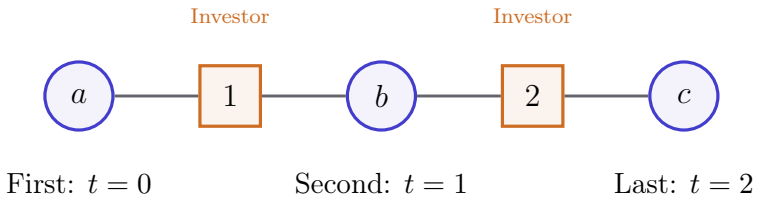
The model therefore separates the valuation of an investor by the public market from the beliefs of private firms about that valuation. IPO decisions do not reveal another firm's signal directly. Rather, they constitute endogenous binary signals generated by the earlier firm's private information.

6.1 Firms, investors, and sequence of decisions

Firms are indexed by $i \in N$, where N is the total number of firms $i \in \{a, b, c, \ldots\}$. Investors are indexed by $k \in K$ where K is the total number of investors and, for expositional convenience, are represented by the numbers $k \in \{1, 2, \ldots\}$.

The model has three periods, $t = 0, 1, 2$. Each firm receives a single IPO opportunity, and firms act sequentially like in a similar work by Hoffmann-Burchardi (2001). The

sequential timing reflects the fact that IPO opportunities do not arise simultaneously across firms. Preparing an IPO requires time and involves regulatory filings, underwriting process, and market-timing considerations, so firms naturally reach the listing decision at different points in time. A firm that acts later can therefore observe earlier IPO outcomes before deciding whether to proceed with its own listing. To illustrate, consider the following three-firm case.



In the previous example we have three sequential IPO opportunities. The corresponding investor sets for each firm are

$$K_a = \{1\}, \qquad K_b = \{1, 2\}, \qquad K_c = \{2\}.$$

Thus firms a and b share investor 1, while firms b and c share investor 2. The order of IPO opportunities is firm a deciding on IPO, then firm b and at last, firm c. Because each firm acts only once and the order of moves is fixed, calendar-time subscripts are suppressed from this point onward. A firm subscript identifies the stage at which that firm receives its IPO opportunity.

6.2 Environment and information

The firm-specific components have prior means $\bar{\theta}_a$, $\bar{\theta}_b$, and $\bar{\theta}_c$, with

$$\bar{\theta}_a > 0, \qquad \bar{\theta}_b > 0, \qquad \bar{\theta}_c > 0.$$

No agent receives additional private information about these components.

For every investor k and period t, ε_t^k denote the public market's valuation of investor k at time t. This valuation evolves over time and can capture, for example, the market's assessment of the investor's reputation, certification ability, or expected contribution to its portfolio firms. Since each firm receives a single IPO opportunity and the firm subscript identifies the stage at which that firm acts, the relevant time is uniquely determined by the firm. I therefore use the shorthand $\varepsilon_i^k \equiv \varepsilon_t^k$ when firm i acts at time t.

The public market observes ε_i^k, whereas the private firm does not observe it directly. I assume

$$\varepsilon_i^k \sim N(0, \sigma_\varepsilon^2).$$

Assumption 2 (Joint normality investor valuations)**.** *For each investor k, the vector of public-market valuation components that corresponds to the firms backed by investor k is multivariate normal.*

In particular, in the three-firm example, $\varepsilon_a^1, \varepsilon_b^1, \varepsilon_b^2, \varepsilon_c^2$ are jointly normally distributed. Also, valuations associated with the same investor are positively related across connected firms:

$$\mathrm{Cov}(\varepsilon_a^1, \varepsilon_b^1) > 0, \qquad \mathrm{Cov}(\varepsilon_b^2, \varepsilon_c^2) > 0.$$

Moreover, for firms backed by the same investor, this covariance is assumed to be weakly decreasing in the number of periods separating their IPO opportunities. Thus, more recent valuations related to a common investor are more informative than valuations observed further apart in time. Investor-specific components are independent across different investors.

A private signal is observed only by the firm receiving it. Thus firm a never observes s_b^1 or s_c^2 firm b never observes s_a^1 nor s_c^2, and firm c never observes either s_b^1 or s_b^2. Each signal for a firm i and investor k is defined as follows

$$s_i^k = \epsilon_i^k + \eta_i^k$$

where $\eta_i^k \sim \mathcal{N}(0, \sigma_\eta^2)$ with the η_i^k's mutually independent and independent of all investor-specific valuation components $\{\varepsilon_j^\ell\}$.

When the firm receives a single IPO opportunity in one period t, it chooses a binary action $d_i \in \{0, 1\}$, where

$$d_i = \begin{cases} 1, & \text{if firm } i \text{ goes public,} \\ 0, & \text{if firm } i \text{ remains private.} \end{cases}$$

IPO decisions are publicly observed.[9] Recall that a firm subscript identifies the stage at which that firm receives its IPO opportunity. Hence, d_i and $\mathcal{I}_i^M$ is the action taken and the market information available at that stage where firm i moves, respectively.

Before the game begins, firms and public-market participants observe characteristics of firms and investors that are publicly available. This information determines the common prior about the firm-specific component θ_i and forms the public market's valuation of the investors backing each firm based on, for example, past investors' reputation, financial support and connections, governance roles. Thus, agents enter the game with prior information about both firms and investors rather than with uninformative priors.

[9]IPO decisions are publicly observable. An IPO is a public and regulated process involving formal disclosure and, if completed, the listing of the firm's shares on a public market. Hence later firms and public-market participants observe whether an earlier firm has completed an IPO.

This information structure is intended as a reduced-form representation of the firm's uncertainty about its potential public-market valuation. Prior to an IPO, firms do not generally know the exact valuation they will obtain in the public market. The IPO literature emphasizes that valuation is progressively revealed during the going-public process, as underwriters collect information from investors and incorporate it into the offer price (Benveniste and Spindt, 1989; Hanley, 1993).

The investor-specific structure of the signals reflects the fact that investors themselves can affect how firms are perceived in public markets. A large literature emphasizes the certification, reputation, and monitoring roles of venture capital investors in IPOs (Megginson and Weiss, 1991; C. Krishnan et al., 2011). Rather than assuming that the firm observes a separate market price that corresponds to each investor, s_i^k should be interpreted as the firm's private assessment of the public-market valuation component associated with investor k. These signals summarize private pre-IPO information available to the firm through its interactions with underwriters, existing investors, and prospective market participants. This investor-specific decomposition allows information revealed by an IPO decision to propagate through shared-investor links. In Appendix F.3, I consider an alternative information structure in which firms receive a noisy signal of their aggregate public-market valuation, incorporating the contributions of all their investors. This signal can be interpreted as a preliminary indication of the potential IPO price once public. The main theoretical predictions remain qualitatively unchanged under this alternative specification.

In principle, firms could learn from the prices realized when connected firms go public, rather than relying exclusively on their IPO decisions. I abstract from this additional source of public information in the baseline model and focus on IPO decisions as the mechanism of learning. This choice also maps directly into the empirical analysis, where peer exposure is constructed from whether connected firms complete an IPO. The IPO decision is therefore the observable public outcome linking the theoretical model to the empirical specification.

The information sets at the time firms act are composed by private signals and IPO decisions [10]:

$$\mathcal{I}_a = \{s_a^1\}, \qquad \mathcal{I}_b = \{s_b^1, s_b^2, d_a\}, \qquad \mathcal{I}_c = \{s_c^2, d_a, d_b\}.$$

Thus, firms first receive their own noisy private signals and act sequentially. Before its own IPO opportunity, each firm observes all preceding public IPO decisions, but never the private signals of earlier firms. Public market participants also observe the realized

[10]The increasing number of observed IPO decisions reflects the sequential ordering of firms. The asymmetric number of private signals in this three-firm chain is only for expositional convenience. For example, in a larger circular network with two investors per firm, each firm could receive the same number of private signals while later-moving firms would still observe a longer history of preceding IPO decisions.

history of IPO decisions. Importantly, however, they already know their own valuation of the relevant investors. Hence an IPO decision provides information to later private firms about how the public market values a common investor.

6.3 Market valuation, posterior beliefs, and IPO decisions

The public market is composed of competitive, risk-neutral outside investors. These investors are distinct from the private investors $k \in K$ that generate the links in the private-firm network. If a firm goes public, outside investors compete to purchase its shares and take the market price as given. Competition therefore implies that, in equilibrium, the IPO price equals the public market's valuation of the firm given the information available at that date. Hence P_i is not a price chosen strategically by the firm or by a particular investor, but the competitive equilibrium price at which the firm's shares would be valued if it listed. The idea behind this is that a firm can decide whether to sell its shares to the public, but it cannot dictate how much public investors are willing to pay for them. Hence, if a firm asks for a price above the market's valuation, competitive investors will not be willing to purchase the shares at that price, whereas, if it offers its shares below their valuation, it unnecessarily gives up proceeds.

The valuation of a firm will be determined by $V_i = \theta_i + \sum_{k \in K_i} \alpha_i^k \epsilon_i^k$. The realization of the firm-specific component θ_i is not observed by either the firm or the public market at the time of the IPO decision, and it does not play an informational role in the baseline model. Both therefore evaluate this component using its prior expectation. I assume that it is independent of the investor-specific. Hence, conditional on the information available at the IPO stage, its contribution to expected firm value is $\bar{\theta}_i$. The prior distribution is $\theta_i \sim N(\bar{\theta}_i, \sigma_\theta^2)$.

Hence, the market valuation of a firm is then based on $\mathbb{E}_M\{V_i | \mathcal{I}_M\} = \bar{\theta}_i + \sum_{k \in K_i} \alpha_i^k \epsilon_i^k$, where $\mathcal{I}_M$ is the public-market information set. Since ε_i^k is already the public market's valuation of the contribution associated with investor k to firm i, the public-market valuation of firm i is

$$P_i = \bar{\theta}_i + \sum_{k \in K_i} \alpha_i^k \varepsilon_i^k, \tag{9}$$

where $\alpha_i^k \geq 0$ is the equity share of investor k in firm i, and it captures the importance of investor k for the valuation of firm i.

Thus, in the three-firm example,

$$P_a = \bar{\theta}_a + \alpha_a^1 \varepsilon_a^1, \qquad P_b = \bar{\theta}_b + \alpha_b^1 \varepsilon_b^1 + \alpha_b^2 \varepsilon_b^2, \qquad P_c = \bar{\theta}_c + \alpha_c^2 \varepsilon_c^2.$$

The private firms do not observe the ε_i^k's. At the time firm i chooses whether to go public, define its posterior about investor k as $\mu_i^k \equiv \mathbb{E}_i \left[\varepsilon_i^k \mid \mathcal{I}_i \right]$. Hence, the relevant

posterior beliefs in the three-firm example are therefore

$$\begin{aligned}
\mu_a^1 &= \mathbb{E}_a[\varepsilon_a^1 \mid s_a^1], \\
\mu_b^1 &= \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a], \qquad (10)\\
\mu_b^2 &= \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2], \qquad (11)\\
\mu_c^2 &= \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a, d_b].
\end{aligned}$$

Equation (10) shows explicitly where firm a's decision enters firm b's posterior. By contrast, under independence across investors, d_a contains no direct information about investor 2, which explains (11).

Before incorporating any informative previous IPO decision, the Gaussian signal structure gives a simple posterior. Since $\mathrm{Cov}(\varepsilon_i^k, s_i^k) = \sigma_\varepsilon^2$ and $\mathrm{Var}(s_i^k) = \sigma_\varepsilon^2 + \sigma_\eta^2$, we obtain the expected posterior

$$\mathbb{E}_i[\varepsilon_i^k \mid s_i^k] = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_i^k.$$

The posterior variance after the private signal is

$$\mathrm{Var}_i(\varepsilon_i^k \mid s_i^k) = \frac{\sigma_\varepsilon^2 \sigma_\eta^2}{\sigma_\varepsilon^2 + \sigma_\eta^2}.$$

The variance across realizations of the corresponding posterior mean is

$$\mathrm{Var}\left(\mathbb{E}_i[\varepsilon_i^k \mid s_i^k]\right) = \frac{\sigma_\varepsilon^4}{\sigma_\varepsilon^2 + \sigma_\eta^2} \qquad (12)$$

Equation (12) is the variance of firms' posterior assessment (i.e. posterior mean) about investor k. Once previous IPO decisions are incorporated, the posterior distribution can additionally depend on the realized public history.

Firm i's expected public-market valuation at its IPO opportunity is

$$\widehat{P}_i \equiv \mathbb{E}_i[P_i \mid \mathcal{I}_i] = \bar{\theta}_i + \sum_{k \in K_i} \alpha_i^k \mu_i^k.$$

In particular,

$$\widehat{P}_b = \bar{\theta}_b + \alpha_b^1 \mu_b^1 + \alpha_b^2 \mu_b^2.$$

Let $W_i > 0$ denote the value of remaining private and let $\kappa_i > 0$ denote the cost of going public. The binary IPO decision is $d_i \in \{0, 1\}$, where $d_i = 1$ denotes that i did an IPO. Firm i chooses

$$d_i = 1 \quad \Longleftrightarrow \quad \widehat{P}_i \geq W_i + \kappa_i. \qquad (13)$$

The public-market valuation in (9) is the price available if the firm lists. If $d_i = 1$, the

valuation is realized as the IPO price. If $d_i = 0$, no IPO transaction price is realized.

Lemma 1 (Current decision and current market valuation)**.** *The current IPO decision does not change the contemporaneous public-market valuation P_i. Therefore, the expected price that the market assigned to firm i when i's takes a decision d_i is*

$$\mathbb{E}_M \left[P_i \mid \mathcal{I}_i^M, d_i \right] = \mathbb{E}_M \left[P_i \mid \mathcal{I}_i^M \right] = P_i. \tag{14}$$

Thus, an observed IPO is informative because it reveals that the firm's private information was sufficiently favorable for its expected valuation to cross the listing threshold, while the IPO decision itself does not mechanically change the contemporaneous market valuation.

6.4 Equilibrium Characterization

The model is a sequential Bayesian game with imperfect information. The equilibrium concept is Perfect Bayesian Equilibrium together with competitive public-market pricing.

Proposition 1 (Equilibrium)**.** *There exists a Perfect Bayesian Equilibrium in cutoff strategies. For each firm $i \in \{a, b, c\}$, the equilibrium IPO decision is*

$$d_i^* = \begin{cases} 1, & \text{if } \widehat{P}_i = \bar{\theta}_i + \sum_{k \in K_i} \alpha_i^k \mu_i^k \geq W_i + \kappa_i, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$\mu_i^k = \mathbb{E}_i \left[\varepsilon_i^k \mid \mathcal{I}_i \right].$$

The equilibrium exists because firms make their IPO decisions sequentially, and earlier firms' payoffs do not depend on subsequent decisions. This allows strategies and beliefs to be constructed recursively using Bayes' rule. Under continuous signals, equilibrium IPO decisions are unique almost surely. Interestingly, the equilibrium shows how IPO decisions reveal information about common investors, allowing information to spread across connected firms and influence their incentives to go public.

6.5 Own-signal monotonicity

Lemma 2 (Own-signal monotonicity)**.** *Holding the publicly observed action history fixed, a firm's posterior assessment of a relevant investor is strictly increasing in its own signal whenever posterior uncertainty remains.*

Corollary 1 (Monotone IPO rule)**.** *Given that $\alpha_i^k > 0$, then $\widehat{P}_i$ is strictly increasing in the firm's own signal s_i^k, holding the rest of its information fixed. Whenever both actions occur with positive probability, the IPO strategy can therefore be represented by a cutoff in the relevant private signal.*

6.6 Direct peer spillovers

Since the investor set for firm a in the example is $K_a = \{1\}$,

$$\widehat{P}_a = \bar{\theta}_a + \alpha_a^1 \mu_a^1.$$

Using (12), the firm's posterior mean is as defined as $\mu_a^1 = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_a^1$. Hence, for $\alpha_a^1 > 0$, firm a follows a cutoff rule

$$d_a = 1 \quad \Longleftrightarrow \quad s_a^1 \geq s_a^{1*}, \tag{15}$$

where

$$s_a^{1*} = \frac{\sigma_\varepsilon^2 + \sigma_\eta^2}{\sigma_\varepsilon^2} \frac{W_a + \kappa_a - \bar{\theta}_a}{\alpha_a^1}. \tag{16}$$

Notice that firm s_a^{1*} depends on the parameters of the model: signal's distribution, remaining private (outside option) value W_a, its prior about its type $\bar{\theta}_a$ and the cost of going public κ_a. Both the cost and the value of staying private will make firm a less likely to go IPO. However, a high prior value of firm a, as well as, increments in the equity stake of the investor makes firm a be more likely to execute the listing decision.

Firm b does not observe s_a^1. It observes only the binary decision d_a. Because the public-market valuations of investor 1 are positively related across firms a and b, this binary decision contains information about ε_b^1.

Proposition 2 (Direct peer spillovers)**.** *In equilibrium, for every fixed realization of firm b's own signal s_b^1,*

$$\mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a = 1] > \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1] > \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a = 0]. \tag{17}$$

Consequently,

$$\widehat{P}_b\big|_{d_a=1} > \widehat{P}_b\big|_{d_a=0}. \tag{18}$$

Moreover, in average posterior terms,

$$\mathbb{E}[\mu_b^1 \mid d_a = 1] > \mathbb{E}[\mu_b^1 \mid d_a = 0].$$

Observing an IPO corresponds to an upper-tail event and therefore raises the posterior expectation of the relevant investor-specific component. Conversely, observing no IPO

corresponds to a lower-tail event and lowers the posterior expectation. Therefore, when I compare the expected public-market valuation of firm b under $d_a = 1$ ($\widehat{P}_b\big|_{d_a=1}$) versus the expected public-market valuation of firm b under $d_a = 0$ ($\widehat{P}_b\big|_{d_a=1}$), the price that is expected by the firm b when $d_a = 1$ will be higher than the one when $d_a = 0$.

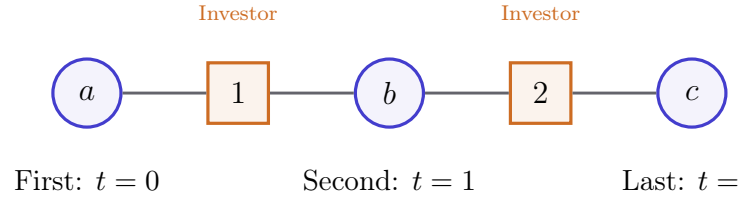
Appendix F reports the corresponding truncated-normal expressions and the properties used in the proofs below.

Corollary 2 (Investor importance)**.** *In equilibrium for a fixed posterior change concerning investor k, the magnitude of the spillover through that investor is increasing in its valuation weight α_b^k.*

If lead investors are associated with a larger valuation weight or with a more informative posterior response, Corollary 2 provides a direct interpretation of larger empirical spillovers through common lead investors. This is an empirical interpretation rather than an additional primitive assumption.

6.7 Second-order exposure

We now return to our example for three firms and two investors:



Firm a's decision directly conveys information about investor 1, but not about investor 2. Nevertheless, it changes firm b's posterior μ_b^1, and hence the conditions under which firm b goes public. Firm c subsequently observes d_b. Since firm b's decision also depends on its private information about investor 2, d_b is informative about the public-market valuation of investor 2 relevant for firm c.

For a given realization of firm b's investor-1 signal s_b^1, let $s_b^{2*}(d_a, s_b^1)$ denote the cutoff value of firm b's investor-2 signal such that firm b chooses to go public if and only if $s_b^2 \geq s_b^{2*}(d_a, s_b^1)$. The first argument records the previously observed IPO decision of firm a, while the second argument is firm b's own signal about investor 1. Thus, $s_b^{2*}(1, s_b^1)$ is the investor-2 signal required for firm b to go public after observing $d_a = 1$, whereas $s_b^{2*}(0, s_b^1)$ is the corresponding cutoff after observing $d_a = 0$.

Proposition 3 (Second-order cutoff effect)**.** *In equilibrium, holding firm b's signal s_b^1 fixed, favorable prior exposure lowers the investor-2 signal required for firm b to go public:*

$$s_b^{2*}(1, s_b^1) < s_b^{2*}(0, s_b^1).$$

Proof is in Appendix F.1.

The intuition behind this result is that holding firm b's own signal about investor 1 fixed, observing $d_a = 1$ raises firm b's posterior assessment of the public-market valuation related to investor 1. This increases b's expected IPO valuation through the investor-1 component. Consequently, firm b requires a less favorable signal about investor 2 in order for an IPO to be optimal. Thus, favorable prior exposure lowers the investor-2 signal cutoff.

Proposition 4 (Second-order attenuation)**.** *Suppose that*

$$\frac{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0)}{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1)} \quad \textit{is increasing in } e. \tag{19}$$

Then observing firm b's IPO produces a smaller posterior update about investor 2 *following* $d_a = 1$ *than following* $d_a = 0$*:*

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 1, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2] < \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 0, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2].$$

Condition (19) can be seen as a monotone likelihood-ratio (MLR) condition. It requires that, as the investor-2 valuation component relevant for firm c increases, an IPO by firm b becomes relatively more likely following $d_a = 0$ than following $d_a = 1$. By Bayes' rule, this likelihood-ratio ordering translates directly into an MLR ordering of firm c's posterior over ε_c^2 after observing $d_b = 1$. Also, Proposition 3 provides the economic intuition for condition (19). Holding s_b^1 fixed, observing $d_a = 1$ raises firm b's expected valuation through investor 1 and therefore lowers the investor-2 signal required for b to go public. As a result, firm b can choose an IPO with less favorable information about investor 2, so that its IPO is less informative about investor 2 following $d_a = 1$.

This cutoff comparison is conditional on s_b^1. Firm c, however, does not observe s_b^1, and therefore evaluates the likelihood of $d_b = 1$ by integrating over its distribution conditional on the observed history. Consequently, the cutoff ordering alone does not guarantee condition (19) because this distribution differs between $d_a = 0$ and $d_a = 1$. Appendix F.1.9 derives the Bayesian posterior comparison and the sufficient likelihood-ratio condition formally. I then evaluate the posterior by deterministic numerical integration under the Gaussian environment. The numerical evaluation delivers:

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 1, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2] < \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 0, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2].$$

Thus, firm b's IPO conveys less new information about investor 2 when firm b has previously observed the favorable IPO of firm a. Consistent with Proposition 4, the relative likelihood in (19) is increasing in e in the benchmark parameterization.

For the numerical evaluation, I use the benchmark parameterization $\sigma_\varepsilon^2 = \sigma_\eta^2 = 1$ and set the covariance of same-investor valuation components across connected firms to 0.6. I set $\alpha_a^1 = 1$ and $\alpha_b^1 = \alpha_b^2 = 0.5$, with $\bar{\theta}_a = \bar{\theta}_b = 1$, and evaluate firm c at its mean signal, $s_c^2 = 0$. The IPO thresholds are set to $W_a + \kappa_a = 2.98$ and $W_b + \kappa_b = 2.40$, which generates unconditional IPO probabilities close to the approximately 0.26% annual IPO frequency observed among connected firm-year observations in the data.

Under this benchmark, the sufficient likelihood-ratio condition for attenuation is satisfied over the evaluated support of ε_c^2. I then vary the investor weights, same-investor covariances, signal precision, and firm-specific components to assess the robustness of the result. The attenuation result continues to hold over the parameter values considered, although its magnitude becomes small when either the upstream or downstream investor component has little influence on firm b's IPO decision. Appendix F.1.10 reports the complete numerical evaluation and sensitivity analysis.

7 Policy Implications

The results suggest that policies affecting firms' access to public markets may have effects beyond the firms directly targeted by the regulation. If IPO activity propagates through common investor networks, a policy that changes the incentives of one group of firms to go public may also affect the IPO decisions of connected private firms. The JOBS Act exercise is consistent with this possibility: predetermined exposure to U.S. peers affected subsequent peer IPO activity and, through it, the IPO propensity of connected firms. Policy evaluations that consider only directly affected firms may therefore miss part of the broader effect operating through firms networks.

More generally, the magnitude of these indirect effects may depend on network structure. Hence, the heterogeneous results depending on investor type, and the attenuation effect suggest that policy spillovers might not affect all firms in the same manner. Then where affected firms are located in the firms network, and the extent to which their connections overlap, will also matter. These results do not establish the welfare consequences of IPO regulation or imply a particular optimal policy. However, they suggest that network spillovers should be taken into account when evaluating policies that alter access to public equity markets. Related to this, firms when considering whether to execute an IPO, do not consider the informational gains that peers will obtain afterwards when observing their decisions. Consequently, IPO decisions generate a positive informational externality, as firms do not fully internalize the benefits of the information they reveal to their peers. Then, IPO activity may be lower than the socially optimal level when these benefits are not taken into account.

8 Conclusion

I study whether IPO decisions propagate across private firms connected through common investors. Using a dynamic network of firm financing relationships, I find that recent IPOs among directly connected peers predict a higher probability that the focal firm goes public. The relationship is concentrated at short lags, is larger through common lead investors, is not confined to same-industry peers, and is substantially weaker when the IPO peer has already been exposed to IPO activity elsewhere in the network. Additional redundancy tests show similar attenuation when realized IPO peers are connected through highly overlapping investor relationships. Peer IPO exposure is also linked to more favorable IPO and financing outcomes.

The theoretical framework provides a parsimonious interpretation of the main empirical patterns. Specifically, an IPO is an endogenous public signal that reveals information to connected private firms about how the public market values their investors. This affects their valuations and hence, the prices they expect to receive if they go public. Given that the informational value of a new IPO depends on prior network exposure, the model rationalizes both positive direct-peer spillovers and their attenuation when previous IPO activity makes subsequent signals more redundant. Taken all this together, the empirical and theoretical results show how shared-investor networks can transmit information between public and private capital markets.

Several extensions are developed in the Appendix in Sections F.6 and F.7. The underpricing extension allows connected firms to learn not only from a peer's decision to go public but also from the market's response to that decision. The multiple-investor extension examines how investor overlap affects the information transmitted through the network and the informational value of successive IPOs. Finally, allowing firms to have private information about their own type introduces an additional signaling channel, as IPO decisions may reveal information about both market conditions and firm quality, connecting the model to the IPO signaling literature (Welch, 1989; Thomas J Chemmanur and Fulghieri, 1999). These extensions suggest further research on how firms' private information interacts with learning from peers and whether firms strategically disclose information in response to changes in market beliefs following peer IPOs. They also motivate a richer empirical analysis of IPO outcomes, examining whether offer prices, first-day returns, and subsequent market performance provide additional signals to connected private firms beyond the IPO decision itself.

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A Appendix

This appendix includes information about the data, robustness tests and proofs.

A.1 Crunchbase source files

Here, I describe the main databases and variables used in the analysis. The funding rounds database contains information on various aspects of firm financing, including variables such as the lead investor UUID, funding round name, firm name, funding amount per round, and the date of the funding round. This dataset can be merged with others using the lead investor UUID, though it does not include investor names. The different types of funding rounds are defined in Crunchbase as the funding types. In here I will summarize the main features of the respective funding names we will encounter.

The first type of funding is defined as the angel investments that provide early funding from individuals or groups to help firms launch. These are usually followed by pre-seed rounds, which involve minimal institutional backing. Seed rounds come next, offering higher amount of money before a firm offers Series A. Venture funding includes Series A and B, while Series C and later rounds includes higher funding for established companies. Equity crowdfunding allows individuals to invest in exchange for shares, whereas product crowdfunding raises capital by offering products in development. Private equity firms or hedge funds provide late-stage funding of $50M or more, while convertible notes serve as interim financing that converts to equity. Then, debt financing involves loans repaid with interest, while secondary market transactions let investors buy shares from existing shareholders. I should also mention grants offer funding without taking equity, and corporate investments come from companies seeking strategic partnerships. We see also Post-IPO funding includes equity investments, loans, and secondary transactions after a company goes public. I also have the type of Non-equity assistance provides mentorship or office space without financial stakes, and a general funding round refers to investments without a specific classification.

The investors database provides details on investors, including their country, name, total funding in USD, founding and closure dates, number of investments (encompassing all funding rounds for a given firm, which raises the question of whether exit rates should be considered), investor type (e.g., angel, venture capital, investment partner), entity type (organization or individual), total funding provided in USD, as well as profile creation and closure dates. However, this database does not contain information on the names of the firms in which investors have invested. In the investors database we find that various entities provide funding and support to firms at different stages. Generally, accelerators offer short-term mentorship and capital in exchange for equity, while incubators take a

more active role in managing firm ideas internally. Angel groups and micro-VCs focus on early-stage funding, whereas venture capital (VC) firms invest from seed to later stages. Corporate venture capital and family investment offices provide funding from corporations and high-net-worth families, respectively. Larger investment entities like private equity firms, hedge funds, and pension funds typically invest in later-stage or post-IPO companies. Investment banks primarily handle stock issuance but may invest in private equity rounds. Secondary purchasers buy shares from existing shareholders rather than the company itself diversifying risks. Other support structures include entrepreneurship programs, university programs, and firm competitions, which provide resources without necessarily taking equity. Government offices may support firms with capital or mentorship, while syndicates forming syndicated loans, and funds of funds pool resources to invest indirectly.

The investment database includes variables such as firm name, investor type (organization or individual), investor name, funding round name, and the investor UUID. I have used as the date where the investment took place the variable in Crunchbase created_at that provides information on the date and time. For analysis purposes, I just focus on the year.

The investment partners database contains information on investment partnerships, including variables such as firm name, funding round name, UUIDs for investors and investment partners, and their respective names. In the same way here, I have used as the date where the investment took place the variable in Crunchbase created_at that provides information on the date and time. For analysis purposes, I just focus on the year. I am counting as investors the investment partners of the investors, this is something that will not alter my results at first because I will consider that two firms are connected if at least one common investors exist among them.

Finally, the organizations database provides information on organizations, including the number of funding rounds they have participated in, the number of exits, organization name, primary role, industry category and size of the firm, among other relevant attributes. For industry category the subgroups were extremely detailed to be able to determine a industry level shock or control for it. Hence, I opted to group the subcategories into coarser categories that determine the industry of each firm. These categories are up to 20, and some of them are, Agriculture, Education, Government or Administration, Financial services, Marketing, Energy, Manufacturing, Sciences, Tech, Transportation, Tourism, Professional Services, Real Estate, among others.

From the final dataset excludes Post-IPO funding rounds for our analysis to be able to examine the network of funding rounds that were before any IPO to see the influence on IPO. Incorporating this exclusion analysis, the final dataset will include information on 208918 firms and 155524 investors. From here I will keep the firms that are connected

to others through at least one common investor. For some firms and investors I will not have information on certain attributes such as money that was invested in the firm , or when it was announced. For the cases where we have no information on the amount invested we will treat it as missing, not to bias our estimates with treating them as zeros. However, for the firms that do not appear with any IPO information are assumed to not have undertaken an IPO. These cases will be set to zero. The same will happen when we do not observe any investment undertaken by an investor in a specific year.

The majority of financing agreements in the data come from angel investors, which together with investment partners make up more than half of the registered agreements in the private market data. Normally, angel investors finance firms that are at the early stages of their life cycle. Likely these deals do not involve very large sums, but because of this, angel investors invest in risky opportunities in exchange of some part, or fraction, of the business' equity. On the other hand, venture capital and private equity firms, which make up almost another 25% percent of the data, are usually involved in financing deals for firms with some track record and high-growth opportunities. We can see this in Table 13. One possible caveat is the relatively small share of hedge fund deals that are contained in the database. It likely that, according to Table 13, in the data we have many young and small firms, with some of them having grown into an equity level such that they have gotten financing from bigger partners.

When focusing on all firms included in the analysis and successfully match them to their corresponding investors, we find that the distribution of investor types is heavily skewed toward venture capital. Overall, VC-related investors dominate the sample: `venture_capital` alone accounts for 30.6% of observations (278,502 out of 910,053), and venture capital also appears frequently in multi-type combinations (e.g., with `accelerator`, `micro_vc`, and `syndicate`). The investor with the maximum number of firms is Y Combinator that possess 69297 firms and it is an accelerator. Accelerators that do add knowledge are usually top-tier programs like Y Combinator, Techstars, or 500 firms. They also provide structured mentorship, fundraising help, product guidance, and network access. They run intensive programs (e.g., 3-month cohorts), and assign experienced mentors or partners. Some like Techstars are in the venture capital group since it is characterized as Venture Capital, Accelerator.

Looking at the detail of the final merged database (excluding Post-IPO funding rounds) of 155,524 investors (some we do not know the type of) and 208,918 firms we get that the database gives us around financing deals since we do not always have this information.

As Table 9 below shows, the average deal size in the data is of about 21.8 US$ B (Billions) with the largest funding deal. Notice that many funding rounds have missing information on raised amount.

Table 9: Summary statistics for raised amount (in millions USD)

Variable	Obs	Mean	Std. Dev.	Min	Max
raised_amount	185,191	25.2	157.0	0	21,800

Note: All monetary values are expressed in millions of USD. The mean corresponds to $25.2 million and the maximum to $21.8 billion.

A clearer picture about the nature of the firms in the data emerges when looking at the type of deals that have been signed. As shown in Figure 1 most of the funding agreements I can identify are seed or pre-seed deals, which usually are done at the very start of a company's life cycle. We can also see that Series A deals are the second-most frequent in the database. Normally, firms do go through many rounds of funding. The stage and scope of the financing needs, and what is given in return to investors, determines whether the financing involves series A, B, or C stocks. Here, Series C will usually involve financing to companies that are already established and seek to increase their size, while Series A funding, involve greater risk for investors as the firm has likely not fully develop its product.

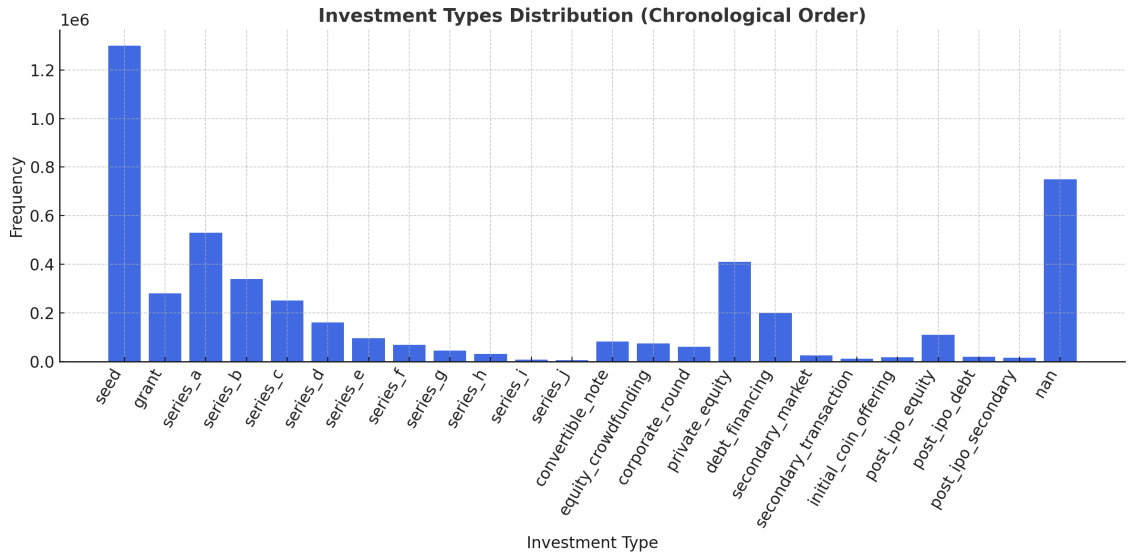


Figure 1: Investment Type

When looking at the behavior of investors, as seen in next Table 10, the sample shows that, on average, firms in the data usually have around 4 different investors across all rounds, which some having as many as 192 investors in their whole time span. It is likely that the same investor pours funds into a firm many times. Some investors seem to be present quite often in the private market, as I have even more than 4000 observations for the same investor across funding rounds for all firms. We see that, on average, the average

rounds per investor at the same firm is above 1, and this is obvious because we are just considering the matched observations that will allow us to have at least an investor.

This means that all firms we consider have an investor in the private market in at least one of their funding rounds. For the cases where the same investor invests in the same type of funding round, but at different dates, we consider as many as funding rounds different dates combinations we have. Hence, this approach takes into account the most detailed and complete information from the investors' side, but also on the firms side. This is going to be important further in the future when we consider the changes in valuation of the firm that incorporates all the investments that each investor undertook in any of the years. Therefore, all firms in here have reached the condition of having investors, but not all of them will be part of my network because of the existence of investors that are not common to any other firm . Then, we have a complete mapping from firms to investors. We can see some descriptives for the database on investors in Table 10

Table 10: Descriptives for investors

Variable	N	Mean	Std. Dev	Min	Max
Investors per firm	208,918	3.94	5.48	1	192
Funding rounds per investor	155,524	6.43	38.00	1	4,915
Funding rounds per investor at same firm	822,943	1.21	0.58	1	22

Now having described how does the private market data look like in my sample, it is important to highlight the frequency and time span of the Initial Public Offerings I have available. Before that, however, one downside of the data for my approach is that there are firms for which I have their IPO date but no match in the investor database. Although these firms make a sizeable share of total IPOs (see right panel of figure 2) in the sample, they are not valid for my network approach and are thus dropped. All in all, the behavior of the full sample and the reduced has some similarities, and have the most of the IPOs from years 2010–2018, being more persistence up to 2020 in the reduced sample of firms.

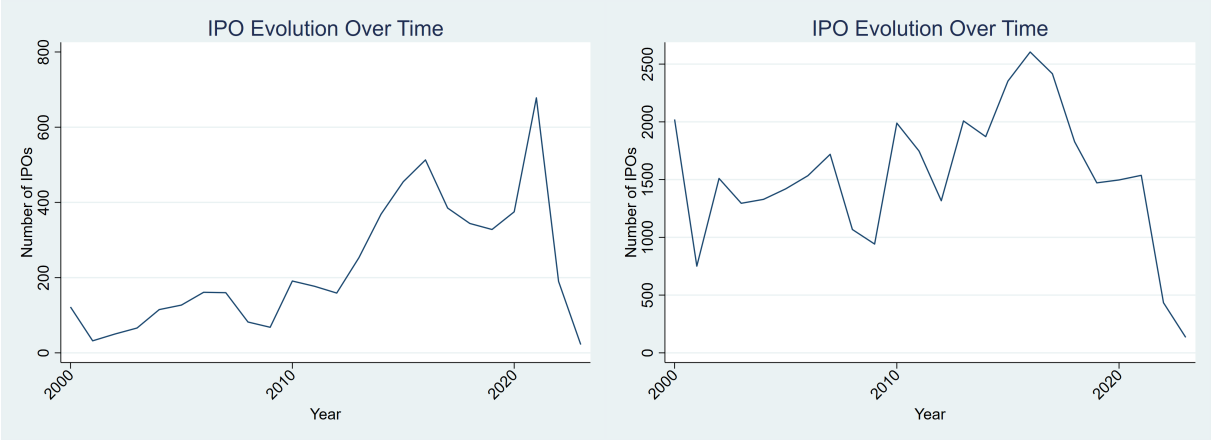


Figure 2: Number of IPOs for matched network firms (left) and number of IPOs for all firms (networked or not) (right)

It is worth mentioning that in the database of IPOs where some firms will not enter in my analysis because we have no information on investment that match the firm with an investor, or because these firms are in isolation. We have that the majority of IPOs are in USA being representing the 21% of the information on IPOs, followed by China with 17% of the information.

Regarding relevant characteristics of investments, we observe that the firms that are invested per investor are on average 5 (Table 12 in Appendix A.2. This is based on the final dataset, which includes all investors and firms with available information on funding rounds. From this dataset, we observe that not all investors remain connected within the network. This is because some investors may only be linked to isolated firms (those not connected to any other). As a result, these investors never appear as common investors between firms and do not serve as linking nodes in the graph.

This implies that whenever two firms are linked, they necessarily share at least one common investor. Conversely, it is possible for an investor who funds multiple firms to be isolated in the corresponding firm network when no other investor participates in any of those same firms. The total number of investors that at the same time invest in at least two firms are 65504 investing on average on 11 firms, and maximum 4084 firms. Hence, the network of firms are formed by 65504 being connected to at least a pair of firms.

Once we exclude the post-IPO funding rounds information we have that each investor invests on average in 2 categories (industries):

Table 11: Summary statistics for `num_categories` (full sample)

	N	Mean	Std. Dev	Min	Max
`num_categories`	158,193	2.04	2.08	1	20

This table includes all investors from the beginning, where some of them we know are not going to appear in the firms network (as common investors) since they do not invest in more than one firm .

We also observe that each investor type invests on average in 8 categories in total. Check Table 14 in Appendix A.2. The most predominant categories in my database for the graph are Sciences and Health and Software and IT.

Among unique combinations of investor type and categories 456 because investor_types have very specific subcategories that I grouped by the previous types in a broader sense (VC, Equity, etc).

A.2 Data and Network Construction

It is important to note that the data I use for this paper does not include funds or acquisitions, even though many investments that are reported in the funding rounds datasets are originated from different types of funds. Additionally, while data on jobs, education, employees, and investors is available, it will not be utilized in this paper at this stage. After matching the corresponding investors' identification, firm and funding rounds we will have an entire dataset that corresponds to investors, investments and funding rounds spanning from 1901 to 2023, with some jumps in time. For the IPO information we will have that the periods that correspond to our final dataset are from 1890 to 2023.

This appendix reports additional summary statistics for the firm network.

Table 12: Statistics on number of firms funded by investor

	N	Mean	Std. Dev	Min	Max
firms per investor	155,524	5.29	29.61	1	4,084

Table 13: Main investor types in private markets

Investor Type	Percentage
Angel Investors	34.51
Investment Partner	29.14
Venture Capital	16.91
Private Equity	8.88
Micro VC	2.47
Accelerators	2.10
Investment Banks	1.47
Incubator	1.33
Family Investment	1.02
Government Office	0.63
Hedge Funds	0.45
Venture Debt	0.37
Syndicates	0.11
Firm Competition	0.07
University Programs	0.24
Entrepreneurship Program	0.24
Pension Funds	0.05
Total Observations	130,501

The distribution of industries (num_categories) is as follows:

Table 14: Summary statistics for `num_categories`

	N	Mean	Std. Dev	Min	Max
`num_categories`	456	7.73	6.42	1	21

First orders peers aggregate distribution per year:

Table 15: Firms with at least one peer by year

Year	Firms with peers
2000	2,287
2001	2,946
2002	3,507
2003	4,101
2004	4,882
2005	5,751
2006	6,907
2007	8,983
2008	10,992
2009	13,259
2010	16,783
2011	19,923
2012	23,982
2013	30,339
2014	42,039
2015	54,015
2016	68,966
2017	85,547
2018	104,373
2019	122,503
2020	143,890
2021	169,602
2022	192,832
2023	198,243

Regarding the sample convention, post-IPO financing rounds are excluded from the construction of the pre-IPO network. Investments before 2000 enter as initial conditions. Missing funding amounts remain missing. Also, a firm without a recorded IPO is coded as not completing an IPO during the sample. The record creation date is used only when the funding announcement date is unavailable, creating possible measurement error in the exact year of link activation.

B Robustness Empirical Results

B.1 Lag structure

For the baseline analysis in 1 , the inclusion of additional time-varying controls may change the estimation sample because of missing values. I therefore conduct a common-sample exercise to distinguish the effect of adding peer controls from the effect of changing the sample. I first estimate the full specification, which includes both focal-firm and peer controls, and retain the observations included in that estimation. I then re-estimate the more parsimonious specification on exactly the same set of firm-year observations.

Table 16 reports the common-sample comparison. Column (1) excludes both focal-firm and peer-level time-varying controls and includes only the lagged peer IPO ratio, in addition to the fixed effects. Column (2) includes both focal-firm and peer-level controls. All time-varying controls are lagged by one year. Holding the estimation sample constant ensures that any difference between the two specifications is due to the inclusion of the additional controls, rather than to a change in sample composition. The similarity of the estimated coefficients indicates that neither the additional controls nor the observations excluded because of their missing values materially drive the estimated relationship.

Table 16: First-Order Peer IPO Exposure and IPO Probability

	IPO	IPO
	(1)	(2)
Lagged peer IPO exposure	0.0170***	0.0168***
	(0.00603)	(0.00608)
Observations	989,915	989,915
Clusters (firm)	154,484	154,484
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
R^2	0.261	0.265
Adjusted R^2	0.110	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO exposure is measured as the share of the focal firm's first-order peers completing an IPO in $t-1$. Column (1) excludes additional controls. Column (2) controls for the lagged focal firm's transformed age, cumulative investments, cumulative unique investors, and average investor score, and the lagged average first-order peer characteristics. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

I examine the temporal persistence of the relationship between peer IPO exposure and a firm's IPO decision by estimating specifications with longer lags of first-order peer IPO

exposure. The distributed-lag specification is

$$IPO_{it} = \alpha + \sum_{\ell=1}^{4} \delta_\ell PeerIPORatio_{i,t-\ell} + X'_{i,t-1}\gamma + \rho_i + \mu_{t\times r\times s} + \varepsilon_{it}.$$

where $PeerIPORatio_{i,t-\ell}$ denotes first-order peer IPO exposure ℓ years before the focal firm's IPO decision. The vector $X_{i,t-1}$ contains the lagged focal-firm and peer controls, ρ_i denotes firm fixed effects, and $\mu_{t\times r\times s}$ denotes year-by-industry-by-country fixed effects. Table 17 reports the results. In the following Table, Columns (1)–(4) introduce each lag separately, and Column (5) includes all four lags simultaneously. All columns are estimated on the common sample selected by the joint distributed-lag specification. This prevents differences across columns from being driven by the loss of observations required to construct the longer lags.

Table 17: Dynamic Effects of First-Order Peer IPO Exposure

	IPO Lag 1	IPO Lag 2	IPO Lag 3	IPO Lag 4	IPO Joint model
Lagged peer IPO ratio (t-1)	0.0130*				0.0148**
	(0.00754)				(0.00740)
Lagged peer IPO ratio (t-2)		0.0150*			0.0167**
		(0.00810)			(0.00830)
Lagged peer IPO ratio (t-3)			0.00139		0.00392
			(0.00584)		(0.00635)
Lagged peer IPO ratio (t-4)				-0.00526	-0.00284
				(0.00577)	(0.00591)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Industry-Country-Year FEs	Yes	Yes	Yes	Yes	Yes
Observations	551155	551155	551155	551155	551155
Clusters (firm)	96151	96151	96151	96151	96151
R-squared	0.300	0.300	0.300	0.300	0.300
Adjusted R-squared	0.135	0.135	0.135	0.135	0.135
P-value: lags 2–4 jointly zero					0.218
P-value: all lags jointly zero					0.035
Cumulative effect: lags 1–4					0.0326
SE of cumulative effect					0.0166
P-value of cumulative effect					0.050
Difference: Lag 1 minus Lag 2					-0.0019
SE of Lag 1 minus Lag 2					0.0117
P-value: Lag 1 equals Lag 2					0.868

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's first-order peers completing an IPO in the corresponding lagged year. Columns (1)–(4) include each lag separately, while Column (5) includes all four lags simultaneously. All specifications control for the focal firm's transformed age, cumulative investments, cumulative unique investors, average investor score, cumulative share of investments involving lead investors, and leave-one-out investor-industry specialization and lagged average First-order peer controls. The difference between the first and second lag coefficients, together with its standard error and corresponding p-value, is calculated from the joint specification. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 17 examines the temporal persistence of the estimated relationship. Columns (1)–(4) introduce each lag of the peer IPO ratio separately, while Column (5) includes all four lags simultaneously. All columns use the common sample required to observe four lags.

The results indicate that the relationship is concentrated in the first two years following a peer IPO. In the joint specification, the coefficients on the one-year and two-year lags are

both positive and statistically significant, at 0.0148 and 0.0167, respectively. Moreover, the two coefficients are statistically indistinguishable from one another: the null hypothesis that the Lag 1 and Lag 2 effects are equal cannot be rejected ($p = 0.868$). By contrast, the coefficients on the third and fourth lags are small and statistically insignificant.

The null hypothesis that the coefficients on Lags 2–4 are jointly equal to zero cannot be rejected ($p = 0.218$), although this joint test should be interpreted together with the individually significant Lag 2 estimate. The cumulative effect over the four years is positive, at 0.0326, and is statistically significant at the 5 percent level ($p = 0.050$). Overall, the estimates suggest that the association between peer IPO exposure and the focal firm's IPO probability persists for approximately two years, with no statistically detectable individual effect at longer horizons.

The main text reports the one-year lag. The specifications including two and more lags show that the relationship is concentrated in the first one or two years and becomes statistically indistinguishable from zero at longer horizons.

B.2 Investment by New Investors

To examine whether peer IPO exposure is followed by investor entry beyond the investors previously involved with the IPO peers, I use an alternative investor-entry outcome. The dependent variable equals one if, in year t, the focal firm receives at least one investor that finances the focal firm for the first time and that, as of $t-1$, had not invested in any of the focal firm's first-order peers that went public in $t-1$. I estimate

$$\text{NewNonIPOInvestor}_{it} = \alpha_i + \lambda_{sct} + \beta\,\text{PeerIPOShare}_{i,t-1} + \gamma' X_{i,t-1} + \delta' Z_{i,t-1} + \varepsilon_{it}, \quad (20)$$

where $X_{i,t-1}$ contains lagged focal-firm characteristics and $Z_{i,t-1}$ contains lagged characteristics of the focal firm's first-order peers. The specification includes firm fixed effects and industry-by-country-by-year fixed effects. Standard errors are clustered at the firm level.

Table 18 examines whether peer IPO exposure is followed by the arrival of new investors that were not previously connected to the peers responsible for that exposure. The dependent variable equals one if, in year t, the focal firm receives at least one investor that finances the firm for the first time and that, as of $t-1$, had not invested in any of its first-order peers that went public in $t-1$. The purpose of this exercise is to assess whether the response to peer IPOs can be explained only by capital being reallocated by investors already connected to the IPO peer. If this were the only mechanism at work, peer IPO exposure should not predict the arrival of investors outside the IPO peers' existing investor base. The results are reported in Table 18. The coefficient on lagged peer

IPO exposure is positive and statistically significant. A 10-percentage-point increase in the share of first-order peers going public in $t-1$ is linked to a 0.287-percentage-point increase in the probability that the focal firm receives such a new investor in year t. This suggests that the response to a peer IPO is not limited to capital being reallocated by investors already involved with the IPO peer.

This exercise does not directly test the learning mechanism developed in Section 6. However, it makes a narrow capital-reallocation explanation less compelling. In the model, the effect of a peer IPO instead operates because connected private firms observe the IPO decision and update their beliefs about the public-market valuation they could expect if they went public themselves. The empirical result is therefore consistent with a response to peer IPOs that extends beyond the investors directly involved in the IPO event.

Table 18: Peer IPO Exposure and the Arrival of Unconnected New Investors

	New investor not connected to peer IPO
	IPO
Lagged first-order peer IPO exposure	0.0287*
	(0.0154)
Lagged focal-firm controls	Yes
Lagged first-order-peer controls	Yes
Firm Fixed Effects	Yes
Industry-Country-Year FEs	Yes
Observations	989,915
Clusters (firm)	154,484
Dependent-variable mean	0.126
R^2	0.411
Adjusted R^2	0.291

Notes: The dependent variable is an indicator equal to one if focal firm i receives at least one new investor in year t that had not previously invested in a first-order peer completing an IPO in year $t-1$, and zero otherwise. Investors that had previously financed the focal firm are not classified as new investors. Lagged first-order peer IPO exposure is the share of firm i's active first-order peers that completed an IPO in year $t-1$. The specification includes lagged focal-firm controls and lagged averages of the corresponding characteristics across the firm's active first-order peers. Standard errors, clustered at the firm level, are reported in parentheses. The model is estimated as a linear probability model. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

B.3 Age heterogeneity

I examine whether the association between peer IPO exposure and a focal firm's IPO probability varies over the firm's life cycle. I divide firms into two groups according to their age: young firms, defined as firms aged five years or less, and mature firms, defined as firms older than five years. I then estimate the baseline specification separately for the two groups. All specifications include the standard focal-firm and first-order peer controls, firm fixed effects, and year-by-industry-by-country fixed effects. Standard errors are clustered at the firm level.

Table 19 examines whether the association between lagged peer IPO exposure and the focal firm's IPO probability varies with firm age. Young firms are defined as firms aged five years or less, while mature firms are those older than five years. The estimated association is positive for both groups. For firms aged five years or less, the coefficient on lagged peer IPO exposure is 0.034, while for firms older than five years it is 0.013. Thus, the point estimate is larger among younger firms, although both groups exhibit a positive relationship between peer IPO exposure and the probability of going public.

Table 19: Peer IPO Exposure and IPO Probability by Firm Age

	IPO Age ≤ 5	IPO Age > 5
Lagged peer IPO ratio	0.034^{***}	0.013^{*}
	(0.011)	(0.007)
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Observations	253,210	708,887
Clusters (firm)	81,765	120,017
Adjusted R-squared	0.095	0.130

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's first-order peers completing an IPO in $t-1$. Young firms are defined as firms aged five years or less in year t, while mature firms are those older than five years. Column (1) restricts the sample to young firms, while Column (2) restricts the sample to mature firms. Both columns include lagged focal-firm controls comprising transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure, together with the corresponding first-order lagged average peer controls. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

One possible interpretation of the results is that younger firms may be more respon-

sive to information generated by IPO activity elsewhere in their firms network, although the age heterogeneity by itself does not identify the underlying mechanism. I therefore interpret this result as descriptive heterogeneity rather than as direct evidence for the learning channel. Also, younger firms face greater uncertainty, even though they may also be farther from the point at which an IPO is feasible. However, when facing more uncertainty the effect of having peers that go IPO might be stronger due to the reduction in this uncertainty. Nevertheless, this heterogeneity does not independently establish the learning mechanism.

B.4 Alternative Measures of Connection Strength

The baseline analysis defines two firms as connected whenever they share at least one investor, treating all first-order connections equally. In this section, I examine whether the results are robust to alternative measures that account for the strength of the investor overlap between connected firms. I consider two different approaches. First, I distinguish between connections involving exactly one common investor and those involving multiple common investors. Second, I construct a weighted network in which peer connections receive greater weight when the focal firm and the peer share a larger fraction of their investor relationships.

Number of common investors. I first distinguish between peers connected through exactly one common investor and peers sharing more than one investor. Stronger investor overlap may facilitate greater information transmission or investor involvement and could therefore be related to larger IPO spillovers.

Table 20 compares peer IPO exposure through connections involving exactly one common investor with exposure through connections involving multiple common investors. The estimated coefficients are positive and statistically significant for both types of connections. Although the point estimate is larger for peers connected through exactly one common investor, the difference between the two coefficients is not statistically significant. I therefore do not interpret this exercise as evidence that the magnitude of the peer IPO spillover varies systematically with the discrete number of shared investors. Rather, the results show that the positive relationship between peer IPO activity and the focal firm's IPO probability is present under both relatively narrower and broader common-investor connections.

Table 20: Peer IPO Spillovers by Common-Investor Intensity

	IPO Both ratios	IPO One common investor	IPO Multiple common investors
Peer IPO ratio: one common investor	0.030***	0.032***	
	(0.007)	(0.007)	
Peer IPO ratio: multiple common investors	0.020***		0.020***
	(0.004)		(0.004)
Firm Fixed Effects	Yes	Yes	Yes
Industry-Country-Year FEs	Yes	Yes	Yes
One common − multiple common	0.010		
p-value of equality test	0.210		
Observations	359,857	359,857	359,857
Clusters (firm)	65,746	65,746	65,746
Adjusted R-squared	0.113	0.113	0.113

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Common-investor intensity is measured by the number of investors shared by the focal firm and each of its first-order peers. The table reports the lagged share of peers completing an IPO in $t-1$, separately for peers connected to the focal firm through exactly one common investor and peers connected through more than one common investor. All specifications include lagged focal-firm controls comprising transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. The specifications also includes the lagged average first-order peer. All specifications use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

In Table 20 the positive relationship is present for peers connected through both one and multiple common investors, and the estimated coefficients are not statistically distinguishable. Because the simple number of shared investors is a coarse measure of connection strength, I also examine continuous measures of investor overlap. These specifications provide clearer evidence that the relevance of peer IPO activity increases with the strength of the direct investor connection.

Weighted investor-overlap network. As an additional robustness exercise, I allow the strength of each first-order connection to vary continuously with the extent of investor overlap. For each directed pair (i, j), I define the edge weight as

$$\omega_{ij,t} = \frac{|\mathcal{K}_{i,t} \cap \mathcal{K}_{j,t}|}{|\mathcal{K}_{i,t}|} \equiv \frac{CI_{ij,t}}{N_{i,t}},$$

where $\mathcal{K}_{i,t}$ and $\mathcal{K}_{j,t}$ denote their respective investor sets, and $CI_{ij,t}$ is the number of common investors shared by firms i and j in year t, and $N_{i,t}$ is the total number of investors connected to firm i in year t. Thus, $\omega_{ij,t}$ measures the share of firm i's investor

base that overlaps with firm j, and notice that it assigns greater weight to peers sharing a larger proportion of the focal firm's investors.

Using these weights, I construct a weighted first-order IPO exposure measure:

$$WeightedIPO_{i,t} = \frac{\sum_{j \in \mathcal{N}_{i,t}} \omega_{ij,t} IPO_{j,t}}{\sum_{j \in \mathcal{N}_{i,t}} \omega_{ij,t}}.$$

This variable captures the weighted fraction of first-order peer exposure linked to irms that go public in year t, assigning greater importance to peers with stronger investor overlap.

Table 21 reports the results. The positive relationship between peer IPO activity and the focal firm 's IPO decision remains when first-order connections are weighted according to the degree of investor overlap. This provides additional evidence that the baseline findings are not driven by the binary treatment of all common-investor connections as equally strong.

Table 21: Weighted Peer IPO Exposure and IPO Decisions

	IPO No peer controls	IPO Peer controls
Weighted peer IPO ratio	0.022***	0.021***
	(0.006)	(0.006)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Observations	989,915	989,915
Clusters (firm)	154,484	154,484
Adjusted R-squared	0.115	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Weighted peer IPO ratio is the weighted share of first-order peers completing an IPO in $t-1$, where edge weights are defined as previously described. Both columns include lagged focal-firm controls: transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (2) also includes lagged average first-order peer controls. Both specifications use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

B.5 Nonlinearity in First-Order Peer IPO Exposure

The baseline specification imposes a linear relationship between the share of first-order peers completing an IPO and the focal firm's probability of going public. I examine whether this restriction is important by augmenting the baseline specification with the square of lagged peer IPO exposure:

$$IPO_{i,t} = \alpha + \beta_1 PeerIPO_{i,t-1} + \beta_2 PeerIPO^2_{i,t-1} + \gamma X_{i,t-1} + \delta \overline{X}_{N_i,t-1} + \rho_i + \mu_{t\times r\times s} + \varepsilon_{i,t},$$

where $PeerIPO_{i,t-1}$ denotes the lagged share of first-order peers that completed an IPO, and the remaining controls and fixed effects are identical to those in the baseline specification.

I then allow the relationship between peer IPO exposure and the focal firm's IPO probability to be nonlinear by including both the lagged peer IPO ratio and its square. The coefficient on the linear term is positive and statistically significant, whereas the coefficient on the squared term is negative and statistically significant. The resulting marginal effect is positive at low and moderate levels of exposure but decreases as the proportion of IPO peers rises. The estimated relationship reaches its maximum at a peer IPO ratio of approximately 0.424.

Table 22: Nonlinear Relationship Between Peer IPO Exposure and IPO Probability

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio	0.046***	0.045***
	(0.008)	(0.008)
Squared lagged peer IPO ratio	-0.054***	-0.053***
	(0.017)	(0.017)
Focal-firm controls	Yes	Yes
First-order controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Estimated turning point	0.422	0.424
Observations	989,915	989,915
Clusters (firm)	154,484	154,484
Adjusted R-squared	0.115	0.115

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio and the Squared lagged peer IPO ratio. It includes both $ratio_ipo_first_{i,t-1}$ and its squared term, $ratio_ipo_first^2_{i,t-1}$. The implied turning point is $-\hat{\beta}_1/(2\hat{\beta}_2)$. The specification includes the lagged focal-firm controls transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. It also includes the the lagged average first-order peer controls. Both use common sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Because the specification is nonlinear in peer exposure, the marginal association of an increase in the peer IPO share is

$$\frac{\partial IPO_{i,t}}{\partial PeerIPO_{i,t-1}} = \beta_1 + 2\beta_2 PeerIPO_{i,t-1}.$$

Because observations with very high peer IPO ratios are relatively uncommon, I interpret this pattern primarily as evidence of diminishing marginal effects rather than as evidence that sufficiently high peer IPO exposure reduces the probability of going public. This result is consistent with the interpretation that additional peer IPOs convey less incremental information when the focal firm has already observed substantial IPO activity among its peers.

B.6 Pure Relevant Cliques Among IPO Peers

The previous exercise measures informational redundancy continuously using the strength of indirect connections surrounding realized IPO peers. I next consider a more stringent measure that captures whether the set of IPO peers forms a fully redundant structure with respect to the investors relevant for the focal firm .

For each focal firm i and year t, I identify the first-order peers j that complete an IPO in year $t-1$. For firm -years with at least two such IPO peers, I construct all unordered pairs (j, k) among the realized IPO peers and classify each pair according to the investors connecting j and k. Specifically, each IPO-peer pair belongs to one of three categories:

1. **Pure relevant pair:** firms j and k are connected, and every investor connecting them is also relevant for focal firm i, in the sense that the investor is shared with i.

2. **Mixed pair:** firms j and k are connected through at least one investor that is not shared with focal firm i.

3. **Disconnected pair:** firms j and k do not share an investor and are therefore not directly connected.

I define $\texttt{pure_clique_all}_{it} = 1$ if every unordered pair of IPO peers associated with focal firm i in year t is a pure relevant pair. Equivalently, all realized IPO peers are mutually connected exclusively through investors that are also relevant for the focal firm. The indicator equals zero whenever at least one IPO-peer pair is either disconnected or connected through an investor outside the focal firm 's relevant investor set.

This definition captures an extreme form of informational redundancy. When the variable for pure clique is equal to one, the IPO peers are not only mutually connected, but their connections are entirely generated by the same investor relationships that link them

to the focal firm. Their IPOs are therefore more likely to reveal overlapping information about a common underlying investor set. By contrast, when the pure clique variable is equal to zero, the realized IPO-peer structure contains either disconnected peers or connections involving additional investors, generating potentially more heterogeneous and less redundant information.

The pure-relevant-clique indicator contains a substantial number of missing observations because it is defined only for firm-years in which the focal firm has at least two IPO peers. With fewer than two IPO peers, there is no pairwise relationship among IPO peers and, consequently, no clique structure to classify. These missing values therefore indicate that the measure is not applicable, rather than missing information about the underlying network. Then, I test whether the effect of peer IPO exposure differs between these two network structures by interacting lagged peer IPO exposure with the pure-clique indicator. The relevant specification is

$$IPO_{i,t} = \beta + \delta_1 IPO_j_share_{i,t-1} + \delta_2 \left(IPO_j_share_{i,t-1} \times \texttt{pure_clique_all}_{it}\right) + \\ + \rho_i + \lambda_{gct} + X'_{i,t-1}\gamma + \varepsilon_{i,t},$$

where ρ_i denotes firm fixed effects, λ_{gct} denotes the fixed effects used in the corresponding specification, and $X_{i,t-1}$ contains the remaining controls. In this specification, δ_1 captures the association between peer IPO exposure and the focal firm's IPO decision in non-pure structures, whereas $\delta_1 + \delta_2$ gives the corresponding effect when all realized IPO peers form a pure relevant clique. A negative interaction coefficient therefore indicates that peer IPO exposure has a weaker effect when the IPO peers provide highly overlapping signals. Table 23 reports the results.

Table 23: Peer IPO Exposure and Pure Relevant Cliques

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio	0.102***	0.110***
	(0.032)	(0.033)
Pure relevant clique	0.001*	0.001
	(0.000)	(0.000)
Lagged peer IPO ratio × pure clique	-0.094***	-0.082**
	(0.033)	(0.033)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Observations	189,440	189,440
Clusters (firm)	43,884	43,884
Adjusted R-squared	0.107	0.107

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the lagged share of the focal firm's first-order peers completing an IPO. Both columns include lagged focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (2) includes the lagged average first-order peer controls. Both use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

The estimates show that the positive peer IPO effect is concentrated in non-pure structures. The coefficient on lagged peer IPO exposure is positive and statistically significant, while its interaction with the pure relevant clique indicator is negative and substantially offsets the baseline effect. This pattern is consistent with informational redundancy: when all IPO peers are mutually connected exclusively through investors that are also relevant for the focal firm, additional peer IPOs convey less incremental information. By contrast, IPO-peer configurations containing disconnected peers or connections through additional investors provide relatively more heterogeneous signals and are connected to a stronger focal-firm response.

As in the preceding exercise, this result should be interpreted as heterogeneity in the informational content of observed peer IPO configurations rather than as an independent causal effect of clique formation.

B.7 Nonlinear specifications

As a supplementary functional-form exercise, Table 24 reports average marginal effects from logit models. Both columns include the standard time-varying focal-firm controls and year fixed effects. Column (1) excludes first-order peer controls, whereas Column (2) includes them. To ensure comparability, both specifications are estimated using the sample from the model with peer controls.

The average marginal effect of the lagged peer IPO ratio is positive and statistically significant in both specifications. Without peer controls, the estimated average marginal effect is 0.010 (SE = 0.001). After including peer controls, it is 0.009 (SE = 0.001). Thus, in the specification with peer controls, a 10-percentage-point increase in the lagged peer IPO ratio rises the probability that the focal firm completes an IPO in an approximately 0.09-percentage-point.

These results provide supplementary evidence that the positive association between peer IPO exposure and the focal firm's IPO probability is also present under a nonlinear probability model. They are not intended to replace the baseline linear probability estimates, since the logit specification does not include firm or year-by-industry-by-country fixed effects. Incorporating firm fixed effects in a nonlinear binary-choice model would retain only firms exhibiting within-firm variation in IPO status. Given the rarity of IPOs, this would exclude most firms and create substantial separation concerns such as the incidental parameter problem in nonlinear models.

Table 24: Peer IPO Exposure: Logit Average Marginal Effects

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio	0.010***	0.009***
	(0.000)	(0.001)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Industry-Country-Year FEs	Yes	Yes
Observations	988,255	988,255
Clusters (firm)	154,484	154,484
Pseudo R-squared	0.124	0.132

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's first-order peers completing an IPO in $t-1$. The table reports average marginal effects from a logit model rather than log-odds coefficients. Each reported estimate is the sample average of the observation-specific change in predicted IPO probability related to a one-unit increase in the corresponding explanatory variable. The model includes lagged focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure, together with the corresponding lagged average first-order peer controls. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 25 reports average marginal effects from the probit specifications. The estimated marginal effect of the lagged peer IPO ratio is positive and statistically significant in both specifications. Without first-order peer controls, the average marginal effect is 0.014, while after including peer controls it is 0.013. Thus, in the specification with peer controls, a 10-percentage-point increase in the lagged share of direct peers completing an IPO is linked to an approximately 0.13-percentage-point increase in the probability that the focal firm completes an IPO. The positive probit estimates provide additional evidence that the relationship between peer IPO exposure and the focal firm's IPO probability is not specific to the linear probability functional form. Nevertheless, the probit results should be interpreted as supplementary evidence rather than as a direct nonlinear counterpart to the baseline estimates. Unlike the baseline linear probability specification, the probit model does not include firm or year-by-industry-by-country fixed effects.

Table 25: Peer IPO Exposure: Probit Average Marginal Effects

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio	0.014***	0.013***
	(0.001)	(0.001)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Industry-Country-Year FEs	Yes	Yes
Observations	988,255	988,255
Clusters (firm)	154,484	154,484
Pseudo R-squared	0.130	0.136

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's first-order peers completing an IPO in $t-1$. The table reports average marginal effects from a probit model rather than coefficients on the latent probit index. Each estimate is the sample average of the observation-specific change in predicted IPO probability corresponding to a one-unit increase in the corresponding explanatory variable. The model includes average focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure, together with the corresponding first-order peer controls. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

C Placebo Networks

This appendix describes the construction of the placebo networks and the corresponding placebo exercise. The purpose is to assess whether the estimated relationship between peer IPO activity and a focal firm's IPO decision can be reproduced after randomly changing peer identities while preserving each firm's number of eligible peers.

For every regression year t, I first construct the observed peer network using all firm–investor relationships observed by $t-1$. Two firms are connected if they share at least one investor. A firm is eligible to serve as a peer in year $t-1$ if it belongs to the private-firm risk set in that year. A firm that goes public in $t-1$ therefore remains eligible and contributes to peer IPO exposure in year t, whereas a firm that went public before $t-1$ is excluded from the eligible peer set.

Let $\mathcal{N}_{i,t-1}$ denote the set of eligible peers of focal firm i. For each placebo draw $b = 1, \ldots, B$, I directly rewire the projected peer network using degree-preserving edge. Eligible-to-eligible links are rewired as an undirected network, while links between focal

firms outside the lagged eligible set and eligible peer firms are rewired preserving their directed structure. The procedure does not create self-links or duplicate links and preserves each focal firm's exact number of eligible peers.

The annual cumulative network snapshots are randomized independently. Consequently, the procedure preserves the relevant degree structure in every year but does not impose that the randomized networks form a single nested history of persistent peer relationships. Finite degree-preserving swaps do not necessarily generate a uniform draw from all networks with the same degree sequence, so the exercise is interpreted as an empirical degree-preserving network placebo.

Given placebo peer set $\mathcal{N}_{i,t-1}^{(b)}$, placebo peer IPO exposure is defined as

$$PeerIPO_{i,t-1}^{(b)} = \frac{\sum_{j \in \mathcal{N}_{i,t-1}^{(b)}} IPO_{j,t-1}}{|\mathcal{N}_{i,t-1}^{(b)}|},$$

whenever $|\mathcal{N}_{i,t-1}^{(b)}| > 0$. IPO outcomes are never randomized.

To isolate the variation generated by changing peer identities, I hold the estimation sample and control variables fixed across placebo regressions. The sample is defined using the observed network. Focal-firm controls and first-order peer controls are kept at their observed-network values in every draw. Thus, only peer IPO exposure is replaced by its placebo-network counterpart.

For each placebo draw b, I estimate

$$IPO_{i,t} = \alpha + \delta^{(b)} PeerIPO_{i,t-1}^{(b)} + \gamma' X_{i,t-1} + \beta' X_{i,t-1}^{N} + \rho_i + \mu_{t \times r \times s} + \varepsilon_{i,t}^{(b)}, \qquad (21)$$

where $X_{i,t-1}$ contains focal-firm controls and $X_{i,t-1}^{N}$ contains the first-order peer controls calculated from the observed network. All specifications include firm fixed effects and year $\times$ industry $\times$ country fixed effects, and standard errors are clustered at the firm level.

I construct $B = 100$ placebo networks. The placebo coefficients are centered close to zero, while the coefficient estimated using the observed network lies above the entire empirical placebo distribution. None of the 100 placebo estimates is as large in absolute value as the observed-network estimate.

Using a +1 finite-sample correction [11], I compute the two-sided empirical placebo tail probability as

$$p_{\text{placebo}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{|\widehat{\delta}^{(b)}| \geq |\widehat{\delta}_{\text{real}}|\right\}}{B+1}.$$

[11] The correction avoids reporting a zero p-value when none of the finite number of placebo networks produces an estimate as extreme as the actual-network estimate.

With 100 placebo draws and no placebo coefficient as large in absolute value as the observed-network estimate, this gives

$$p_{\text{placebo}} = \frac{1}{101} \approx 0.0099.$$

The rewiring changes approximately 90.3% of focal-peer assignments, while the number of eligible peers of every focal firm remains unchanged. Thus, the exercise substantially changes peer identities without changing firms' eligible-peer counts.

Figure 3 presents the empirical distribution of the 100 placebo coefficients and the corresponding observed-network estimate.

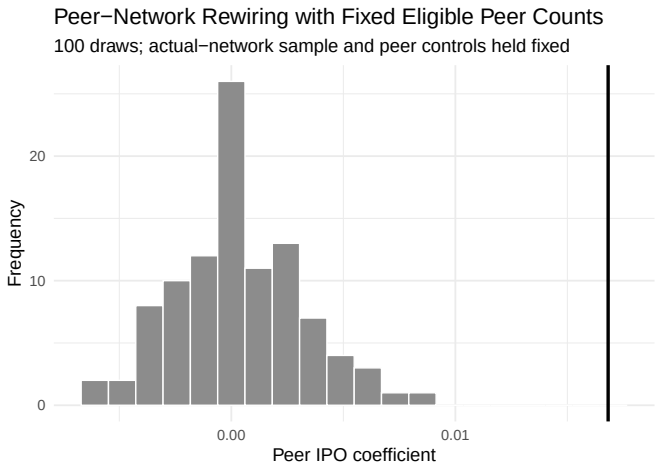


Figure 3: Distribution of peer IPO coefficients across 100 placebo networks generated by degree-preserving rewiring of the projected firm–firm network. The vertical line denotes the observed-network estimate. All regressions are estimated on the same common sample. Peer IPO exposure and first-order peer controls are reconstructed in each placebo network.

Overall, the placebo coefficients are tightly concentrated around zero, whereas the coefficient obtained from the observed shared-investor network lies outside the empirical placebo distribution. The estimated association is therefore not reproduced by randomly changing peer identities while holding constant firms' eligible peer counts, the estimation sample, and the observed controls. This finding supports the empirical relevance of the observed shared-investor network structure. Nevertheless, the placebo exercise does not by itself establish causality or distinguish learning from coordination, liquidity, or other mechanisms operating through the network.

D Investor and Network Fixed Effects

This section presents additional tests addressing the possibility that the baseline relationship is driven by unobserved investor characteristics or by the structure of the common-

investor network. I first introduce fixed effects based on firms' cumulative sets of lead investors.

D.1 Cumulative Leader-Investor-Set Fixed Effects

I first introduce fixed effects based on the exact cumulative set of lead investors involved in each firm. The construction includes all investors identified as leaders, regardless of whether they are classified as venture capital investors.

Let $\mathcal{L}_{it}$ denote the cumulative set of lead investors observed for firm i up to and including year t. The set evolves according to $\mathcal{L}_{it} = \mathcal{L}_{i,t-1} \cup \mathcal{L}_{it}^{\text{new}}$, where $\mathcal{L}_{it}^{\text{new}}$ contains the lead investors newly observed for firm i in year t.

This is a cumulative construction. When a new lead investor appears, it is added to the existing set, while previously observed lead investors remain in the set. Hence, a new leader does not replace the firm's previous leaders. The set remains unchanged in years without a new lead investor and when a previously observed leader appears again.

After sorting the investor identifiers, I assign a global categorical identifier, denoted by `leader_set_cum_id`, to every distinct cumulative leader set. Two firm-year observations receive the same identifier if and only if their exact cumulative leader sets coincide:$s_{it}^{L} = s_{j\tau}^{L} \iff \mathcal{L}_{it} = \mathcal{L}_{j\tau}$. The numerical value of the identifier has no cardinal or ordinal interpretation. It changes for a firm only when a previously unobserved lead investor is added to its cumulative set.

Table 26 augments the baseline specification with cumulative leader-set fixed effects:

$$IPO_{it} = \beta\, PeerIPO_{i,t-1} + X'_{i,t-1}\gamma + \alpha_i + \delta_{g(i),c(i),t} + \lambda_{s_{it}^{L}} + \varepsilon_{it},$$

where α_i denotes firm fixed effects, $\delta_{g(i),c(i),t}$ denotes industry–country–year fixed effects, and $\lambda_{s_{it}^{L}}$ denotes cumulative leader-set fixed effects.

The cumulative leader-set fixed effects are not interacted with calendar-year indicators. They absorb unobserved factors associated with the identity and exact composition of the firm's historical lead-investor set. They are nevertheless dynamic at the firm level because a firm's fixed-effect category changes whenever a previously unobserved lead investor enters its cumulative set. Firms that never have an observed lead investor remain in this category throughout the sample period. For convenience, the empty-set category is coded as `leader_set_cum_id` $= 0$.

Table 26: Peer IPO Spillovers with Cumulative Leader-Set Fixed Effects

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio	0.010*	0.010*
	(0.006)	(0.006)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Cumulative leader-set FEs	Yes	Yes
Observations	977,952	977,952
Clusters (firm)	154,008	154,008
Adjusted R-squared	0.137	0.137

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's first-order peers completing an IPO in $t-1$. The cumulative leader-investor set contains all investors that have invested in the focal firm and have been classified as lead investors by year t. Both columns include focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (2) includes the lagged average first-order peer controls. Both use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

The coefficient on lagged peer IPO exposure remains positive and statistically significant after including the cumulative leader-set fixed effects. This result indicates that the baseline relationship is not driven solely by unobserved characteristics regarding the identity or composition of the firm's lead investors. Because lead investors are likely to be particularly involved in monitoring, governance, and IPO preparation, controlling for their exact cumulative composition constitutes a demanding robustness test.

D.2 Cumulative VC Lead-Investor-Set Fixed Effects

I next apply the same cumulative construction to the subset of investors that are simultaneously identified as lead investors and classified as venture capital investors. Thus, the construction excludes lead investors that are not VCs as well as VC investors that do not act as leaders.

Let $\mathcal{K}_{it}^{VL}$ denote the cumulative set of VC lead investors related to firm i by year t. As

in the preceding construction, a newly observed VC lead investor is added to the existing set, while previously observed VC lead investors remain in the set. The set therefore remains unchanged in years without a new VC leader, when a previously observed VC leader appears again, or when a new investor is not simultaneously classified as both a VC and a leader. I assign a global categorical identifier to every distinct cumulative VC-leader set. Two firm-year observations receive the same identifier if and only if their exact cumulative VC-leader sets coincide: $s_{it}^{VL} = s_{j\tau}^{VL} \iff \mathcal{K}_{it}^{VL} = \mathcal{K}_{j\tau}^{VL}$. In the data, this identifier is denoted by `vcst_gl`. Its numerical value has no cardinal or ordinal interpretation. It changes for a firm only when a previously unobserved VC lead investor is added to its cumulative set.

Firm-years preceding the first observed VC lead investor are assigned to a common empty-set category. Firms that never have an observed VC lead investor remain in this category throughout the sample period. For convenience, the empty-set category is coded as `vcst_gl` $= 1$.[12]

Table 27 first augments the baseline specification with the cumulative VC-leader-set identifier as an additive fixed effect:

$$IPO_{it} = \beta\, PeerIPO_{i,t-1} + X'_{i,t-1}\gamma + \alpha_i + \delta_{g(i),c(i),t} + \lambda_{s_{it}^{VL}} + \varepsilon_{it},$$

where α_i denotes firm fixed effects, $\delta_{g(i),c(i),t}$ denotes industry–country–year fixed effects, and $\lambda_{s_{it}^{VL}}$ denotes cumulative VC-leader-set fixed effects.

This specification does not interact the VC-leader-set identifier with calendar-year indicators. It absorbs unobserved factors that are related to the identity and the composition of the firm's cumulative VC lead-investor set. The identifier is nevertheless dynamic at the firm level because a firm's fixed-effect category changes whenever a previously unobserved VC lead investor enters its cumulative set.

[12] An investor with missing VC classification is not included in the cumulative VC-leader set, since inclusion requires the investor to be positively identified as both a VC and a lead investor.

Table 27: Peer IPO Spillovers with Cumulative VC Lead-Investor-Set Fixed Effects

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio	0.015**	0.015**
	(0.006)	(0.006)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Cumulative VC-set FEs	Yes	Yes
Observations	984,903	984,903
Clusters (firm)	154,265	154,265
Adjusted R-squared	0.135	0.135

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is measured as the share of the focal firm's first-order peers completing an IPO in $t-1$. The cumulative VC lead-investor set contains investors that have invested in the focal firm by year t and are simultaneously classified as venture capital investors and identified as lead investors. All specifications include lagged focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Notice that the coefficient on lagged peer IPO exposure also remains positive and significant after including the cumulative VC-leader-set fixed effects. This result indicates that the baseline relationship is not driven solely by persistent unobserved characteristics related to the identity or composition of the firm's VC and lead investors since the positive effect persists.

D.3 Restriction to More Recent Firm Cohorts

A potential concern is that the sample includes some firms founded many years before the beginning of the sample period. These older private firms may differ systematically from more recently founded firms and may have a lower propensity or weaker incentives to undertake an IPO. Their inclusion could therefore affect both the composition of the estimation sample and the measurement of peer IPO exposure.

To address this concern, I reconstruct the common-investor network after restricting it to firms founded in or after 1990. This restriction is imposed on both sides of each network connection: focal firms founded before 1990 are excluded from the estimation

sample, and firms founded before 1990 are also removed from the focal firm's peer set. Consequently, older firms enter neither the numerator nor the denominator of the peer IPO exposure measure.

More formally, letting f_i denote firm i's founding year and $\mathcal{N}_{it}$ its original set of first-order peers in year t, I define the restricted peer set as $\mathcal{N}_{it}^{1990} = \{j \in \mathcal{N}_{it} : f_j \geq 1990\}$, and retain focal firms satisfying $f_i \geq 1990$. As in the baseline network, a peer that completed an IPO before year t is excluded from $\mathcal{N}_{it}^{1990}$. A peer that goes public during year t remains in the year-t peer set and is removed from subsequent years. I then reconstruct the peer IPO exposure measure as

$$\text{Peer IPO Exposure}_{it}^{1990} = \frac{\sum_{j \in \mathcal{N}_{it}^{1990}} \mathbf{1}\{\text{IPO}_{jt} = 1\}}{|\mathcal{N}_{it}^{1990}|},$$

whenever the restricted peer set is nonempty, and re-estimate the baseline specification using this reconstructed network. An important remark is that a total of 176,675 firms are less than 34 years old, where firm age is measured as the firm's maximum age by 2023.

Table 28 reports the results. The estimated coefficient on lagged peer IPO exposure remains positive and statistically significant and is similar in magnitude to the baseline estimate. The baseline relationship is therefore not driven by the inclusion of substantially older private firms or by their presence in the peer-exposure denominator.

Table 28: Peer IPO Spillovers in the Network of Firms Founded in or after 1990

	IPO No peer controls	IPO Peer controls
Lagged peer IPO ratio: firms founded since 1990	0.022***	0.021***
	(0.006)	(0.006)
Focal-firm controls	Yes	Yes
First-order peer controls	No	Yes
Firm Fixed Effects	Yes	Yes
Industry-Country-Year FEs	Yes	Yes
Observations	929,694	929,694
Clusters (firm)	145,729	145,729
Adjusted R-squared	0.114	0.114

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. The common-investor network is reconstructed after restricting both focal firms and their potential first-order peers to firms founded in or after 1990. Lagged peer IPO ratio is the share of eligible first-order peers in this restricted network that complete an IPO in $t-1$. The specification includes lagged focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Both specifications use the same sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

D.4 Conditional Heterogeneity Among Observed Peer IPO Events

The previous results examine whether IPO activity among connected firms predicts the focal firm's IPO decision. I next study the network structure surrounding the peers that actually complete an IPO. Specifically, I ask whether, among firm-years with at least one first-order peer IPO, the focal firm's IPO probability is systematically related to the strength of the direct and indirect common-investor connections associated with those realized IPO events.

This analysis is conditional on observing at least one first-order peer IPO. It therefore does not estimate the effect of having an IPO peer relative to having no IPO peers. Instead, it compares realized peer IPO events according to the network structure surrounding the firms that go public.

For each firm i and year t, I identify the subset of first-order neighbors $j \in N_{i,t-1}$ that complete an IPO in year $t-1$. For these IPO peers, I measure the strength of their connections to second-order neighbors k using common-investor overlap. Let $c_{jk,t-1}$ denote the number of common investors linking first-order peer j and second-order peer k. I construct the average indirect overlap related to the peers that did IPO as

$$\text{AvgCJK_IPO}_{i,t-1} = \frac{\sum_{j \in N_{i,t-1}} IPO_{j,t-1} \left(\frac{1}{|N_{j,t-1} \setminus \{i\}|} \sum_{k \in N_{j,t-1} \setminus \{i\}} c_{jk,t-1} \right)}{\sum_{j \in N_{i,t-1}} IPO_{j,t-1}}.$$

Thus, the measure is constructed only from those first-order peers that actually complete an IPO and captures how strongly these realized IPO peers are connected to the surrounding second-order network. To separate the intensity of these connections from differences in network size, I include network-type fixed effects. Network types are constructed by discretizing by the number of first-order neighbors of focal firm i and by the total number of indirect (j,k) links generated by all of its first-order neighbors, denoted n_{jk}. These bins are interacted with year indicators. Importantly, n_{jk} is computed using all first-order neighbors, irrespective of whether they complete an IPO. Network type therefore captures the overall scale of the direct and indirect network independently of the realized IPO event.

The distinction between the two measures is important. The type fixed effects compare firm -years with similar direct and indirect network size, whereas $\text{AvgCJK_IPO}_{i,t-1}$ varies with the strength of the indirect connections specifically linked to the peers that actually go public. The coefficient on this variable therefore captures whether realized peer IPOs are associated with a stronger or weaker focal-firm response depending on how redundant the surrounding network is.

Figure 4 illustrates the construction. All indirect (j,k) links enter the measure of total indirect network size, n_{jk}, whereas only links originating from first-order peers that

complete an IPO contribute to AvgCJK_IPO$_{i,t-1}$.

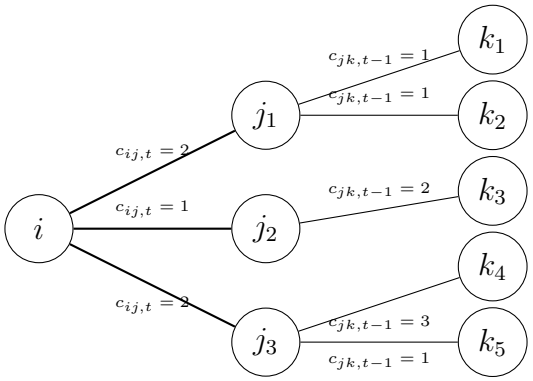


Figure 4: Construction of indirect network overlap. All first-order neighbors j and their second-order neighbors k enter the measure of total indirect network size. Common-investor overlap associated with the IPO peers is used to construct AvgCJK_IPO$_{i,t-1}$.

For example, suppose that j_1 and j_3 complete an IPO whereas j_2 does not. The indirect links originating from all three first-order peers contribute to n_{jk}, but only the j_1–k and j_3–k connections contribute to the IPO-conditioned common-investor overlap measure:

Table 29: Illustration of IPO Conditioning in the Indirect-Overlap Measure

j	$IPO_{j,t-1}$	$c_{jk,t-1}$ values	Used in AvgCJK_IPO$_{i,t-1}$?
j_1	1	1, 1	Yes
j_2	0	2	No
j_3	1	3, 1	Yes

To separate connection intensity from differences in overall network size, I include network-type fixed effects. Network types are constructed by discretizing the number of first-order neighbors of the focal firm and the total number of indirect (j,k) links generated by all its first-order neighbors. The latter is calculated using all first-order neighbors, regardless of whether they complete an IPO. The network-type fixed effects therefore control flexibly for the overall scale of the direct and indirect network, while the IPO-conditioned overlap measures vary with the connections specifically associated with the peers that go public. I estimate the following regression

$$IPO_{i,t} = \alpha + \delta\, IPOShare_{i,t-1} + \beta_1 \log\big(\bar{c}^{IPO}_{ij,i,t-1}\big) + \beta_2 \log\big(\bar{c}^{IPO}_{jk,i,t-2}\big) + X'_{i,t-1}\theta + Z'_{i,t-1}\pi + \mathcal{F}_{i,t} + \varepsilon_{i,t}, \tag{22}$$

where $IPOShare_{i,t-1}$ is the share of first-order peers completing an IPO, $\bar{c}^{IPO}_{ij,i,t-1}$ measures the average direct common-investor overlap between the focal firm and its IPO peers the

previous year, and $\bar{c}^{IPO}_{jk,i,t-2}$ measures the average indirect common-investor overlap between those IPO peers and their second-order neighbors two previous years. The vector $X_{i,t-1}$ contains the focal-firm controls: firm age, cumulative investments, cumulative unique investors, the firm's investor score, the share of lead investors, and the leave-one-out investor similarity measure. The vector $Z_{i,t-1}$, included in selected specifications, contains the average age, investor score, cumulative number of investors, cumulative number of investments, industry similarity, and investor overlap of the peers completing an IPO. The term $\mathcal{F}_{i,t}$ denotes the relevant set of fixed effects (industry-year-country, firm and network FEs).

Table 30: Network Structure Conditional on Observed Peer IPO Events

	IPO No controls	IPO Focal controls	IPO All controls
Lagged peer IPO ratio	0.044***	0.048***	0.049***
	(0.015)	(0.016)	(0.016)
Lagged log indirect common-investor overlap	-0.003*	-0.000	0.003
	(0.002)	(0.002)	(0.002)
Lagged log direct common-investor overlap	0.016***	0.010***	0.011***
	(0.001)	(0.001)	(0.001)
Focal-firm controls	No	Yes	Yes
Average IPO-peer controls	No	No	Yes
Firm Fixed Effects	Yes	Yes	Yes
Network-type Fixed Effects	Yes	Yes	Yes
Industry-Country-Year FEs	Yes	Yes	Yes
Observations	321,210	303,266	297,179
Clusters (firm)	68,594	64,893	63,922
Adjusted R-squared	0.088	0.101	0.102
Direct minus indirect overlap	0.019	0.010	0.007
p-value of equality test	0.000	0.000	0.015

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t.Lagged peer IPO ratio is the proportion of the focal firm's first-order peers completing an IPO in $t-1$. Direct overlap is the log average common-investor overlap between the focal firm i and its first-order IPO peers j, measured in $t-1$. Indirect overlap is the log average common-investor overlap between those IPO peers j and their indirect neighbors k, measured in $t-2$. The reported coefficient difference is the direct-overlap coefficient minus the indirect-overlap coefficient. Column (1) excludes additional time-varying controls. Column (2) includes lagged focal-firm controls of transformed firm age, cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure. Column (3) controls for the lagged average controls of the first-order peers completing an IPO. The focal-firm and IPO-peer controls are measured at $t-1$. Network types are constructed as reported above. All columns use the same estimation sample. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 30 reports three specifications using the same estimation sample. Column (1) ex-

cludes both focal-firm and IPO-peer controls. Column (2) includes the focal-firm controls, while Column (3) additionally includes the average characteristics of the peers completing an IPO. All columns absorb firm, network-type, and year-by-industry-by-country fixed effects. Standard errors are clustered at the focal-firm level.

The coefficient on the lagged peer IPO share is positive, statistically significant, and highly stable across the three specifications, ranging from 0.096 to 0.097. In the fully controlled specification, a 10-percentage-point increase in the share of first-order peers completing an IPO rises by an approximately 0.97-percentage-the focal firm's IPO probability within the conditional sample. Direct and indirect common-investor overlap exhibit markedly different patterns. Direct overlap between the focal firm and its IPO peers is positive and statistically significant in every specification. Its coefficient declines from 0.014 in Column (1) to 0.009 after the controls are included, but remains precisely estimated. By contrast, indirect overlap between IPO peers and their second-order connections is small and statistically insignificant in all three columns. Its sign also changes across specifications, providing no evidence of a robust positive or negative association with the focal firm's IPO probability. The direct-overlap coefficient is significantly larger than the indirect-overlap coefficient in every specification. In the fully controlled model, the estimated difference is 0.007, and equality of the two coefficients is rejected at the 5 percent level ($p = 0.019$). These results indicate that common-investor overlap linking the focal firm directly to its IPO peers has a stronger association with the focal firm's IPO decision than the overlap linking those peers to second-order firms. The results do not, however, establish that stronger indirect overlap makes the peer IPO signal noisier or reduces its informational content. Such an interpretation would require a robustly negative coefficient on indirect overlap or a negative interaction between indirect overlap and peer IPO exposure, neither of which is demonstrated by this table. The evidence is therefore more appropriately interpreted as showing the greater relevance of direct, relative to indirect, common-investor connections.

As an alternative specification, Table 31 replaces the year-by-industry-by-country fixed effects with separate year fixed effects, while retaining firm and network-type fixed effects, and with no controls:

Table 31: Network Structure Conditional on Observed Peer IPO Events: Alternative Fixed Effects

	IPO No additional controls
Lagged peer IPO ratio	0.031**
	(0.015)
Lagged log indirect common-investor overlap	-0.004**
	(0.002)
Lagged log direct common-investor overlap	0.016***
	(0.001)
Focal-firm controls	No
Average IPO-peer controls	No
Firm Fixed Effects	Yes
Year Fixed Effects	Yes
Network-type fixed effects	Yes
Industry-Country-Year FEs	No
Observations	329,228
Clusters (firm)	70,229
Adjusted R-squared	0.085
Direct minus indirect overlap	0.019
p-value of equality test	0.000

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. Lagged peer IPO ratio is the share of first-order peers completing an IPO in $t-1$. Direct overlap is the log of the average common-investor overlap between the focal firm i and its first-order IPO peers j, measured in $t-1$. Indirect overlap is the log of the average common-investor overlap between those IPO peers j and their second-order neighbors k, measured in $t-2$. firm, year, and network-type fixed effects. No focal-firm or peer controls are included. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

E JOBS Act Instrument

This appendix provides the results and a detail explanation for the instrumental-variable strategy described in Section 5. The design exploits the 2012 JOBS Act as a policy-induced source of variation in IPO activity among firms connected through the pre-policy firm network. The JOBS Act introduced the Emerging Growth Company (EGC) classification and reduced disclosure and compliance requirements for eligible firms accessing US public

markets. Because revenue-based EGC eligibility is not observed for private firms in the data, I use US domicile as an intent-to-treat measure of exposure to the reform. The instrument should therefore be interpreted as exposure to the JOBS Act regulatory regime rather than confirmed firm-level eligibility or take-up.[13]

The excluded instrument is constructed using the number of US-domiciled firms in firm i's first-order network observed in 2011, interacted with the post-JOBS Act period. The lagged value of this interaction instruments the lagged first-order peer IPO rate. Freezing the network in 2011, the last pre-policy cross-section, prevents post-policy link formation from affecting exposure to the instrument. The identifying variation therefore comes from differences across firms in their predetermined exposure to US peers before the policy rather than from endogenous changes in network composition after the reform.

The 2011 network contains peers that remained private immediately before the policy change. Firms that had already completed an IPO before 2012 do not enter the relevant private peer set and consequently do not contribute to the policy-exposure measure. Thus, the instrument captures exposure through firms that were still potential IPO candidates when the regulatory change occurred.

The preferred IV specification controls for the focal firm 's lagged cumulative number of investments, lagged cumulative number of unique investors, age in logarithms, and lagged average investor IPO-exit score. I additionally control for the predetermined average characteristics of the peers in the frozen network. This variable captures differences in the cohort composition of the pre-policy peer group that may independently predict subsequent IPO activity. I do not include the full set of time-varying peer characteristics used in the baseline OLS specification, but the majority. Such variables may themselves respond to the same financing and IPO processes affected by the policy and therefore might not be valid controls in the IV specification. Moreover, conditioning on the full set of peer controls absorbs a substantial amount of the policy-induced variation in peer IPO activity: when these controls are included, the Kleibergen–Paap statistic falls to approximately 7.61.

E.0.1 Instrumental-variables specification

The endogenous variable is the lagged share of first-order peers completing an IPO in the dynamic network, $\sum_{j \in N_{i,t-1}} \widetilde{g}_{ij,t-1} IPO_{j,t-1}$, where $\widetilde{g}_{ij,t-1}$ denotes the row-normalized first-order network weight.

The instrument exploits predetermined exposure to U.S. peers around the JOBS Act. Let N_i^{2011} denote firm i's first-order neighborhood in the network frozen as of 2011. I

[13]Using US firms classification avoids using revenue-based EGC eligibility for private firms for which the required financial information is generally unavailable. The resulting instrument therefore captures policy exposure rather than confirmed EGC status.

define

$$Z_{i,t-1} = PostJOBS_{t-1} \sum_{j \in N_i^{2011}} 1\{j \text{ is U.S.-domiciled}\}.$$

Thus, the instrument interacts post-JOBS Act exposure with the number of U.S. peers connected to firm i through the pre-policy 2011 network. Freezing the network before the policy change prevents subsequent link formation from affecting the instrument.

The first-stage specification is

$$\sum_{j \in N_{i,t-1}} \widetilde{g}_{ij,t-1} IPO_{j,t-1} = \pi Z_{i,t-1} + \gamma' X_{i,t-1}^{IV} + \beta \sum_{j \in N_{i,t-1}} \widetilde{g}_{ij,t-1} X_{j,t-1} + \rho_i + \mu_{t \times r} + u_{i,t}, \tag{23}$$

where $_{i,t-1}$ contains the dynamic controls of firm i included in the IV specification, ρ_i denotes firm fixed effects, and $\mu_{t \times r \times s}$ denotes year–industry–country fixed effects. The coefficient π measures whether firms with greater predetermined exposure to U.S. peers experienced a larger increase in peer IPO activity following the JOBS Act.

The second-stage specification is

$$IPO_{i,t} = \alpha + \delta^{IV} \left(\widehat{\sum_{j \in N_{i,t-1}} \widetilde{g}_{ij,t-1} IPO_{j,t-1}} \right) + \boldsymbol{\gamma}' \mathbf{X}_{i,t-1}^{IV} + \beta \sum_{j \in N_{i,t-1}} \widetilde{g}_{ij,t-1} X_{j,t-1} + \rho_i + \mu_{t \times s} + \mu_c + \varepsilon_{i,t}. \tag{24}$$

where $\widehat{PeerIPO}_{i,t-1}$ is the fitted value from equation (23). The coefficient δ^{IV} therefore captures the effect of peer IPO exposure using only the variation induced by predetermined exposure to U.S. peers around the JOBS Act. The vector $X_{i,t-1}$ contains the focal-firm controls: cumulative investments, cumulative unique investors, average investor score, the cumulative share of investments involving lead investors, and the leave-one-out investor-industry specialization measure, and the focal firm's direct exposure to the JOBS Act. On the peer side, I control for lagged average characteristics of first-order peers, excluding cumulative investments and the number of unique investors, as these variables may respond to the policy and therefore constitute potentially endogenous controls. The latter is calculated using the current dynamic network rather than the frozen 2011 network, which is used only to construct the instrument. Both stages include firm, ρ_i, country μ_c, and year-industry fixed effects, $\mu_{t \times s}$. Standard errors are clustered at the firm level. Because the endogenous peer-IPO measure is lagged, the policy exposure entering the instrument is also measured at $t-1$. Thus, the instrument first becomes active for IPO outcomes in 2013, when peer IPO activity in 2012 enters the lagged exposure measure.

The first stage is strong in the preferred full-sample specification. The coefficient on predetermined JOBS Act peer exposure is positive and statistically significant. The Kleibergen–Paap rk Wald F-statistic is 115.00, and the Kleibergen–Paap underidentification test strongly rejects underidentification ($p < 0.001$). For reference, the corresponding Cragg–Donald statistic is 176.27. Weak-instrument-robust inference also rejects a zero coefficient on the endogenous peer IPO variable: the Anderson–Rubin test has $p = 0.001$. These diagnostics provide no indication that the preferred IV estimate is driven by a weak first stage. The second-stage coefficient on lagged peer IPO exposure is 1.38 with a standard error of 0.45 ($p = 0.001$). The IV estimate is substantially larger than the baseline OLS coefficient. Given that the IV exploits only the component of peer IPO exposure shifted by the JOBS Act through the predetermined network, the two estimates might not identify the same average effect. The difference could reflect heterogeneous treatment effects or measurement error in observed peer exposure and/or differences in measures given that the endogenous control is in ratio, but the IV is in levels. For this reason, I use the OLS specification as the main specification.

Table 32: Instrumental-Variables Estimate of Peer IPO Exposure

	OLS	IV
Lagged peer IPO ratio	0.018***	1.388***
	(0.006)	(0.449)
Kleibergen–Paap rk Wald F-statistic		115.00
Observations	996,253	996,253
Clusters (firm)	154,946	154,946
Focal-firm controls	Yes	Yes
Peer-age control	Yes	Yes
Additional peer controls	Yes	Yes
Firm Fixed Effects	Yes	Yes
Industry-Year FEs	Yes	Yes
Country Fixed Effects	Yes	Yes

Notes: The dependent variable is an indicator equal to one if the focal firm completes an IPO in year t. The endogenous regressor is the lagged share of first-order peers completing an IPO. This variable is instrumented using the lagged policy exposure of U.S. peers to the JOBS Act, constructed from the common-investor network fixed as of 2011. The specification controls for the focal firm's lagged cumulative investments, cumulative unique investors, and lagged average investor score, as well as an indicator for a U.S. focal firm in the post-JOBS Act period and the lagged average controls of its first-order peers. The reported coefficient is the second-stage linear-probability estimate. The weak-instrument statistic is the Kleibergen–Paap rk Wald F-statistic. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

A central concern is that the JOBS Act can affect a US-domiciled focal firm directly as well as indirectly through its peers. In the preferred full-sample specification I therefore include `own_US_post_jobs`, equal to one for US-domiciled focal firms during the post-policy period. This variable absorbs the average direct post-2012 shift in IPO propensity for US focal firms. Its estimated coefficient is -0.0069 and statistically significant in the preferred full-sample specification. The sign of this coefficient is not itself central to the identification strategy; its role is to separate the focal firm's own policy exposure from variation generated by the predetermined composition of its peer network.

Importantly, including this control does not mechanically establish the exclusion restriction. The identifying assumption is that, conditional on the focal firm's own policy exposure, firm fixed effects, year-by-industry fixed effects, and the included controls, the predetermined number of US peers affects the focal firm's IPO decision through its effect on peer IPO activity rather than through another post-2012 channel. This assumption could fail, for example, if firms with greater predetermined exposure to US peers also experienced differential changes in access to underwriters, investor capital, financing conditions, or investor portfolio strategy after the reform. For this reason, the JOBS Act design is a complement.

As an additional robustness check, I add more peer controls and replace the year-by-industry fixed effects with country-by-industry-by-year fixed effects. These fixed effects absorb country-specific changes in IPO conditions within industries, including the average direct effect of the JOBS Act on US firms. In this specification, `own_US_post_jobs` is absorbed by the fixed effects and is therefore not separately included. The estimated peer effect remains similar at 1.39, while the Kleibergen–Paap first-stage statistic remains approximately 123.60. The similarity of the estimate provides evidence that the preferred result is not driven solely by broad country-specific post-policy IPO conditions. Nevertheless, this specification produces a rank-deficiency warning given that there exists sparse fixed-effect cells and is therefore treated as a supplementary robustness exercise rather than the preferred specification.

F Theoretical Model Proofs

F.1 Results

F.1.1 Proof of Lemma 1

Proof. Recall that the competitive public-market price is

$$P_i = \bar{\theta}_i + \sum_{k \in K_i} \alpha_i^k \varepsilon_i^k.$$

By construction, ε_i^k is the public market's valuation of investor k in firm i. Hence, when firm i receives its IPO opportunity, each ε_i^k entering P_i is already known to the public market. Since $\bar{\theta}_i$ and the weights α_i^k are also common knowledge, P_i is measurable with respect to the market information set $\mathcal{I}_i^M$.

It follows immediately that

$$\mathbb{E}_M \left[P_i \mid \mathcal{I}_i^M \right] = P_i.$$

Moreover, adding the firm's current action to the market's conditioning set cannot change the expectation of an object that is already known:

$$\mathbb{E}_M \left[P_i \mid \mathcal{I}_i^M, d_i \right] = P_i.$$

To see the economic intuition, firm i does not observe the ε_i^k's directly. Instead, before choosing d_i, it receives the noisy signal about each investor k $s_i^k = \epsilon_i^k + \eta_i^k$.

The IPO decision therefore depends on the firm's perception of the market valuation. From the public market's perspective, however, the ε_i^k's are already known. Conditional on $\mathcal{I}_i^M$, the additional variation in the firm's signals comes from the perception errors η_i^k.

Thus observing d_i may reveal whether the firm received relatively favorable or unfavorable signals about a valuation that the market already knows, but it does not reveal a new value-relevant component to the public market. Formally, for every $k \in K_i$,

$$\mathbb{E}_M \left[\varepsilon_i^k \mid \mathcal{I}_i^M, d_i \right] = \varepsilon_i^k = \mathbb{E}_M \left[\varepsilon_i^k \mid \mathcal{I}_i^M \right].$$

Consequently, d_i determines whether the already-determined competitive market valuation is realized as an IPO price; it does not change that contemporaneous valuation.

F.1.2 Proof Proposition 1

Proof. Before any IPO decision, each firm receives its own private signals.

Firm a observes s_a^1, forms posterior belief

$$\mu_a^1 = \mathbb{E}_a[\varepsilon_a^1 \mid s_a^1],$$

and therefore

$$\widehat{P}_a = \bar{\theta}_a + \alpha_a^1 \mu_a^1.$$

It chooses d_a according to (13). Corollary 1 implies a cutoff strategy.

Firm b has already observed its private signals (s_b^1, s_b^2). After observing the public

decision d_a, but not s_a^1, it forms

$$\mu_b^1 = \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a]$$

and

$$\mu_b^2 = \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2].$$

It then forms

$$\widehat{P}_b = \bar{\theta}_b + \alpha_b^1 \mu_b^1 + \alpha_b^2 \mu_b^2$$

and chooses d_b.

Firm c has already observed s_c^2. After observing the public decisions (d_a, d_b), but neither of firm b's private signals, it forms

$$\mu_c^2 = \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a, d_b].$$

Its expected public-market valuation is

$$\widehat{P}_c = \bar{\theta}_c + \alpha_c^2 \mu_c^2,$$

and it chooses d_c according to (13).

At each information set reached with positive probability, posterior beliefs are obtained from Bayes' rule and firms' actions maximize their expected payoffs given those beliefs. Public-market prices are determined by (9). Lemma 1 ensures that the firm's current IPO decision determines whether the contemporaneous valuation is realized as an IPO price rather than altering that valuation.

Thus the strategies are sequentially optimal, beliefs are Bayes-consistent, and public-market valuations are competitive.

To establish uniqueness of the equilibrium IPO decisions, consider first firm a. Since $\alpha_a^1 > 0$ and its private signal has a continuous, nondegenerate Gaussian distribution, its expected IPO valuation is strictly increasing in s_a^1. Therefore, its optimal IPO decision is uniquely determined except at its indifference cutoff, which occurs with probability zero.

Given firm a's strategy, Bayesian updating uniquely determines the posterior beliefs of firm b following each public decision d_a. Moreover, holding s_b^1 and d_a fixed, firm b's expected IPO valuation is strictly increasing in s_b^2, provided $\alpha_b^2 > 0$. Since the conditional distribution of this signal is continuous and nondegenerate, indifference occurs with probability zero. Thus, firm b's optimal IPO decision is uniquely determined almost surely.

The same argument applies to firm c. Given the strategies of firms a and b, Bayesian updating uniquely determines firm c's posterior beliefs. Since $\alpha_c^2 > 0$, its expected IPO

valuation is strictly increasing in its private signal s_c^2, conditional on the observed public history. Therefore, firm c's optimal IPO decision is unique almost surely.

Under the nondegenerate Gaussian information structure, strictly positive relevant investor weights, and finite IPO thresholds, each possible public action history occurs with positive probability. Bayesian beliefs are therefore uniquely determined along these histories.

Consequently, the equilibrium IPO strategies are uniquely determined up to zero-probability indifference events, and the equilibrium IPO decisions are essentially unique.

Given that firms observe their own private signals but not the private signals of earlier firms, later decisions are taken at non-singleton information sets. Observing an earlier IPO decision therefore requires Bayesian updating over the private information that could have generated that decision.

F.1.3 Proof Proposition 2

Proof. Firm b never observes s_a^1. The signal appears in the proof only as the latent variable generating firm a's decision.

Conditional on firm b's own signal s_b^1, consider the pair (ε_b^1, s_a^1). Its conditional covariance is

$$\mathrm{Cov}(\varepsilon_b^1, s_a^1 \mid s_b^1) = \mathrm{Cov}(\varepsilon_b^1, s_a^1) - \frac{\mathrm{Cov}(\varepsilon_b^1, s_b^1)\,\mathrm{Cov}(s_b^1, s_a^1)}{\mathrm{Var}(s_b^1)}. \tag{25}$$

Using

$$s_a^1 = \varepsilon_a^1 + \eta_a^1, \qquad s_b^1 = \varepsilon_b^1 + \eta_b^1,$$

together with independence of the perception errors,

$$\mathrm{Cov}(\varepsilon_b^1, s_a^1) = \mathrm{Cov}(\varepsilon_b^1, \varepsilon_a^1),$$

$$\mathrm{Cov}(\varepsilon_b^1, s_b^1) = \sigma_\varepsilon^2,$$

$$\mathrm{Cov}(s_b^1, s_a^1) = \mathrm{Cov}(\varepsilon_b^1, \varepsilon_a^1),$$

and

$$\mathrm{Var}(s_b^1) = \sigma_\varepsilon^2 + \sigma_\eta^2.$$

Therefore

$$\mathrm{Cov}(\varepsilon_b^1, s_a^1 \mid s_b^1) = \mathrm{Cov}(\varepsilon_b^1, \varepsilon_a^1)\frac{\sigma_\eta^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} > 0.$$

The conditional variance of the latent signal of firm a is

$$\mathrm{Var}(s_a^1 \mid s_b^1) = \sigma_\varepsilon^2 + \sigma_\eta^2 - \frac{\mathrm{Cov}(\varepsilon_a^1, \varepsilon_b^1)^2}{\sigma_\varepsilon^2 + \sigma_\eta^2}.$$

Its conditional mean is

$$\mathbb{E}[s_a^1 \mid s_b^1] = \frac{\mathrm{Cov}(\varepsilon_a^1, \varepsilon_b^1)}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_b^1.$$

Define, only for this proof,

$$z_a = \frac{s_a^{1*} - \mathbb{E}[s_a^1 \mid s_b^1]}{\sqrt{\mathrm{Var}(s_a^1 \mid s_b^1)}}.$$

Conditional Normality gives

$$\mathbb{E}[\varepsilon_b^1 \mid s_b^1, s_a^1] - \mathbb{E}[\varepsilon_b^1 \mid s_b^1] = \frac{\mathrm{Cov}(\varepsilon_b^1, s_a^1 \mid s_b^1)}{\mathrm{Var}(s_a^1 \mid s_b^1)} \left(s_a^1 - \mathbb{E}[s_a^1 \mid s_b^1]\right).$$

Firm b does not observe s_a^1. If $d_a = 1$, it only learns that $s_a^1 \geq s_a^{1*}$. Integrating over the corresponding upper-truncated normal distribution gives

$$\mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a = 1] - \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1] = \frac{\mathrm{Cov}(\varepsilon_b^1, s_a^1 \mid s_b^1)}{\sqrt{\mathrm{Var}(s_a^1 \mid s_b^1)}} \frac{\phi(z_a)}{1 - \Phi(z_a)} > 0. \tag{26}$$

The $\frac{\phi(z_a)}{1-\Phi(z_a)} = \lambda(z(a))$ is the inverse mills ratio or hazard rate of the standard normal distribution. In this case, it is the hazard rate of the s_a^1 that is random for firm b.
If instead $d_a = 0$, firm b learns that $s_a^1 < s_a^{1*}$, which gives

$$\mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a = 0] - \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1] = -\frac{\mathrm{Cov}(\varepsilon_b^1, s_a^1 \mid s_b^1)}{\sqrt{\mathrm{Var}(s_a^1 \mid s_b^1)}} \frac{\phi(z_a)}{\Phi(z_a)} < 0. \tag{27}$$

This proves (17), and shows that 26 is not linear in s_b^1 because it enters in the hazard rate .

Because d_a concerns investor 1 and investors are independent across k, it does not change

$$\mu_b^2 = \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2].$$

Equation (13) therefore implies (18).

Finally, recall that

$$\mu_b^1 = \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a].$$

Hence, by the law of iterated expectations,

$$\mathbb{E}[\mu_b^1 \mid d_a] = \mathbb{E}\left[\mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a] \mid d_a\right] = \mathbb{E}[\varepsilon_b^1 \mid d_a]. \tag{28}$$

Since $d_a = 1$ corresponds to the upper tail of s_a^1 and $\text{Cov}(\varepsilon_b^1, s_a^1) = \text{Cov}(\varepsilon_b^1, \varepsilon_a^1) > 0$, we have $\mathbb{E}[\varepsilon_b^1 \mid d_a = 1] > \mathbb{E}[\varepsilon_b^1 \mid d_a = 0]$. Combining this with (28) proves (19).

The posterior changes induced by observed IPO decisions arise from the truncation of the underlying Gaussian distribution. As in the direct-spillover result, observing an IPO corresponds to an upper-tail event and therefore raises the posterior mean through the upper-tail inverse Mills ratio (or hazard rate),

$$\lambda(z) \equiv \frac{\phi(z)}{1 - \Phi(z)}. \tag{29}$$

Conversely, observing no IPO corresponds to a lower-tail event and shifts the posterior mean downward through the corresponding term $\phi(z)/\Phi(z)$. The upper-tail inverse Mills ratio $\lambda(z)$ is strictly increasing in z, while $\phi(z)/\Phi(z)$ is strictly decreasing. These properties will also be useful below when comparing the informational content of a peer IPO across different prior histories.

F.1.4 Proof Hazard Rate Increasing in z

Proof. For the upper tail,

$$\lambda'(z) = \lambda(z)[\lambda(z) - z].$$

If $z \leq 0$, then $\lambda(z) > 0 \geq z$. If $z > 0$,

$$1 - \Phi(z) = \int_z^\infty \phi(x)\, dx < \int_z^\infty \frac{x}{z}\phi(x)\, dx = \frac{\phi(z)}{z},$$

which implies $\lambda(z) > z$. Hence $\lambda'(z) > 0$.

For the lower tail,

$$\frac{d}{dz}\left(\frac{\phi(z)}{\Phi(z)}\right) = -\frac{\phi(z)}{\Phi(z)}\left[z + \frac{\phi(z)}{\Phi(z)}\right] < 0.$$

F.1.5 Proof of Lemma 2

Proof. Consider firm i and investor k. Let the publicly observed history be held fixed. Conditional on that history, the firm has some prior density for ε_i^k and observes

$$s_i^k = \varepsilon_i^k + \eta_i^k, \qquad \eta_i^k \sim N(0, \sigma_\eta^2).$$

Bayes' rule gives a posterior density proportional to

$$f(\varepsilon_i^k) \exp\left\{-\frac{(s_i^k - \varepsilon_i^k)^2}{2\sigma_\eta^2}\right\}.$$

Differentiating the posterior mean with respect to the signal gives

$$\frac{\partial \mathbb{E}_i[\varepsilon_i^k \mid \mathcal{I}_i]}{\partial s_i^k} \equiv \frac{\partial \mu_i^k}{\partial s_i^k} = \frac{\mathrm{Var}_i(\varepsilon_i^k \mid \mathcal{I}_i)}{\sigma_\eta^2} > 0. \tag{30}$$

Hence μ_i^k is strictly increasing in the firm's own signal whenever posterior uncertainty remains.

F.1.6 Proof Corollary 1

Proof. From (13),

$$\frac{\partial \widehat{P}_i}{\partial s_i^k} = \alpha_i^k \frac{\mathrm{Var}_i(\varepsilon_i^k \mid \mathcal{I}_i)}{\sigma_\eta^2} > 0.$$

The IPO condition compares this increasing object with the fixed threshold $W_i + \kappa_i$.

F.1.7 Proof Corollary 2

Proof. The contribution of investor k to firm b's expected public-market valuation is $\alpha_b^k \mu_b^k$. Hence a fixed change in μ_b^k changes $\widehat{P}_b$ proportionally to α_b^k.

To study how the importance of the common investor for the peer affects the information conveyed by its IPO decision, consider firm a as being backed by more than one investor. Suppose that investor 1 is the investor shared with firm b, while investor 2 is independent of investor 1. Holding the total investor equity stake fixed, $\alpha_a^1 + \alpha_a^2 = \text{constant}$, allows the relative importance of the common investor for firm a to vary.

Under the normal signal structure, the investor-related part of firm a's expected public-market valuation is

$$\frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2}\left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2\right).$$

Since investor 2 is independent of investor 1,

$$\mathrm{Cov}\left(\varepsilon_b^1, \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2}\left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2\right) \middle| s_b^1\right) = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} \alpha_a^1 \,\mathrm{Cov}\left(\varepsilon_b^1, s_a^1 \mid s_b^1\right).$$

Because

$$\mathrm{Cov}\left(\varepsilon_b^1, s_a^1 \mid s_b^1\right) > 0,$$

a larger α_a^1 increases the covariance between the valuation component relevant for firm b and the information entering firm a's IPO decision. Thus, holding total investor ownership fixed, increasing the relative importance of the common investor makes firm a's IPO decision more closely related to the common-investor component that is informative for firm b.

Firm b, however, observes only firm a's binary IPO decision rather than its underlying signals. Therefore, the magnitude of the posterior update also depends on the variance and truncation associated with firm a's IPO rule. The covariance result above establishes that a larger α_a^1 increases the loading of firm a's IPO decision on the common-investor component, but it does not by itself establish that the posterior update following $d_a = 1$ is globally increasing in α_a^1. This will only occurs when the variance and mean are kept constant or when the variance or mean decreases, so z increases.

F.1.8 Proof Proposition 3

Proof. Firm b's expected public-market valuation is

$$\widehat{P}_b = \bar{\theta}_b + \alpha_b^1 \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a] + \alpha_b^2 \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2].$$

For investor 2, (12) gives

$$\mathbb{E}_b[\varepsilon_b^2 \mid s_b^2] = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_b^2.$$

Thus the IPO condition is

$$\bar{\theta}_b + \alpha_b^1 \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a] + \alpha_b^2 \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_b^2 \geq W_b + \kappa_b.$$

The corresponding cutoff is

$$s_b^{2*}(d_a, s_b^1) = \frac{\sigma_\varepsilon^2 + \sigma_\eta^2}{\alpha_b^2 \sigma_\varepsilon^2} \left[W_b + \kappa_b - \bar{\theta}_b - \alpha_b^1 \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a] \right]. \tag{31}$$

Proposition 2 implies

$$\mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a = 1] > \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1, d_a = 0].$$

Hence the bracketed term in (31) is smaller after $d_a = 1$, which proves

$$s_b^{2*}(1, s_b^1) < s_b^{2*}(0, s_b^1).$$

F.1.9 Bayes' rule and the posterior ordering

This subsection provides additional details on the Bayesian updating argument behind Proposition 4. The objective is to compare what firm c learns from observing $d_b = 1$ depending on whether firm b had previously observed $d_a = 0$ or $d_a = 1$.

Let e denote a possible realization of ε_c^2. For a given history $d_a = d$, where $d \in \{0, 1\}$, Bayes' rule gives

$$f\left(e \mid s_c^2, d_a = d, d_b = 1\right) = \frac{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = d\right) f\left(e \mid s_c^2, d_a = d\right)}{\Pr\left(d_b = 1 \mid s_c^2, d_a = d\right)}. \tag{32}$$

The expression has the usual Bayesian interpretation. The term $f\left(e \mid s_c^2, d_a = d\right)$ is firm c's density over the investor-2 valuation component before observing firm b's IPO decision. The term $\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = d\right)$ is the likelihood of observing b go public when the investor-2 component relevant for firm c takes value e. Finally, $\Pr\left(d_b = 1 \mid s_c^2, d_a = d\right)$ is the normalizing term that ensures that the posterior density integrates to one. In particular,

$$\Pr\left(d_b = 1 \mid s_c^2, d_a = d\right) = \int_{\mathbb{R}} \Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = d\right) \times f\left(e \mid s_c^2, d_a = d\right) de. \tag{33}$$

Because the investor-1 and investor-2 components are independent, observing d_a does not directly change firm c's belief about ε_c^2

$$f\left(e \mid s_c^2, d_a = 0\right) = f\left(e \mid s_c^2, d_a = 1\right) = f\left(e \mid s_c^2\right) \tag{34}$$

Bayes' rule can therefore be written separately for the two histories as

$$f\left(e \mid s_c^2, d_a = 0, d_b = 1\right) = \frac{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0\right) f\left(e \mid s_c^2\right)}{\Pr\left(d_b = 1 \mid s_c^2, d_a = 0\right)}, \tag{35}$$

$$f\left(e \mid s_c^2, d_a = 1, d_b = 1\right) = \frac{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1\right) f\left(e \mid s_c^2\right)}{\Pr\left(d_b = 1 \mid s_c^2, d_a = 1\right)} \tag{36}$$

Dividing equation (35) by equation (36), the common prior density $f(e \mid s_c^2)$ cancels.

This gives

$$\frac{f\left(e \mid s_c^2, d_a = 0, d_b = 1\right)}{f\left(e \mid s_c^2, d_a = 1, d_b = 1\right)} = \frac{\Pr\left(d_b = 1 \mid s_c^2, d_a = 1\right)}{\Pr\left(d_b = 1 \mid s_c^2, d_a = 0\right)} \times \frac{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0\right)}{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1\right)}. \tag{37}$$

The first ratio on the right-hand side of equation (37) does not depend on e. It only rescales the posterior density so that it integrates to one. Therefore, the way in which the ratio of the two posterior densities changes with e is determined entirely by

$$\frac{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0\right)}{\Pr\left(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1\right)}. \tag{38}$$

If the ratio in equation (38) is increasing in e, then the ratio of posterior densities in equation (37) is also increasing in e.

Firm c does not observe either of firm b's private signals. When firm c reaches its IPO opportunity, it observes its own signal s_c^2 and the earlier public decisions (d_a, d_b), and therefore integrates over firm b's unobserved information. Because the investor-1 and investor-2 components are independent, d_a does not directly change firm c's belief about ε_c^2, then $f(\varepsilon_c^2 = e \mid s_c^2, d_a) = f(\varepsilon_c^2 = e \mid s_c^2)$. After observing firm b's IPO, Bayes' rule implies

$$f(e \mid s_c^2, d_a, d_b = 1) = \frac{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a) f(e \mid s_c^2)}{\mathbb{P}(d_b = 1 \mid s_c^2, d_a)}. \tag{39}$$

Accordingly,

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a, d_b = 1] = \frac{\displaystyle\int_{\mathbb{R}} e\, \mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a) f(e \mid s_c^2)\, de}{\mathbb{P}(d_b = 1 \mid s_c^2, d_a)}. \tag{40}$$

Because $\mathrm{Cov}(\varepsilon_b^2, \varepsilon_c^2) > 0$, higher realizations of ε_c^2 shift the conditional distribution of firm b's investor-2 signal upward. Since firm b's equilibrium IPO decision is increasing in that signal,

$$\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a)$$

is increasing in e. Hence observing $d_b = 1$ shifts posterior weight toward higher realizations of ε_c^2, so that

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a, d_b = 1] > \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2].$$

Thus, firm b's IPO conveys positive information about investor 2 under either prior history.

F.1.10 Proof of Proposition 4

Proof. For $d_a \in \{0, 1\}$, Bayes' rule gives

$$f(e \mid s_c^2, d_a, d_b = 1) = \frac{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a) f(e \mid s_c^2)}{\mathbb{P}(d_b = 1 \mid s_c^2, d_a)},$$

where I use the fact that d_a contains no direct information about investor 2, so that $f(e \mid s_c^2, d_a) = f(e \mid s_c^2)$. Hence,

$$\frac{f(e \mid s_c^2, d_a = 0, d_b = 1)}{f(e \mid s_c^2, d_a = 1, d_b = 1)} = \frac{\mathbb{P}(d_b = 1 \mid s_c^2, d_a = 1)}{\mathbb{P}(d_b = 1 \mid s_c^2, d_a = 0)} \frac{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0)}{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1)}.$$

The first term on the right-hand side is constant in e. Therefore, if

$$\frac{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0)}{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1)}$$

is increasing in e, then the ratio of the two posterior densities is also increasing in e.

Thus, the posterior following $(d_a, d_b) = (0, 1)$ places relatively more weight on high realizations of ε_c^2 than the posterior following $(d_a, d_b) = (1, 1)$. Therefore, using Equation 40

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 0, d_b = 1] > \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 1, d_b = 1].$$

Finally, since

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 0] = \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 1] = \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2],$$

both posterior updates start from the same prior mean. Hence,

$$\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 0, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2] > \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 1, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2].$$

Therefore, firm b's IPO generates a smaller posterior update following $d_a = 1$ than following $d_a = 0$.

Numerical evaluation of second-order attenuation Proposition 4 establishes that a sufficient condition for second-order attenuation is that is that Equation 38 is increasing in e. Given that firm c does not observe firm b's private signals, the relevant IPO probabilities integrate over firm b's unobserved information. Under the Gaussian environment, I evaluate these probabilities using deterministic numerical integration rather than simulation.

For each prior history $d_a \in \{0, 1\}$, the calculation uses the conditional distribution of firm b's investor-1 signal s_b^1. Importantly, this distribution is conditioned separately on

$d_a = 0$ and $d_a = 1$, so that observing firm a's decision changes the distribution over the investor-1 information that firm b may have received.

Firm c does not observe s_b^1. Then, conditional on a hypothetical realization $s_b^1 = x$, firm b's equilibrium IPO rule determines the minimum investor-2 signal that is needed for an IPO. The likelihood relevant for firm c therefore integrates this conditional IPO probability over the unobserved signal:

$$\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = d) = \int \mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = d, s_b^1 = x) f(s_b^1 = x \mid d_a = d)\, dx.$$

Thus, s_b^1 is used only as an integration variable and is not part of firm c's information set. The resulting likelihood is then used to compute firm c's posterior assessment of investor 2 under each prior history.

The numerical exercise is intended to illustrate the equilibrium information mechanism rather than to provide a structural estimation of the model. I nevertheless discipline the benchmark parameterization using its empirical counterparts whenever possible. I normalize $\sigma_\varepsilon^2 = 1$ and evaluate firm c at its mean signal, $s_c^2 = 0$. The IPO thresholds are chosen to reproduce the low frequency of IPOs observed in the firm-year data. The thresholds between firm a and b differ because the two firms face different valuation. Because the data do not report investor equity stakes, I use the symmetric benchmark $\alpha_b^1 = \alpha_b^2 = 0.5$ and vary their relative importance in the comparative-static exercise below. This variation has a direct empirical counterpart in the heterogeneity of shared-investor connections, including lead-investor status and investor overlap. Finally, the persistence of same-investor valuations and signal precision are not directly observed in the data. I therefore specify these parameters in the benchmark calibration and examine the role of investor importance through a comparative-static exercise Starting with the first Figure, it plots

$$\frac{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0)}{\mathbb{P}(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1)}.$$

Recall that, for the numerical illustration, I calibrate the IPO thresholds to match the annual IPO rate observed in the data (approximately 0.26%), while keeping the remaining parameters illustrative.

Below Figure 5 shows that this ratio is increasing over the evaluated support of ε_c^2, consistent with the sufficient condition for second-order attenuation in Proposition 4.

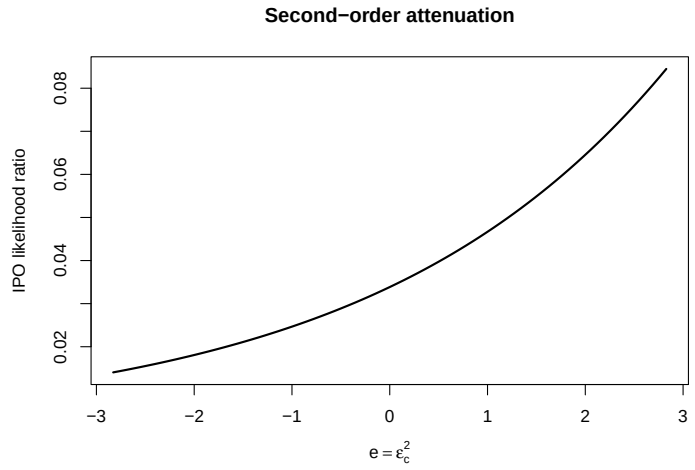


Figure 5: **IPO ratio when b IPOs and second-order attenuation.** The benchmark parameterization sets $\sigma_\varepsilon^2 = \sigma_\eta^2 = 1$, $\text{Cov}(\varepsilon_a^1, \varepsilon_b^1) = \text{Cov}(\varepsilon_b^2, \varepsilon_c^2) = 0.6$, $\alpha_a^1 = 1$, $\alpha_b^1 = \alpha_b^2 = 0.5$, $\bar{\theta}_a = \bar{\theta}_b = 1$, $W_a + \kappa_a = 2.98; W_b + \kappa_b = 2.4$, and $s_c^2 = 0$.

As we see in Figure 5, the sufficient condition for second-order attenuation is satisfied over the evaluated support in the benchmark parameterization. Hence, an IPO by firm b is relatively more selective with respect to investor 2 when $d_a = 0$ than when $d_a = 1$. Observing $d_b = 1$ therefore shifts firm c's posterior toward high realizations of the investor-2 component more strongly following $d_a = 0$. In the benchmark parameterization, the numerical evaluation therefore implies that firm b's IPO conveys less new information about investor 2 after firm b has previously observed the favorable IPO decision of firm a.

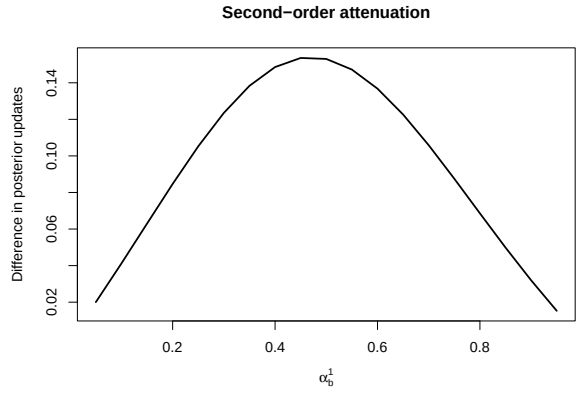


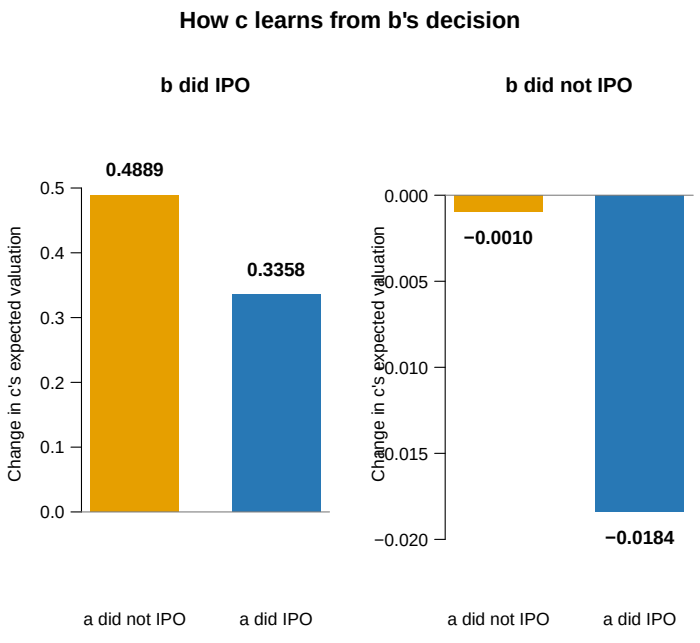
Figure 6: **Second-order attenuation and investor importance.** The figure varies α_b^1 from 0.05 to 0.95 and sets $\alpha_b^2 = 1 - \alpha_b^1$, thereby holding the total investor weight in firm b fixed. The vertical axis reports $\mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 0, d_b = 1] - \mathbb{E}_c[\varepsilon_c^2 \mid s_c^2, d_a = 1, d_b = 1]$. Positive values correspond to second-order attenuation. The remaining parameters are $\sigma_\varepsilon^2 = \sigma_\eta^2 = 1$, $\text{Cov}(\varepsilon_a^1, \varepsilon_b^1) = \text{Cov}(\varepsilon_b^2, \varepsilon_c^2) = 0.6$, $\alpha_a^1 = 1$, $\bar{\theta}_a = \bar{\theta}_b = 1$, $W_a + \kappa_a = 2.98; W_b + \kappa_b = 2.4$, and $s_c^2 = 0$.

Figure 6 illustrates the role of the two information channels. When α_b^1 becomes small, investor 1 has little influence on firm b's IPO decision, so firm a's prior decision has little effect on the informational content of d_b. At the theoretical boundary $\alpha_b^1 = 0$, this source of attenuation disappears. Conversely, when α_b^2 becomes small, firm b's IPO contains little information about investor 2; at $\alpha_b^2 = 0$, second-order learning itself disappears. Thus, second-order attenuation is an interior phenomenon that requires both the upstream and downstream investor components to matter for firm b's IPO decision.

The theoretical network also has a direct empirical counterpart. In the model, firms i and j are connected through investor k whenever $\alpha_i^k > 0$ and $\alpha_j^k > 0$.

A common investor therefore enters the valuation of both firms and creates the channel through which firm j's IPO can convey information relevant for firm i. Empirically, I define two firms as connected when they share at least one private investor. The shared-investor network in the data is therefore the empirical counterpart of the common-investor links in the model.

At last, we can examine the probability of firm c going IPO under each possible history of IPO events. This is shown by the changes in the expected valuation of firm c below.



Note: the two panels use different vertical scales.

Figure 7: **Second-order attenuation for all cases**. Parameters: $\sigma_\epsilon^2 = \sigma_\eta^2 = 1$, $\rho_1 = \rho_2 = 0.60$, $\alpha_a^1 = 1$, $\alpha_b^1 = \alpha_b^2 = 0.50$, $\bar{\theta}_a = \bar{\theta}_b = 1$, $W_a + \kappa_a = 2.98; W_b + \kappa_b = 2.4$, and $s_c^2 = 0$.

F.2 New investors

Investor entry is not modeled explicitly, since investors' financing decisions are outside the theoretical analysis. Nevertheless, the learning mechanism provides a simple interpretation for the empirical result on new investors. If a peer IPO leads the focal firm and potential investors to expect a higher public-market valuation for the firm, then, holding investment terms fixed, the firm becomes more attractive to outside investors. This can increase the likelihood that new investors provide financing. I therefore view the new-investor result as consistent with the valuation channel developed in the model, rather than as a separate prediction of an endogenous investor-entry decision

F.3 Alternative information structure: a single noisy price signal

The baseline model assumes that firms receive private noisy information about the investor-specific components of their potential public-market valuation. This structure is useful because it maps information directly into the shared-investor network. The basic learning mechanism, however, does not rely on firms observing separate signals for each investor.

To see this, consider an alternative information structure in which each firm receives a single noisy signal of its potential public-market valuation,

$$\widetilde{s}_i = P_i + \nu_i,$$

where ν_i is normally distributed noise, independent across firms and independent of the valuation components. Firm i therefore receives an imperfect assessment of the overall price it could obtain in the public market, rather than separate information about each investor-specific component.

Because the public-market valuations of firms sharing an investor are positively correlated, the noisy price signals of connected firms are also positively correlated. Consider firms a and b sharing investor 1. Conditional on firm b's own signal, observing $d_a = 1$ reveals that firm a's private information was sufficiently favorable for it to cross its IPO threshold. By the positive correlation between their potential public-market valuations, this raises firm b's posterior expectation of P_b:

$$\mathbb{E}[P_b \mid \widetilde{s}_b, d_a = 1] > \mathbb{E}[P_b \mid \widetilde{s}_b] > \mathbb{E}[P_b \mid \widetilde{s}_b, d_a = 0].$$

Hence, the positive direct-peer spillover derived in the baseline model continues to hold with a single noisy price signal.

The second-order mechanism also remains. In the three-firm chain a–b–c, observing $d_a = 1$ raises firm b's expected public-market valuation for any given realization of $\widetilde{s}_b$.

Firm b therefore requires a lower realization of its own noisy price signal to choose an IPO:

$$\widetilde{s}_b^*(1) < \widetilde{s}_b^*(0).$$

An IPO by firm b is consequently less selective following an earlier IPO by firm a. Since P_b and P_c are positively correlated through their shared investor, this reduces the incremental information that firm c can obtain from observing $d_b = 1$.

As in the baseline model, translating this conditional cutoff ordering into an ordering of firm c's posterior after integrating over firm b's unobserved private information requires the corresponding relative-selectivity condition. Under this condition, the second-order attenuation result follows in the same way as Proposition 4.

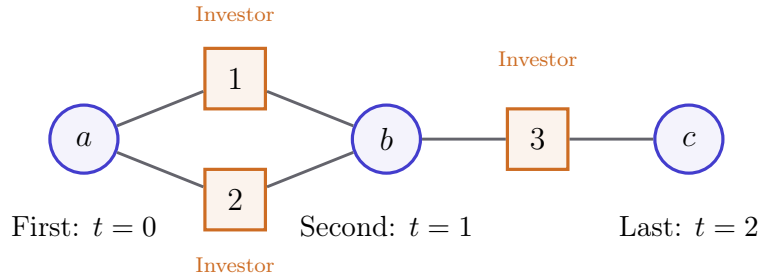
Thus, the main learning mechanism does not depend on firms receiving separate investor-specific signals. The investor-specific information structure used in the baseline model provides a more direct mapping between information and shared-investor links, while a standard single noisy signal of the firm's potential public-market price generates the same direct learning and second-order effects.

F.4 Extension: Multiple common investors

The baseline three-firm example assumes that two directly connected firms share one investor. The same logic extends when two firms share several investors. The important point is that the earlier firm's IPO remains a single observed action. Hence, when several investors are common across two firms, the later firm learns that the weighted combination of signals that entered the earlier firm's IPO decision was sufficiently favorable, without observing the individual signals that generated that decision.

In the empirical results: more common investors by direct investor overlap increases more the IPO of peers on my IPO.

Example 1: Firms a and b share two investors. Consider first the following network:



Here,

$$K_a = \{1, 2\}, \qquad K_b = \{1, 2, 3\}, \qquad K_c = \{3\}.$$

Since firm a acts first, its posterior for each investor is

$$\mu_a^k = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_a^k, \qquad k \in \{1, 2\}.$$

Last expression holds whenever we have identical distributions for ϵ_i^k.

Therefore, firm a goes public if and only if

$$\bar{\theta}_a + \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} \left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2\right) \geq W_a + \kappa_a,$$

or equivalently,

$$d_a = 1 \quad \Longleftrightarrow \quad \alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 \geq \frac{\sigma_\varepsilon^2 + \sigma_\eta^2}{\sigma_\varepsilon^2} \left(W_a + \kappa_a - \bar{\theta}_a\right).$$

Thus, after observing $d_a = 1$, firm b does not observe s_a^1 or s_a^2 separately. It learns that their weighted combination was sufficiently favorable for firm a to cross its IPO threshold.

Because firms a and b share investors 1 and 2, firm b updates both corresponding valuation components:

$$\mu_b^1 = \mathbb{E}_b\left[\varepsilon_b^1 \mid s_b^1, s_b^2, d_a\right], \qquad \mu_b^2 = \mathbb{E}_b\left[\varepsilon_b^2 \mid s_b^1, s_b^2, d_a\right].$$

However, investor 3 is not shared with firm a, so under independence across investors:

$$\mu_b^3 = \mathbb{E}_b\left[\varepsilon_b^3 \mid s_b^3\right] = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_b^3.$$

Firm b therefore goes public whenever

$$\bar{\theta}_b + \alpha_b^1 \mu_b^1 + \alpha_b^2 \mu_b^2 + \alpha_b^3 \mu_b^3 \geq W_b + \kappa_b.$$

For $\alpha_b^3 > 0$, this implies the investor-3 cutoff

$$s_b^{3*}(d_a, s_b^1, s_b^2) = \frac{\sigma_\varepsilon^2 + \sigma_\eta^2}{\alpha_b^3 \sigma_\varepsilon^2} \left[W_b + \kappa_b - \bar{\theta}_b - \alpha_b^1 \mu_b^1 - \alpha_b^2 \mu_b^2\right].$$

Hence, when $d_a = 1$ raises firm b's posterior assessment of the investors it shares with firm a, their combined contribution to $\widehat{P}_b$ increases and firm b requires a less favorable signal about investor 3 to go public.

Conditional on firm b's private signals, define

$$z_1 = \frac{\frac{\sigma_\varepsilon^2+\sigma_\eta^2}{\sigma_\varepsilon^2}\left(W_a+\kappa_a-\bar{\theta}_a\right)-\mathbb{E}_b\left[\alpha_a^1 s_a^1+\alpha_a^2 s_a^2 \mid s_b^1, s_b^2\right]}{\sqrt{\mathrm{Var}_b\left(\alpha_a^1 s_a^1+\alpha_a^2 s_a^2 \mid s_b^1, s_b^2\right)}}.$$

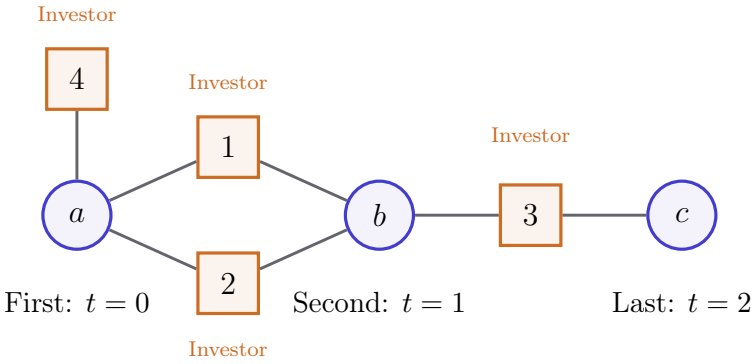
After observing $d_a = 1$, firm b updates its belief about investor 1:

$$\mu_b^1 = \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1] + \frac{\alpha_a^1 \,\mathrm{Cov}_b\left(\varepsilon_b^1, s_a^1 \mid s_b^1\right)}{\sqrt{\mathrm{Var}_b\left(\alpha_a^1 s_a^1+\alpha_a^2 s_a^2 \mid s_b^1, s_b^2\right)}}\lambda(z_1),$$

and similarly for investor 2,

$$\mu_b^2 = \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2] + \frac{\alpha_a^2 \,\mathrm{Cov}_b\left(\varepsilon_b^2, s_a^2 \mid s_b^2\right)}{\sqrt{\mathrm{Var}_b\left(\alpha_a^1 s_a^1+\alpha_a^2 s_a^2 \mid s_b^1, s_b^2\right)}}\lambda(z_1),$$

Example 2: Firms a and b share two investors, while firm a has an additional investor. Now consider



In this case,

$$K_a = \{1, 2, 4\}, \qquad K_b = \{1, 2, 3\}, \qquad K_c = \{3\}.$$

Firm a's IPO rule becomes

$$d_a = 1 \quad \Longleftrightarrow \quad \alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \geq \frac{\sigma_\varepsilon^2+\sigma_\eta^2}{\sigma_\varepsilon^2}\left(W_a+\kappa_a-\bar{\theta}_a\right).$$

Hence, observing $d_a = 1$ tells firm b that the weighted combination of firm a's three investor signals was sufficiently favorable. Firm b does not observe which of the three components generated the favorable IPO decision.

Because firm b shares only investors 1 and 2 with firm a, it uses d_a to update the components that are relevant for both firms:

$$\mu_b^1 = \mathbb{E}_b\left[\varepsilon_b^1 \mid s_b^1, s_b^2, d_a\right], \qquad \mu_b^2 = \mathbb{E}_b\left[\varepsilon_b^2 \mid s_b^1, s_b^2, d_a\right].$$

Investor 4 affects firm a's IPO decision but does not enter firm b's valuation. Consequently, the presence of investor 4 makes d_a a less direct signal about the components shared by firms a and b: firm a may have crossed its IPO threshold partly because its information about investor 4 was favorable.

Firm b nevertheless chooses according to

$$\bar{\theta}_b + \alpha_b^1 \mu_b^1 + \alpha_b^2 \mu_b^2 + \alpha_b^3 \mu_b^3 \geq W_b + \kappa_b,$$

with

$$\mu_b^3 = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + \sigma_\eta^2} s_b^3.$$

Thus, its investor-3 cutoff is again

$$s_b^{3*}(d_a, s_b^1, s_b^2) = \frac{\sigma_\varepsilon^2 + \sigma_\eta^2}{\alpha_b^3 \sigma_\varepsilon^2} \left[W_b + \kappa_b - \bar{\theta}_b - \alpha_b^1 \mu_b^1 - \alpha_b^2 \mu_b^2 \right].$$

The difference relative to the first example is that d_a now reflects information about an investor that firm b does not share. Therefore, although d_a still provides information about investors 1 and 2, the observed action is not generated exclusively by information about the investors common to firms a and b.

The corresponding standardized cutoff is

$$z_2 = \frac{\frac{\sigma_\varepsilon^2 + \sigma_\eta^2}{\sigma_\varepsilon^2} \left(W_a + \kappa_a - \bar{\theta}_a \right) - \mathbb{E}_b \left[\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2 \right]}{\sqrt{\mathrm{Var}_b \left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2 \right)}}.$$

Since investor 4 is independent of investors 1 and 2 and has zero mean, $\mathbb{E}_b[s_a^4 \mid s_b^1, s_b^2] = 0$. Hence, the numerator of z_2 is the same as the numerator of z_1.

However,

$$\mathrm{Var}_b \left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2 \right) = \mathrm{Var}_b \left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 \mid s_b^1, s_b^2 \right) + (\alpha_a^4)^2 \left(\sigma_\varepsilon^2 + \sigma_\eta^2 \right).$$

Therefore, the conditional variance is larger in Graph 2.

After observing $d_a = 1$, firm b updates investor 1 according to

$$\mu_b^1 = \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1] + \frac{\alpha_a^1 \, \mathrm{Cov}_b \left(\varepsilon_b^1, s_a^1 \mid s_b^1 \right)}{\sqrt{\mathrm{Var}_b \left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2 \right)}} \lambda(z_2),$$

and investor 2 according to

$$\mu_b^2 = \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2] + \frac{\alpha_a^2 \, \mathrm{Cov}_b \left(\varepsilon_b^2, s_a^2 \mid s_b^2 \right)}{\sqrt{\mathrm{Var}_b \left(\alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2 \right)}} \lambda(z_2).$$

Comparison. Investor 4 does not enter the covariance term relevant for updating either investor 1 or investor 2. Hence,

$$\mathrm{Cov}_b\left(\varepsilon_b^1, \alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2\right) = \alpha_a^1 \mathrm{Cov}_b\left(\varepsilon_b^1, s_a^1 \mid s_b^1\right),$$

and

$$\mathrm{Cov}_b\left(\varepsilon_b^2, \alpha_a^1 s_a^1 + \alpha_a^2 s_a^2 + \alpha_a^4 s_a^4 \mid s_b^1, s_b^2\right) = \alpha_a^2 \mathrm{Cov}_b\left(\varepsilon_b^2, s_a^2 \mid s_b^2\right).$$

Thus, relative to Graph 1, the covariance terms are unchanged while the conditional variance determining firm a's IPO is larger.

Moreover, if $z_1, z_2 > 0$, and the weights on the common investors α_a^1 and α_a^2 are held fixed across the two graphs, then $0 < z_2 < z_1$.

Since

$$\lambda(z) = \frac{\phi(z)}{1 - \Phi(z)}$$

is increasing in z,

$$\lambda(z_2) < \lambda(z_1).$$

Therefore, both the larger conditional variance and the smaller inverse Mills ratio reduce the posterior response of firm b. In particular,

$$\mu_b^1\big|_{\text{Graph 2}} - \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1] < \mu_b^1\big|_{\text{Graph 1}} - \mathbb{E}_b[\varepsilon_b^1 \mid s_b^1],$$

and

$$\mu_b^2\big|_{\text{Graph 2}} - \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2] < \mu_b^2\big|_{\text{Graph 1}} - \mathbb{E}_b[\varepsilon_b^2 \mid s_b^2].$$

Thus, the presence of an additional investor that is not shared with firm b makes firm a's IPO less informative about each of the investors that the two firms share.

F.5 A Binary-State Illustration of Informational Attenuation

Consider three firms a, b, c and two investors, with $K_a = \{1\}$, $K_b = \{1, 2\}$, and $K_c = \{2\}$. Each investor has an unobserved type $\varepsilon^k \in \{\varepsilon_L, \varepsilon_H\}$, where $\varepsilon_H > \varepsilon_L$. Investor types are independent, with prior probability $1/2$ of being high. Firms receive conditionally independent binary private signals, which correctly reveal investor type with probability $1 - \delta$, where $\delta \in (0, 1/2)$.

Firms decide whether to go public by comparing their expected valuation with a fixed exogenous IPO cost κ_i, as in the main model.

Suppose firm a goes public if and only if it receives a high signal about investor 1.

Observing its IPO raises the posterior probability of high investor quality:

$$\Pr(\varepsilon^1 = \varepsilon_H \mid d_a = 1) = \frac{(1-\delta)/2}{(1-\delta)/2 + \delta/2} = 1 - \delta > \frac{1}{2}.$$

Firm b is exposed to both investors. Its IPO decision satisfies

$$d_b = 1 \iff \bar{\theta}_b + \alpha_b^1 \mathbb{E}_b[\varepsilon^1] + \alpha_b^2 \mathbb{E}_b[\varepsilon^2] \geq \kappa_b.$$

Consider a nonempty region of fixed IPO costs in which, following $d_a = 0$, firm b goes public only when its signal about investor 2 is high, whereas following $d_a = 1$, it goes public when at least one of its two private signals is high. Such a region exists[14] when $\alpha_b^2 > \alpha_b^1/[2((1-\delta)^2 + \delta^2)]$, with the IPO cost chosen between the corresponding valuation thresholds.

When $d_a = 0$, observing $d_b = 1$ reveals a favorable signal about investor 2. Hence, $\Pr(\varepsilon^2 = \varepsilon_H \mid d_a = 0, d_b = 1) = 1 - \delta$.

When $d_a = 1$, firm b may instead go public because of favorable information about investor 1. In particular, the probability that its signal about investor 1 is high is $r = \Pr(s_b^1 = H \mid d_a = 1) = (1-\delta)^2 + \delta^2$. Conditional on investor 2 being high, firm b goes public with probability $r + (1-r)(1-\delta)$. Conditional on investor 2 being low, the corresponding probability is $r + (1-r)\delta$.

Applying Bayes' rule,

$$\Pr(\varepsilon^2 = \varepsilon_H \mid d_a = 1, d_b = 1) = \frac{[r + (1-r)(1-\delta)]/2}{[r + (1-r)(1-\delta)]/2 + [r + (1-r)\delta]/2} = \frac{r + (1-r)(1-\delta)}{1+r}.$$

Comparing the two histories yields $(1-\delta) - \frac{r+(1-r)(1-\delta)}{1+r} = \frac{r(1-2\delta)}{1+r} > 0$. Therefore, firm b's IPO conveys less favorable information about investor 2 after firm a has already gone public. Its IPO may now be explained by information about investor 1, reducing the posterior update about investor 2 and attenuating the valuation spillover to firm c.

F.6 Extension: IPO underpricing

The baseline model of three firms a, b, c and two investors $1, 2$ abstracts from the distinction between the IPO offer price and the public-market valuation after listing. In particular, the firm-specific component of value does not play an informational role in the baseline: the analysis focuses on how private firms learn about the public market's assessment of their investors. This subsection introduces a second, firm-specific source

[14] Investor 2's contribution to firm b's valuation must be sufficiently important relative to investor 1's contribution.

of information in order to distinguish the valuation at the offer stage from the valuation after the firm lists.

Recall that the firm-specific component of value is denoted by θ_b. For this extension, assume

$$\theta_b \sim N(\bar{\theta}_b, \sigma_\theta^2),$$

independently of the investor-specific valuation components. The prior mean $\bar{\theta}_b$ is firm specific and summarizes the information about firm b's fundamentals that is already incorporated before the IPO. Thus, the market does not treat firms as ex ante identical: firms that are expected to have stronger fundamentals can already receive higher valuations before listing. The uncertainty in θ_b instead captures residual firm-specific information that has not yet been resolved.

The informational role of θ_b differs from that of ε_b^k. As in the baseline model, the public market observes the relevant investor-specific valuation components ε_b^k, while firm b observes them imperfectly and learns about them through its signals and previous peer actions. In this extension, θ_b is the firm's own underlying type, capturing its quality and growth prospects. The firm knows the distribution of its type at the offer stage, but its exact realization is resolved only before the final IPO completion decision. Once the firm lists, this information becomes publicly observable.

This timing creates an additional selection effect. Conditional on the offer-stage information, firm b completes the IPO only when its realized fundamental component is sufficiently favorable. Hence, observing that the IPO is completed conveys information about θ_b. Moreover, because a previous connected-peer IPO raises firm b's expected valuation through the investor-learning channel established in the baseline model, it lowers the realization of θ_b required for firm b to complete the IPO. The IPO decision is therefore less selective with respect to firm-specific fundamentals following favorable peer exposure. This implication will generate the underpricing result below.

Let me use my model timing and focus on firm b. This firm first observes its baseline information defined as $\mathcal{I}_b = \{s_b^1, s_b^2, d_a\}$. I assume that the issuer and underwriter set the IPO offer price using the information available to the firm at that stage. Then, the price reflects the expected firm-specific component and the firm's posterior beliefs about how the public market values its investors. I abstract from additional information revealed during bookbuilding and assume that the offer price (set by issuer and underwriters conditional on issuer's information) cannot be revised after the firm learns its type.

$$P_b^O = \bar{\theta}_b + \alpha_b^1 \mu_b^1 + \alpha_b^2 \mu_b^2, \tag{41}$$

This offer price is exactly the price in my main model, where

$$\mu_b^1 = \mathbb{E}_b\left[\varepsilon_b^1 \mid s_b^1, d_a\right], \qquad \mu_b^2 = \mathbb{E}_b\left[\varepsilon_b^2 \mid s_b^2\right].$$

After the offer price has been determined, the realization of θ_b becomes known to firm b before its final decision to complete the IPO. This is of the differences with my baseline model (θ is private information), and the second difference is that now I examine price at listing as the only measure to determine whether underpricing exists. Firm b therefore completes the IPO whenever

$$\theta_b + \alpha_b^1\mu_b^1 + \alpha_b^2\mu_b^2 \geq W_b + \kappa_b. \tag{42}$$

Then the firm takes into account its type when deciding on IPO.

Equivalently, the IPO is completed if $\theta_b \geq \theta_b^*(d_a, s_b^1, s_b^2)$, where

$$\theta_b^*(d_a, s_b^1, s_b^2) = W_b + \kappa_b - \alpha_b^1\mu_b^1 - \alpha_b^2\mu_b^2. \tag{43}$$

Once the firm lists, θ_b becomes publicly observed. As in the baseline model, the public market observes the investor-specific valuation components. The public-market valuation after listing is set competitively as in the benchmark:

$$P_b = \theta_b + \alpha_b^1\varepsilon_b^1 + \alpha_b^2\varepsilon_b^2. \tag{44}$$

Proposition 5 (Peer IPO exposure and IPO underpricing)**.** *Fix firm b's own signals (s_b^1, s_b^2). Observing the IPO of connected firm a lowers the firm-specific fundamental required for firm b to complete its IPO:*

$$\theta_b^*(1, s_b^1, s_b^2) < \theta_b^*(0, s_b^1, s_b^2).$$

Under the previous result, conditional on firm b completing its IPO, expected percentage underpricing is strictly lower following $d_a = 1$ than following $d_a = 0$. Aligned with Aghamolla and Thakor (2022) reduction in underpricing is seen as a key consequence of learning across firms.

Proof. The baseline direct-learning result implies, holding s_b^1 fixed, $\mu_b^1\big|_{d_a=1} > \mu_b^1\big|_{d_a=0}$. Since $\alpha_b^1 > 0$, equation (43) therefore implies

$$\theta_b^*(1, s_b^1, s_b^2) < \theta_b^*(0, s_b^1, s_b^2).$$

Because θ_b is normally distributed, conditional on completing the IPO,

$$\mathbb{E}_b\left[\theta_b \mid \mathcal{I}_b, d_b = 1\right] - \bar{\theta}_b = \sigma_\theta \lambda\left(\frac{\theta_b^*(d_a, s_b^1, s_b^2) - \bar{\theta}_b}{\sigma_\theta}\right), \tag{45}$$

where $\lambda\dot{()}$ is the upper-tail inverse Mills ratio or hazard rate.

By the standard monotonicity property of the normal hazard rate, $\lambda(\cdot)$ is strictly increasing. Hence

$$\theta_b^*(1, s_b^1, s_b^2) < \theta_b^*(0, s_b^1, s_b^2)$$

implies

$$\mathbb{E}_b\left[\theta_b \mid s_b^1, s_b^2, d_a = 1, d_b = 1\right] - \bar{\theta}_b < \mathbb{E}_b\left[\theta_b \mid s_b^1, s_b^2, d_a = 0, d_b = 1\right] - \bar{\theta}_b. \tag{46}$$

Realized percentage underpricing is $\frac{P_b - P_b^O}{P_b^O}$. Then, conditional on $\mathcal{I}_b$ and on completing the IPO, the completion decision reveals information about θ_b but no additional information about the investor-specific components beyond $\mathcal{I}_b$. Therefore,

$$\mathbb{E}_b\left[\frac{P_b - P_b^O}{P_b^O}\middle| \mathcal{I}_b, d_b = 1\right] = \frac{\mathbb{E}_b\left[\theta_b \mid \mathcal{I}_b, d_b = 1\right] - \bar{\theta}_b}{P_b^O}. \tag{47}$$

Finally, the baseline learning result also implies

$$P_b^O\big|_{d_a=1} > P_b^O\big|_{d_a=0}.$$

Thus, following $d_a = 1$, the numerator in (47) is strictly smaller while the denominator is larger. Therefore, the expected percentage underpricing is strictly lower following the connected peer IPO.

The result highlights two channels. First, peer IPO learning raises the offer price and lowers the minimum firm-specific type required to complete an IPO, reducing positive selection on firm quality and expected percentage underpricing. Second, given that IPO completion also reveals information about firm quality, the extension introduces an additional informational channel without explicitly modeling strategic signaling.

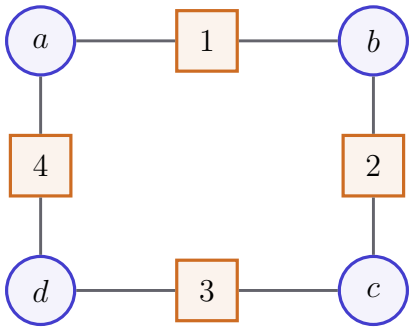
Notice that this section introduces information revelation where high-quality firms are more prone to go IPO. When θ_i are uncorrelated across firms, there is no potential learning from types, but learning from ϵ_i^k gains become smaller.

An empirical analysis of underpricing is not feasible with the available data because I observe the firm's first-day market price but not its IPO offer price. Consequently, the observed price outcome captures the firm's public-market valuation after listing rather than the difference between the offer price and the first-day market price that conventionally

defines IPO underpricing (information on should be in Pitchbook and SDC Platinum).

F.7 Extension: Four Firm Network

Going back to my model, with θ being unknown for firms and public market, I will now explore the model under four-firm network. Firms are $i = \{a, b, c, d\}$ and investors are $k = \{1, 2, 3, 4\}$.



Now each firm obtains a lower informational gain whenever a peer did IPO. This is related to the fact that now $Var(s_i^k|s_j^k)$, where i, j are peers and get their IPO opportunities at different times, is greater due to having an extra investor.

However, when two peers receive their IPO opportunity at the same t, in case they choose to IPO, the firm i will get a higher learning than, keeping everything constant, having only one peer while the peer having always the same number of investors. This is related to the investor overlap result in F.4.

Under this example, now we need to define the covariance structure whenever the peer can go IPO the same year, one year before, or one year after you do. We have $t = 0, 1, 2$ and valuations associated with the same investor are positively correlated across connected firms, but this covarianceweakens with the time separating their IPO opportunities.

Let t_i denote the period in which firm i receives its IPO opportunity. For any two firms i and j backed by the same investor k, assume

$$\text{Cov}(\varepsilon_i^k, \varepsilon_j^k) = \sigma_\varepsilon^2 \rho(|t_i - t_j|), \tag{48}$$

where $\rho(h) > 0, \rho(h+1) \leq \rho(h), \rho(0) = 1$. Thus, the public-market valuations associated with the same investor remain positively related across portfolio firms, but the relationship becomes weaker the further apart the firms are in time. Investor-specific components are independent across different investors.

A general network has no closed form solutions when more than one have the IPO opportunity at the same period t given the truncated normal distributions.

Now consider, without loss of generality, a configuration in which firm a's three peers, firms b, c, and d, receive their IPO opportunities simultaneously at $t = 0$, while firm a

receives its IPO opportunity at $t = 1$. The same information mechanism can generate diminishing responses to multiple peer IPOs. The three peers are also connected to one another through a common investor. Because their IPO decisions are simultaneous, they do not condition on one another's contemporaneous action. However, their decisions are statistically related because they are partly driven by information about the same underlying investor component.

I evaluate this extension computationally using deterministic numerical integration. For each possible realization of the common investor component, I compute the probability of each peer completing an IPO under its equilibrium decision rule. Firm a does not observe the peers' private signals, so I integrate over this unobserved information to obtain firm a's posterior assessment of the common investor conditional on the number of observed peer IPOs.

Figure **??** reports the incremental posterior gain associated with each additional peer IPO. The information gain is positive but decreasing: the first peer IPO produces the largest update, while the second and third IPOs provide progressively less new information. Intuitively, once one connected peer has gone public, part of the information contained in subsequent IPOs has already been revealed.

This diminishing-information pattern provides a theoretical interpretation for two related empirical findings. First, in the nonlinear specification, the coefficient on the peer IPO ratio is positive while the coefficient on its square is negative. This is consistent with a positive but declining marginal response to additional peer IPO exposure: as more connected peers go public, each additional IPO contributes less new information.

Second, the clique results point in the same direction. When the relevant IPO peers form a pure clique; so that the IPO peers are all mutually connected through investors that are also relevant to the focal firm, the marginal association between peer IPO exposure and the focal firm's IPO decision is substantially smaller. In such a configuration, the observed peer IPOs are more likely to contain overlapping information, so that an additional IPO conveys less incremental news than when the IPO peers are less interconnected.

Taken together, the negative curvature in peer IPO exposure and the attenuated peer effect within fully connected IPO cliques are consistent with the model's prediction that overlapping network signals generate diminishing informational content. The numerical exercise is illustrative rather than a structural calibration, so I interpret these correspondences qualitatively rather than as predictions of the empirical coefficient magnitudes. In the following Figure, the three peers are connected through the common investor and make their IPO decisions simultaneously. The declining increments illustrate informational redundancy: successive connected-peer IPOs remain favorable news but convey progressively less new information.

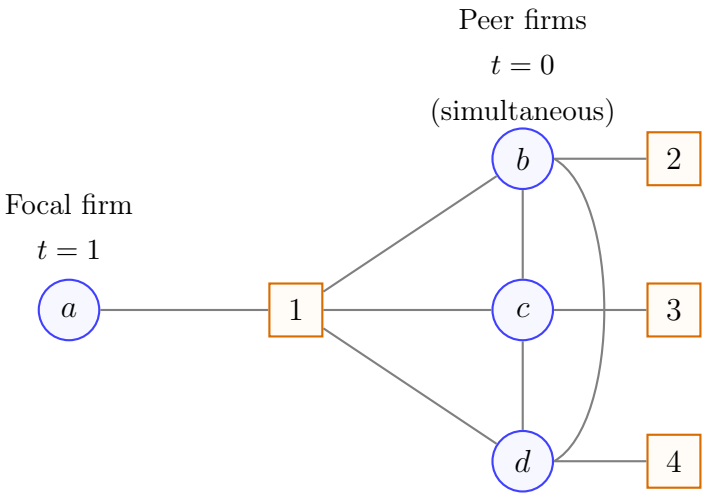


Figure 8: **Common-investor network with simultaneous peer IPO opportunities.** Circles represent firms and squares represent investors. Firms b, c, and d receive their IPO opportunities simultaneously at $t = 0$, while focal firm a receives its opportunity at $t = 1$. Investor 1 is common to all four firms; investors 2, 3, and 4 are specific to firms b, c, and d, respectively. Lines between firms highlight peer connections induced by investor 1, rather than additional investor relationships.

Each peer firm assigns weight $\alpha_{\text{common}} = 0.50$ to investor 1, who is shared by all peers and the focal firm a, and weight $\alpha_{\text{unique}} = 0.50$ to its own peer-specific investor.

F.8 Alternative: Market Learning from Firms

An alternative approach is a mechanism in which firms having more information than the market. Under the same assumption on θ being unknown for the public market and firms, we define now ϵ_i^k as the unobserved investor ability k in firm i. This can be related to certification and reputation roles of investors that have been widely mentioned in papers like Megginson and Weiss (1991) and C. Krishnan et al. (2011). In this setup, firms have imperfect information about investor ability, and the market has a noisier signal for each investor. Hence, when the market, given its information set, define the price that is willing for offer to the firm, it needs to consider that, if a peer did IPO, this speaks positively about how good the investors are. Consequently, private peer firms will expect a higher price by the market once peers IPOs are observed. However, under this setting firms face a fixed-point problem and a signaling by IPO choices. Hence, IPO decisions in a period should affect how the market values your firm. Consequently, IPO and market prices need to be jointly determined in equilibrium.

To limit adverse-selection concerns, it is needed to assume that the public market observes the same underlying signal about investor ability as private firms, but with additional mean-zero noise. Consequently, firms receiving more favorable signals also expect the market to assign higher valuations to their investors, making them more willing

to go public. This preserves a positive relationship between private information and expected market valuations, while allowing the public market to learn from IPO decisions.